Turing Machines

FOCS, Fall 2020

Until now

- Decision functions
- Formal languages
- Finite state machines

A decision function $f : \Sigma^* \rightarrow \{true, false\}$ is FA-computable if there exists a finite state machines M such that f(u) = true exactly when M accepts string u

- f is FA-computable if the associated set A_f is regular

Not every language is regular — including some that correspond to easily computable functions (for us)

These languages are not regular

$\{a^n b^n \mid n \geq 0\} = \{ \varepsilon, ab, aabb, aaabbb, aaaabbbb, \ldots. \}$

$\{a^n b^n c^n \mid n \geq 0\}$

$\{a^n b^m \mid m \geq n \geq 0\}$

$\{u \in \{a,b\}^* \mid \#_a(u) = \#_b(u) \}$ $\#_x(u)$ = number of x in u

$\{u \in \{a,b\}^* \mid u = rev(u)\}$ rev(u) = u reversed

Intuitively, finite state machines cannot remember an arbitrary amount of information (natural number = arbitrary amount of information - think digits)

Why is $\{a^n b^n \mid n \geq 0\}$ not regular?

We argue by contradiction. Assume $\{a^n b^n \mid n \geq 0\}$ is regular. We derive a contradiction from that fact.

If $\{a^n b^n \mid n >= 0\}$ is regular, there is a deterministic finite state machine that accepts it. Call it M. Machine M has some number K of states.

Consider the string $a^{K+1}b^{K+1}$. This string is accepted by M. When following a^{K+1} you go through K+1 states - so two of those states must be the same state S. So you have a loop of length L going from S to S. That loop forces M to also accept $a^{K+1+L}b^{K+1}$ by going around the loop one time more. And $a^{K+1+L}b^{K+1}$ is not in the language. This contradicts M accepting the language $\{a^n b^n \mid n \geq 0\}$.

Our assumption was wrong: $\{a^n b^n \mid n \geq 0\}$ is not regular.

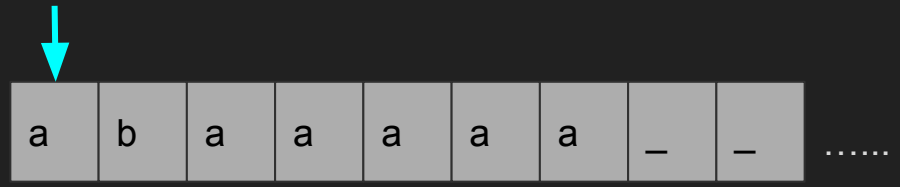
Adding memory to finite state machines

Let's give finite state machines some storage space (memory).

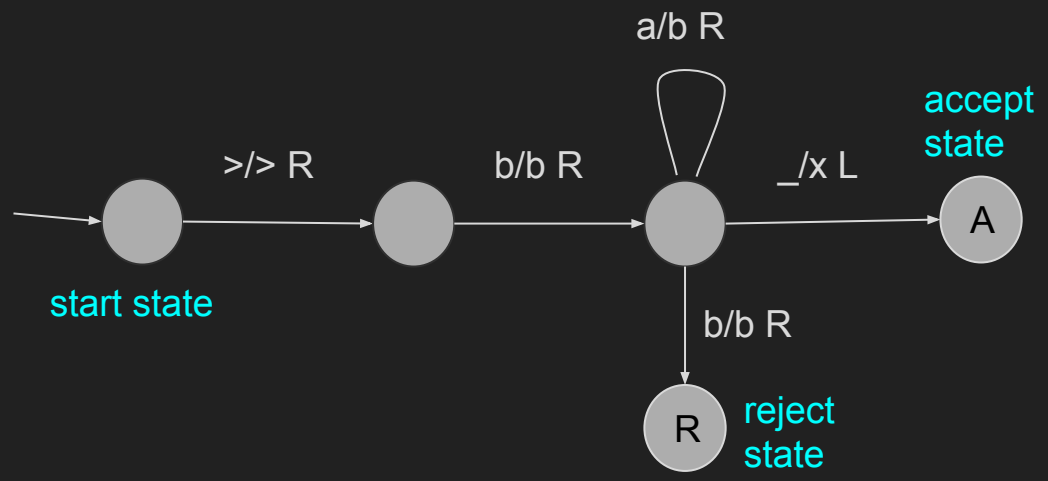
We're used to memory being a large array that you can index into (address):

a	b	a	a	a	a	a	b	b	.....
0	1	2	3	4	5	6	7	8	

That requires numbers, and numbers are complicated. Instead, we use a pointer that we can move left or right to go to a cell that we want to read/modify:



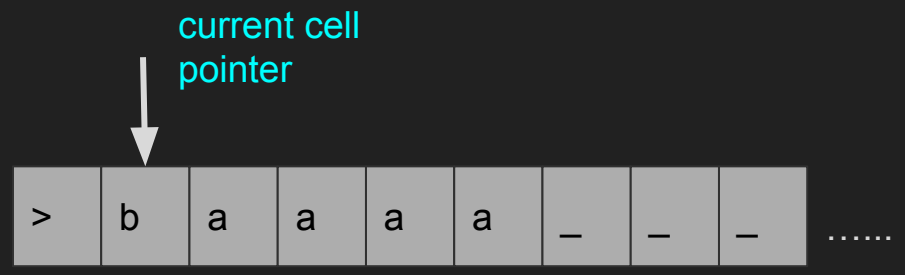
Turing Machines



Finite set of states with labeled transitions between them

Labels of the form x/y D
"when on x, rewrite into y and move in direction D = L or R"

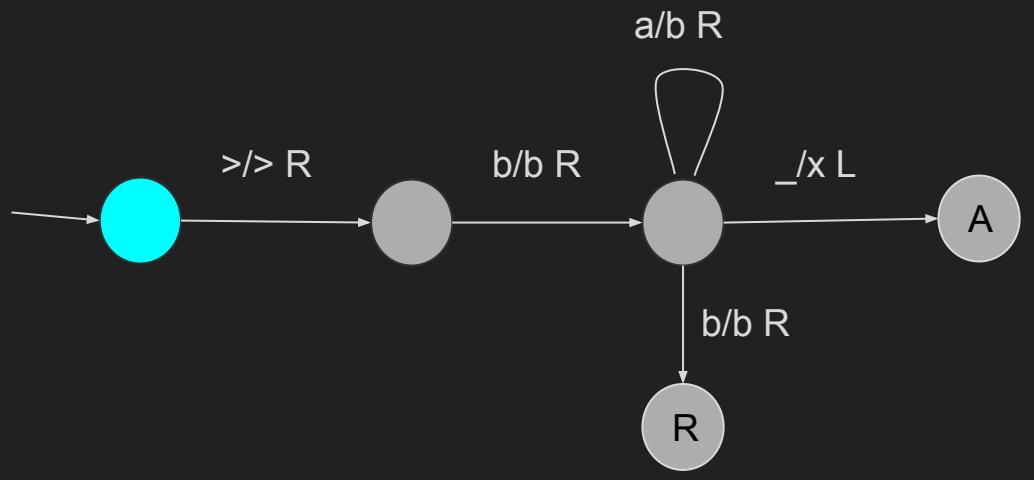
Deterministic - one transition with a given symbol out of every state

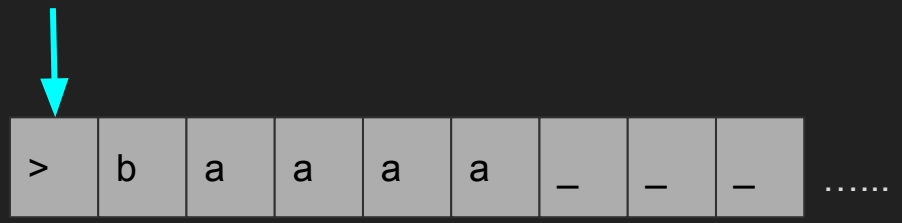


tape initialized with > followed by input string

convention: if no transition with a given symbol, transition to reject state

Running the Turing machine





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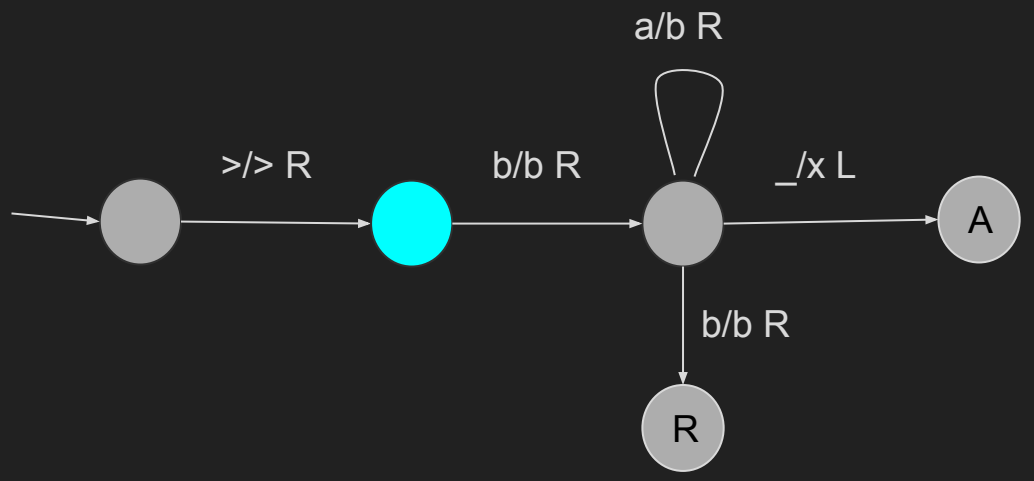

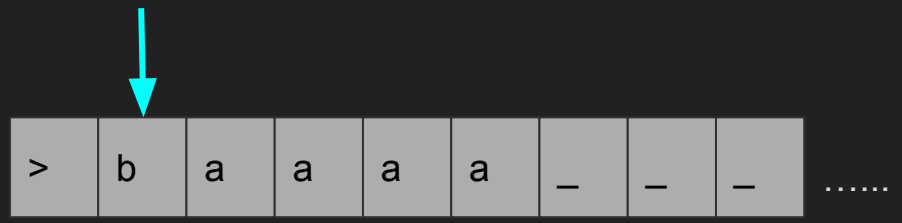

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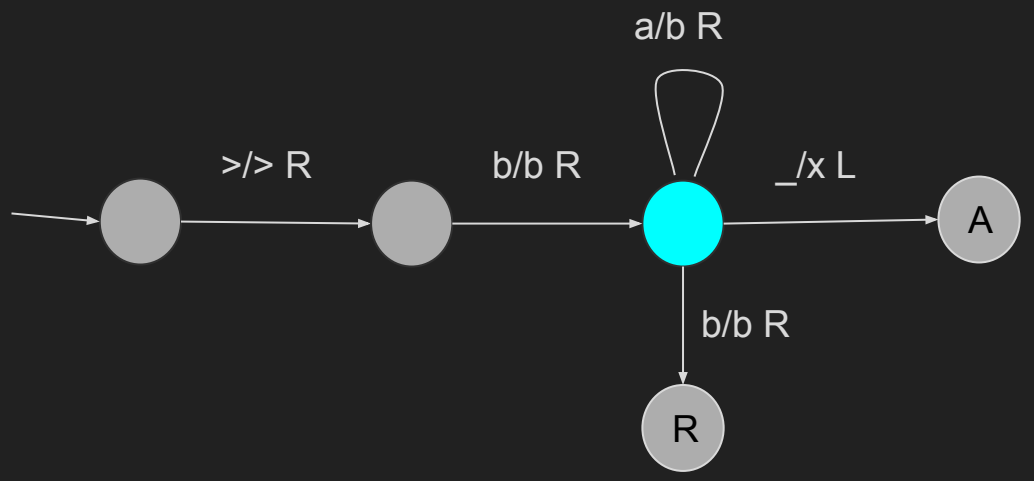

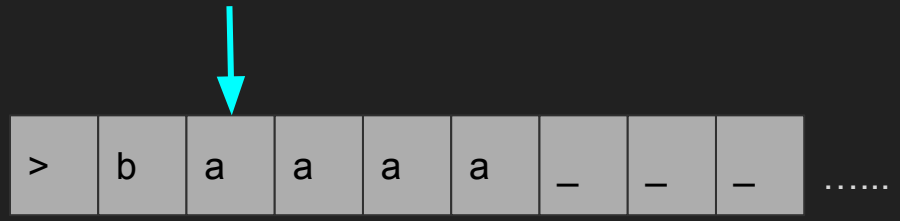

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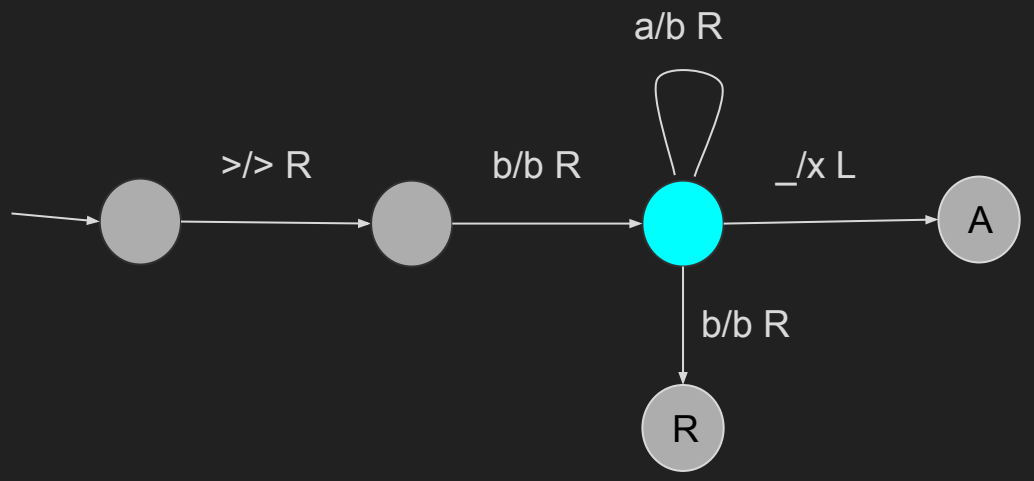

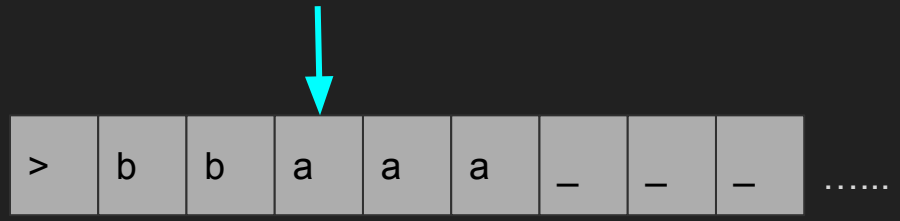

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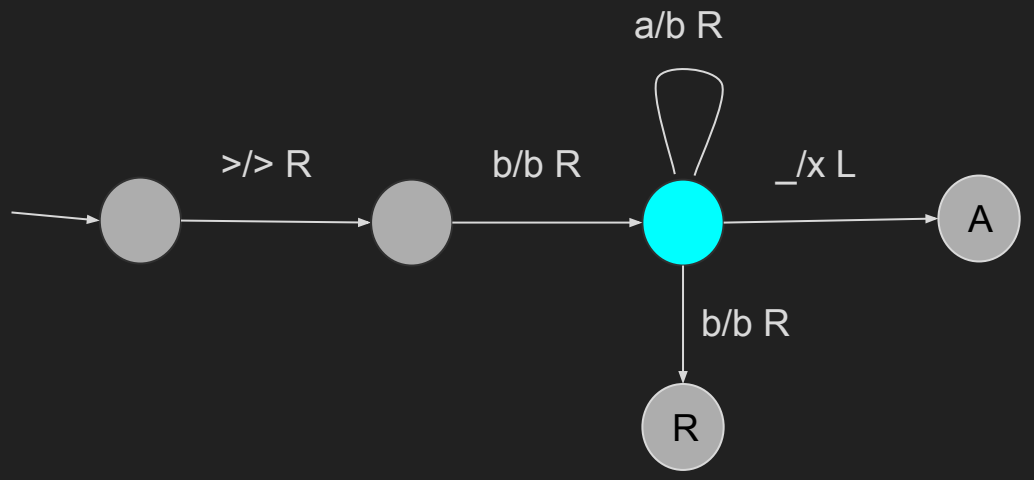

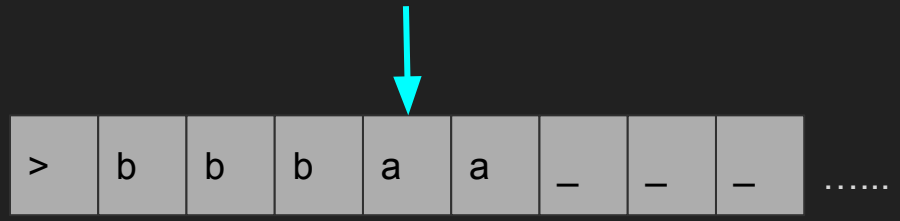

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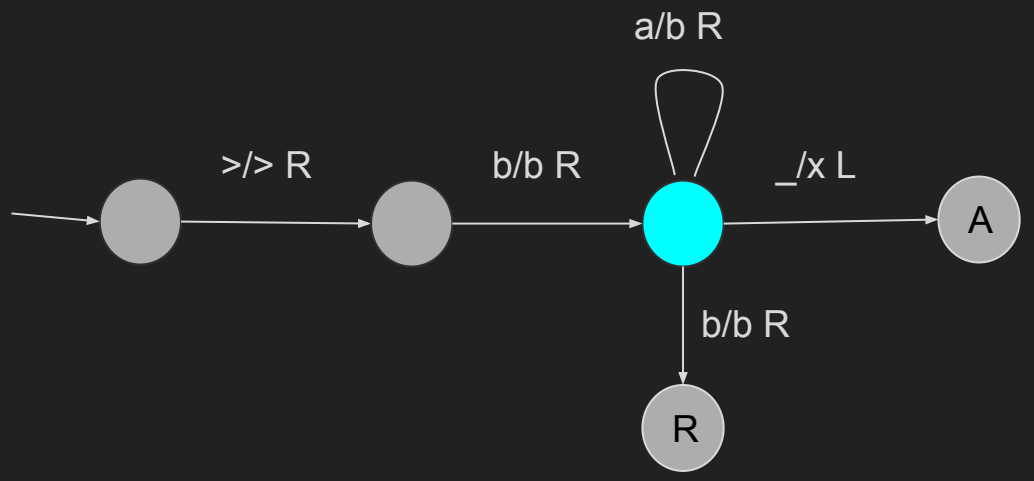

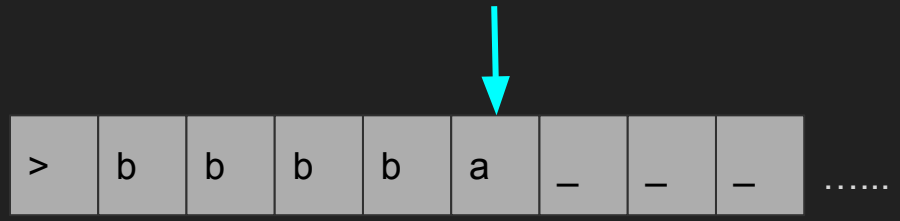

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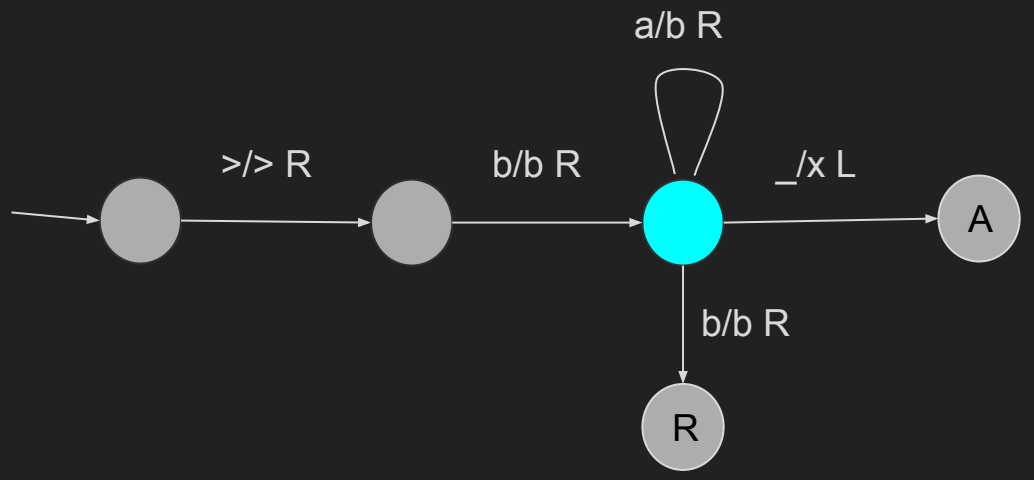

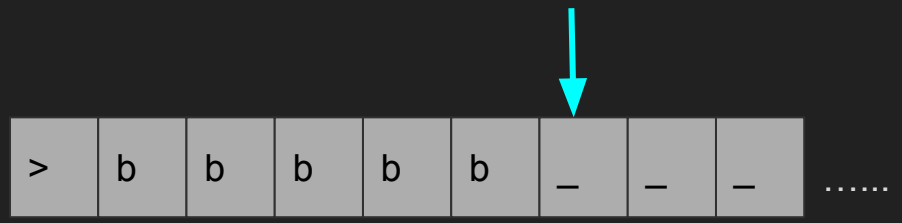

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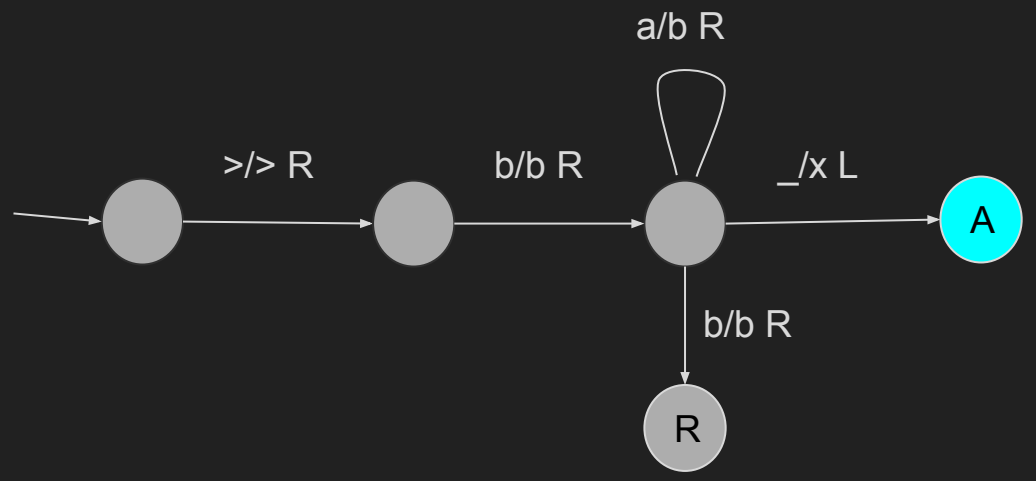
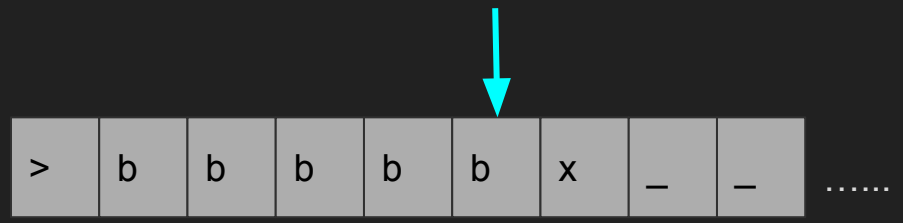


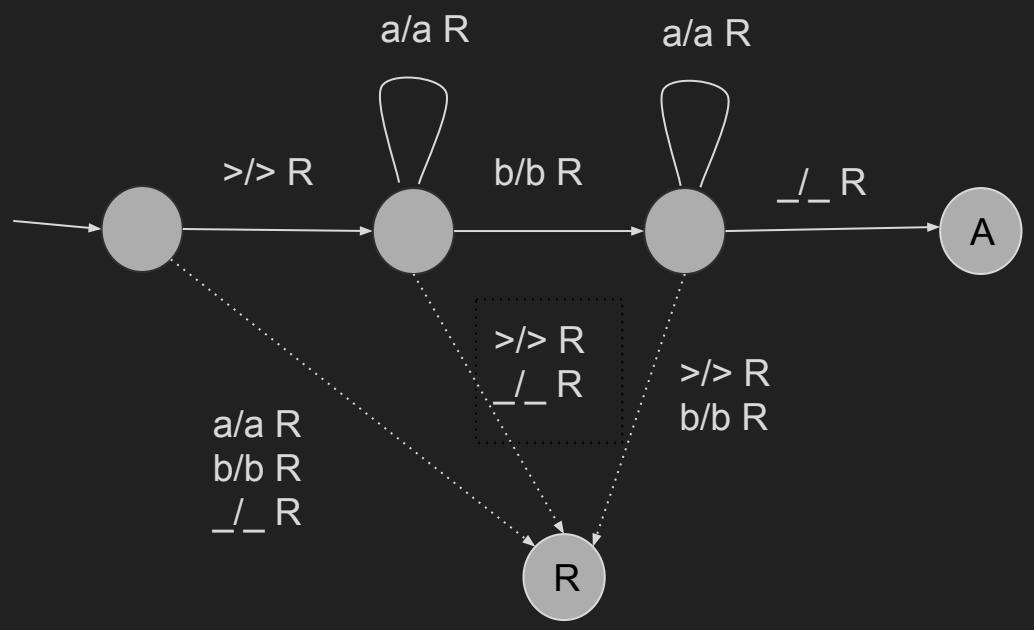


If you ever reach the accept state, ACCEPT the original input string

If you ever reach the reject state, REJECT the original input string

You may never reach the accept or reject state — you keep going forever

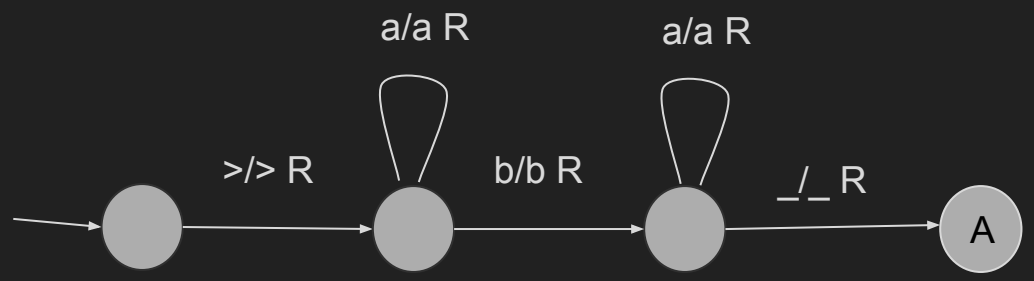
Example — a*ba*



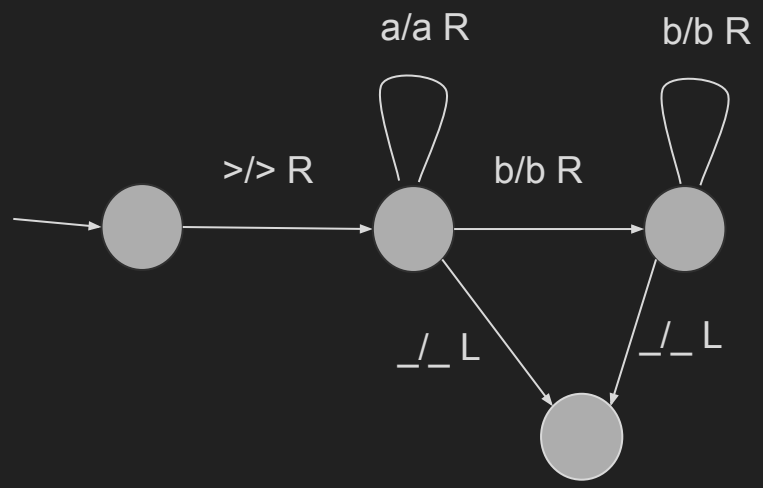
If a language is regular, you can build a TM that accepts all strings in the language by mimicking a deterministic finite state machine that accepts the language

That TM doesn't need to change the tape, and can always move the pointer Right

Example — a*ba*



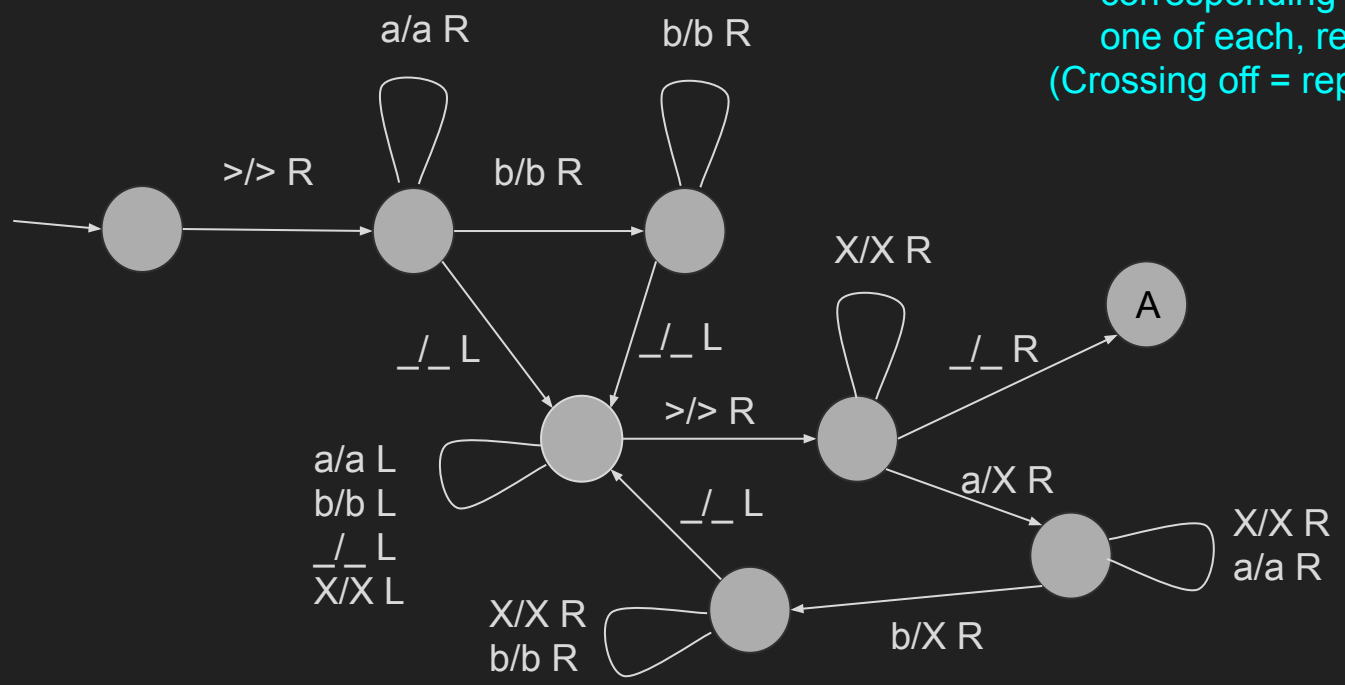
Example — $\{a^n b^n \mid n \geq 0\}$

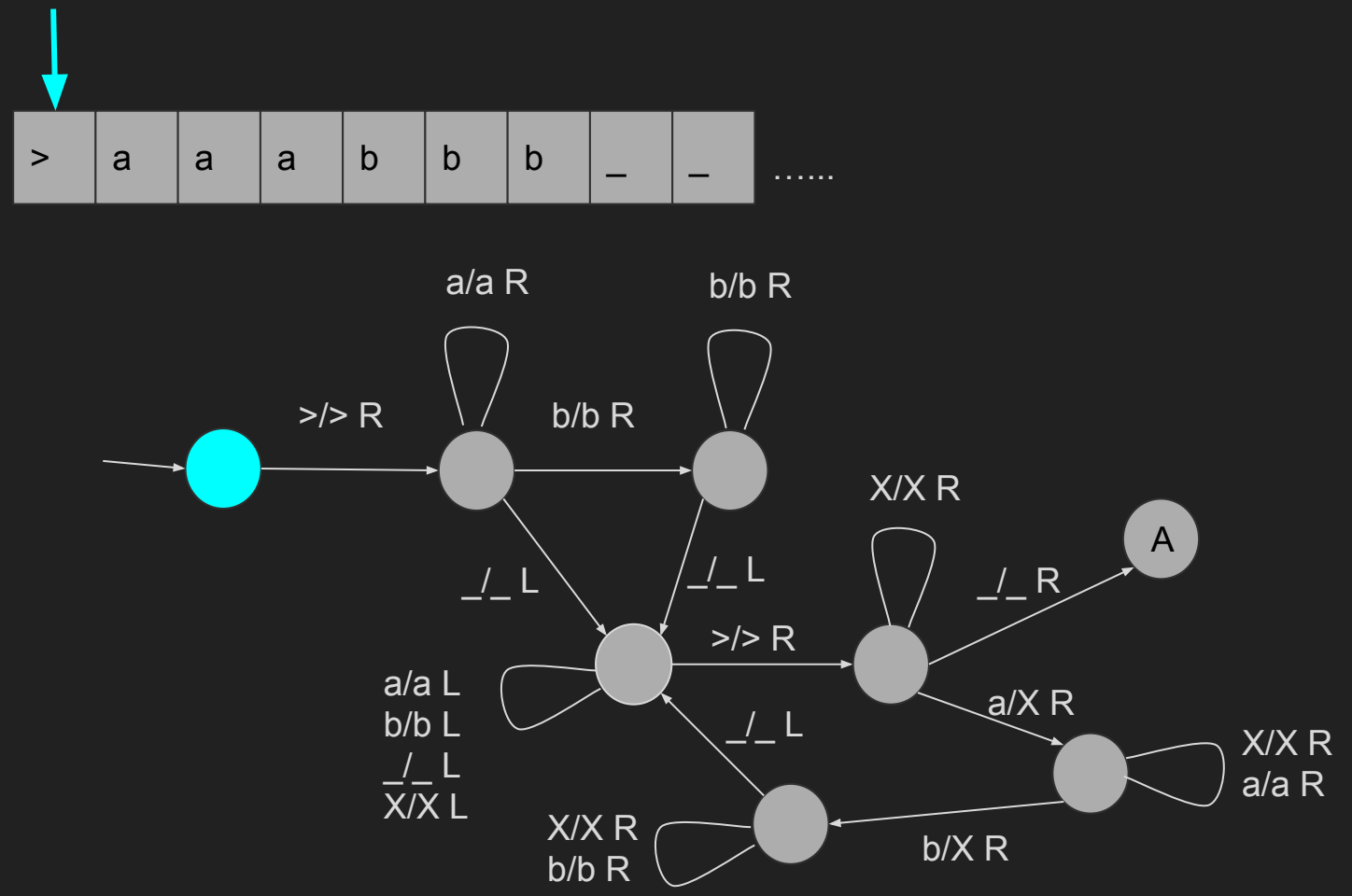


Intuition:
(1) check that we have a*b*

Example — $\{a^n b^n \mid n \geq 0\}$

Intuition:
(1) check that we have a*b*
(2) check that every a has a corresponding b by crossing off one of each, repeatedly
(Crossing off = replace by X)

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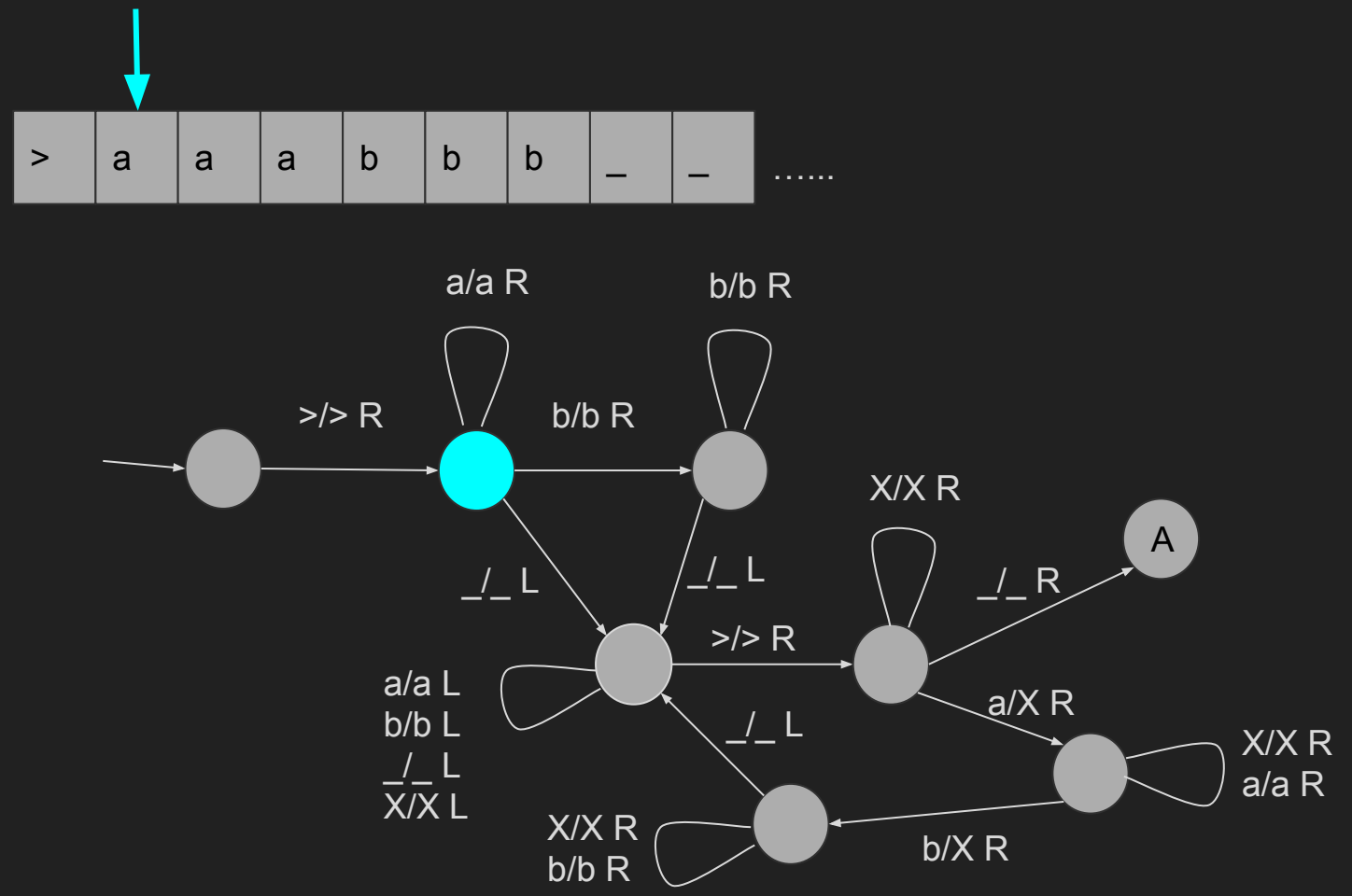
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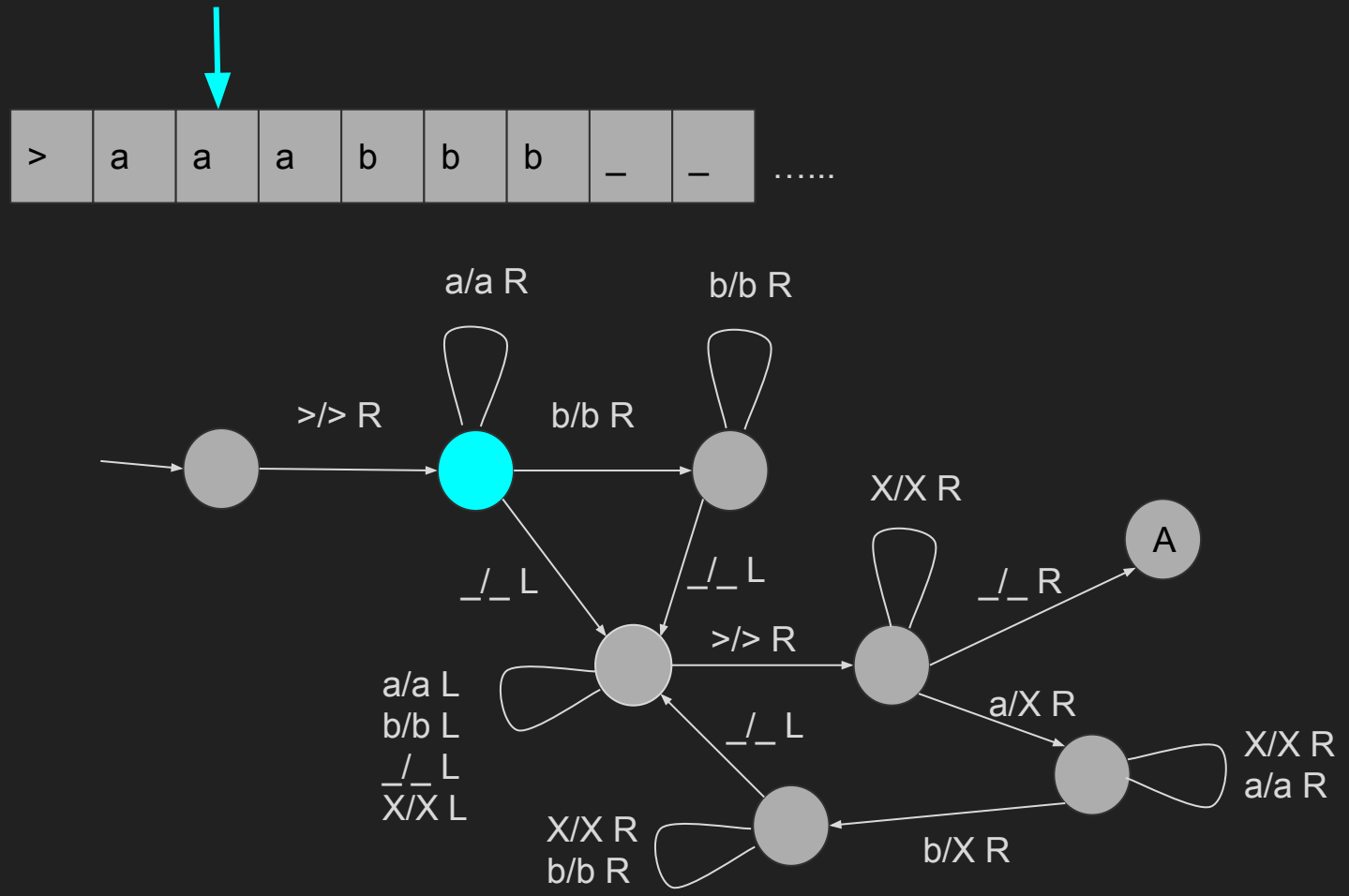
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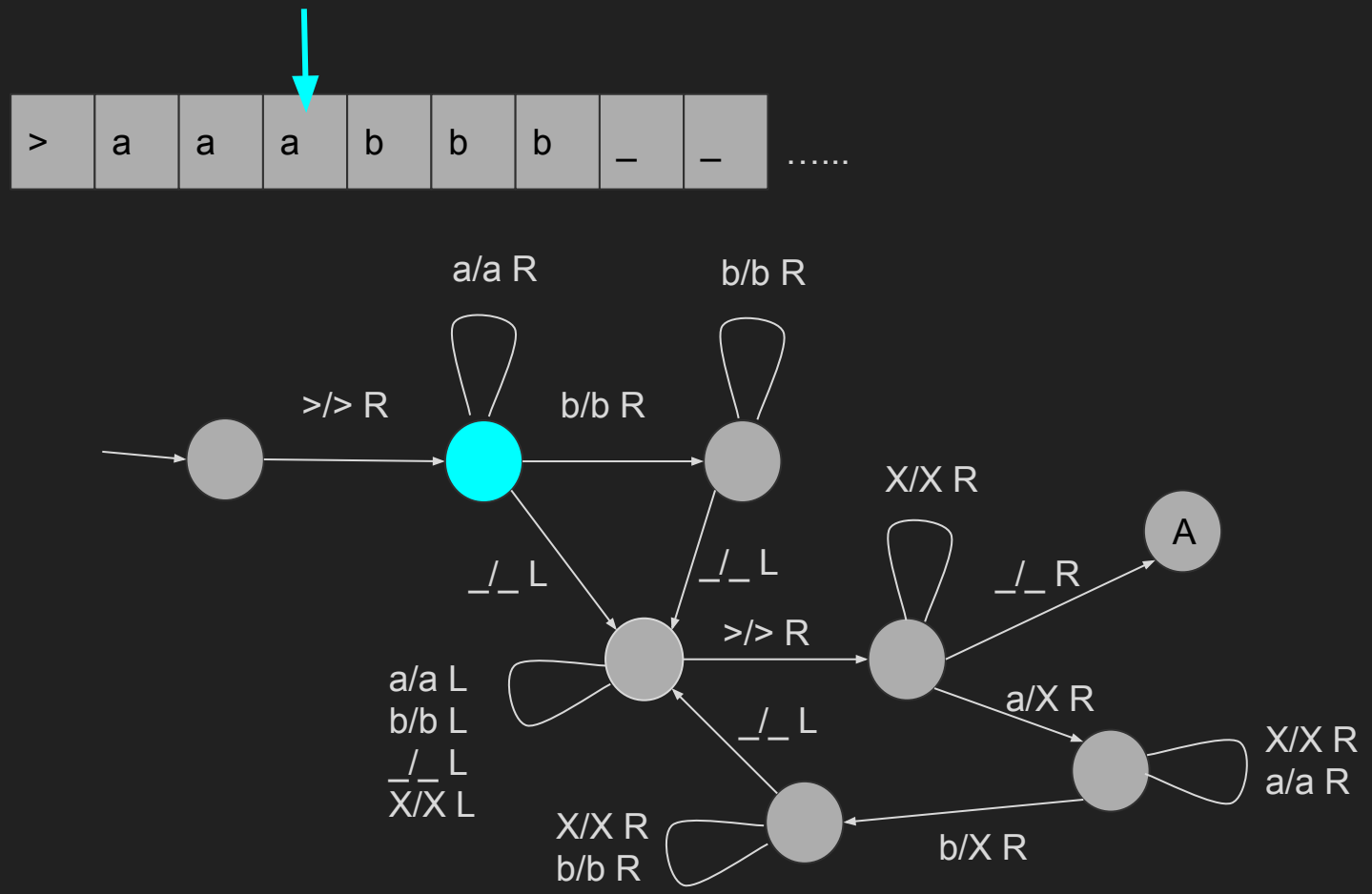
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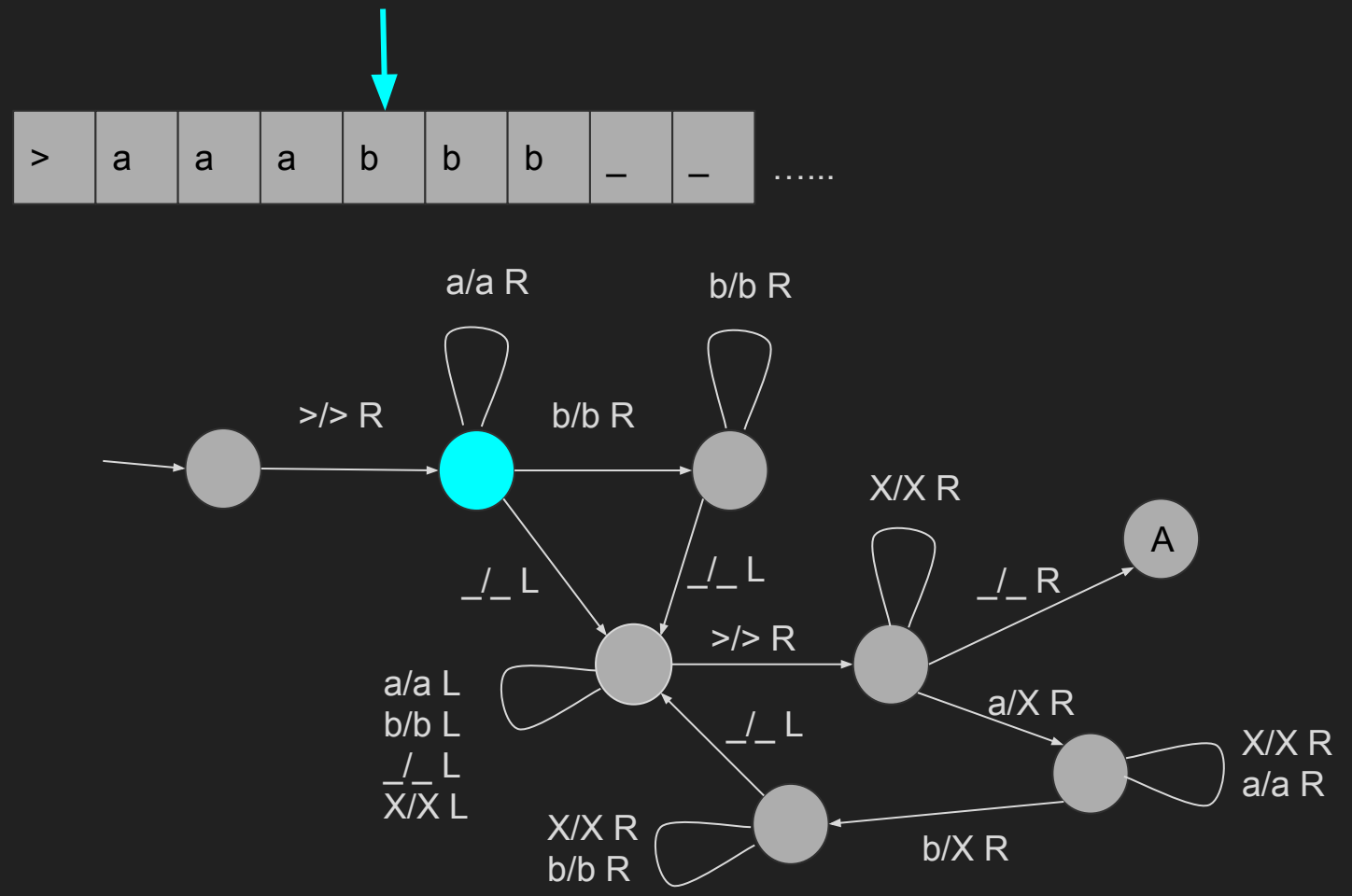
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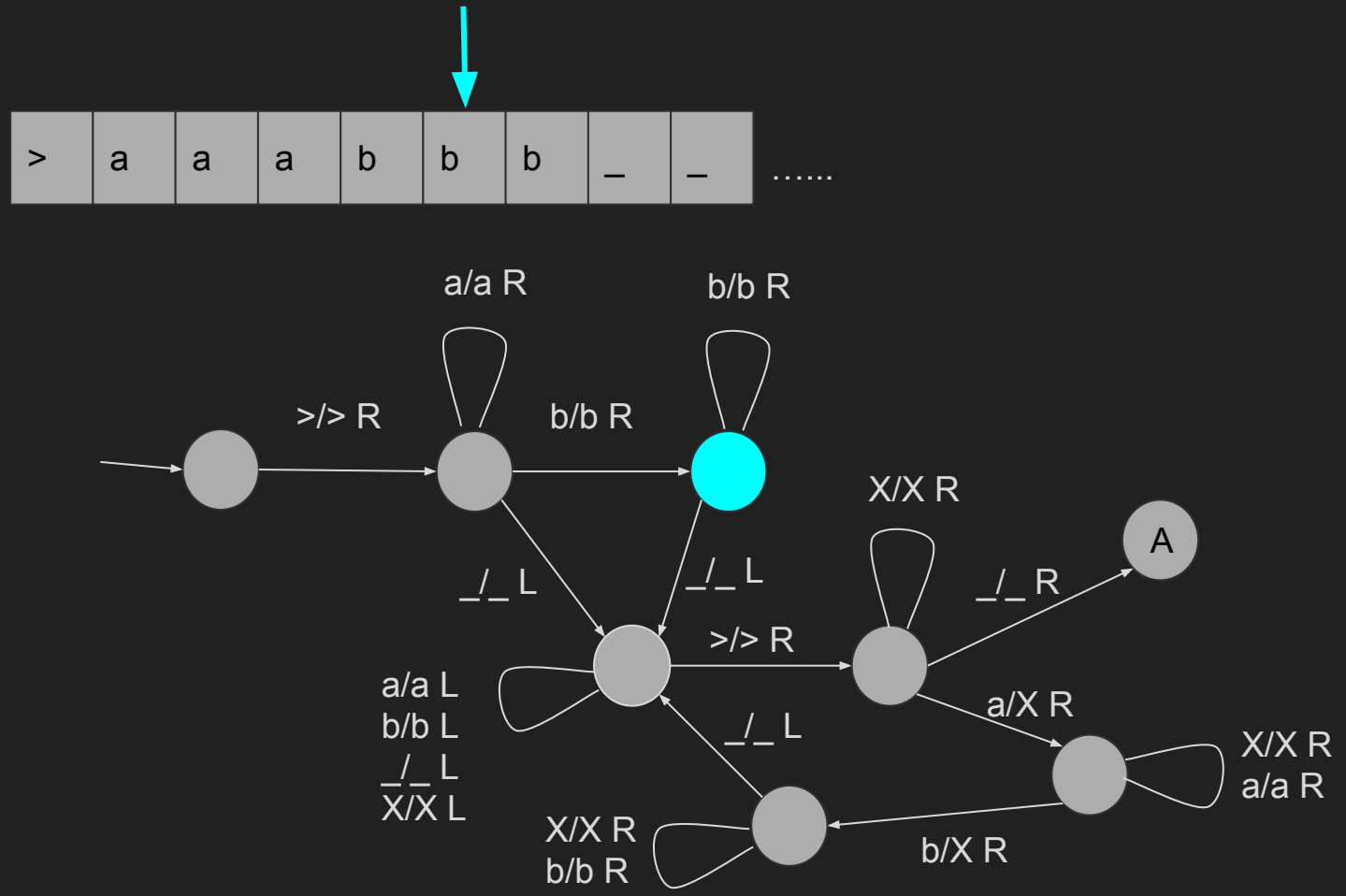
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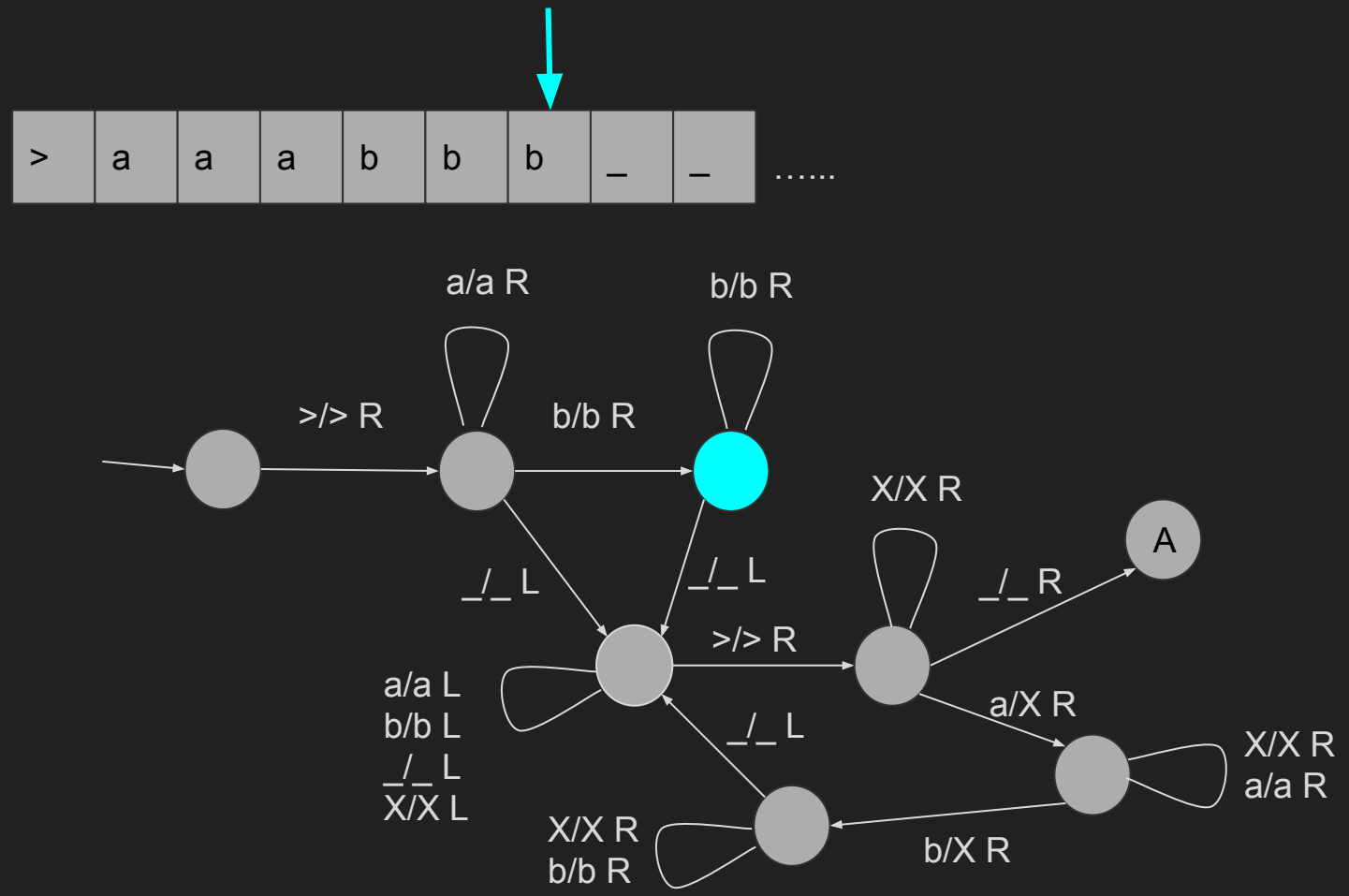
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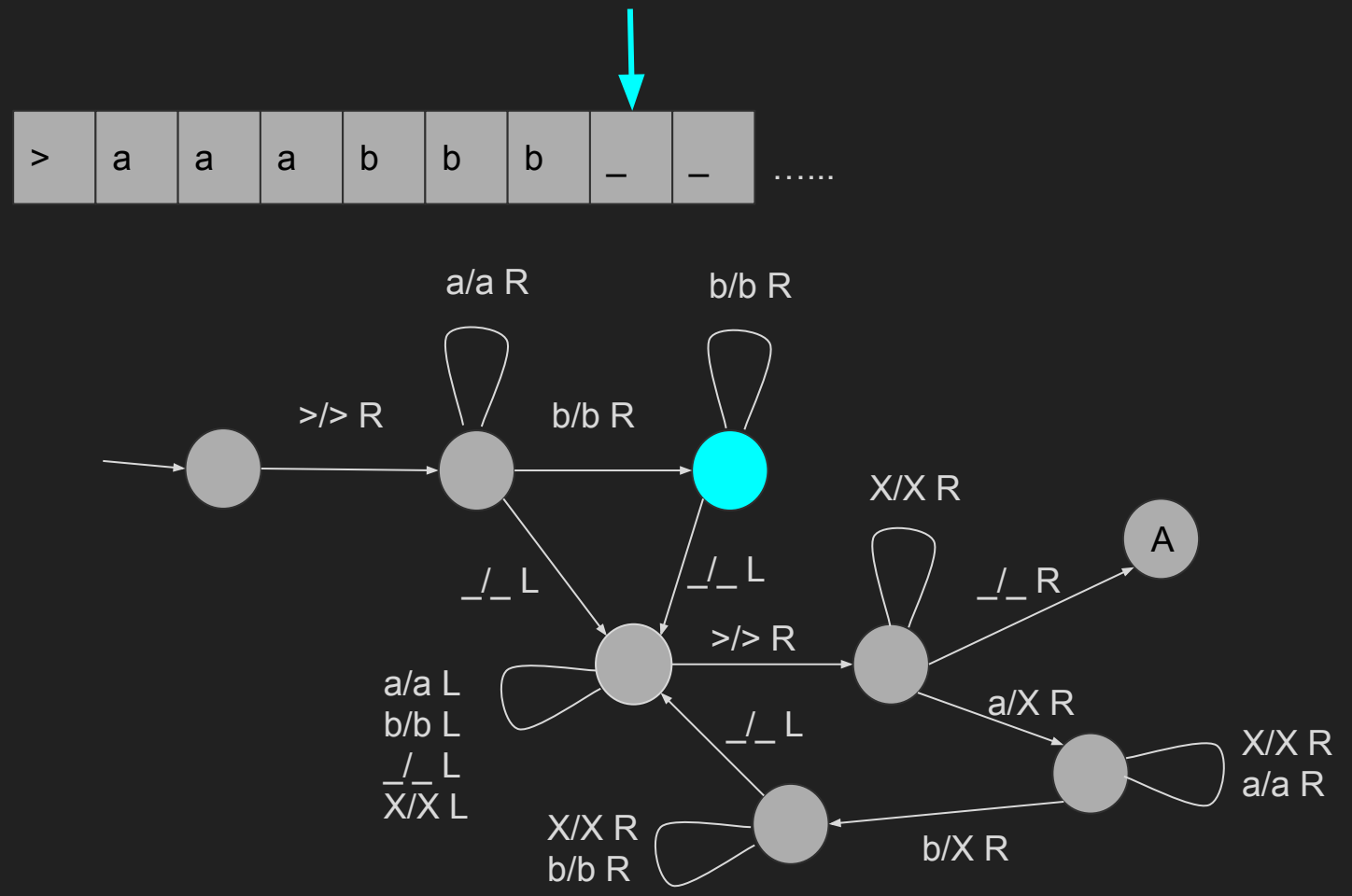
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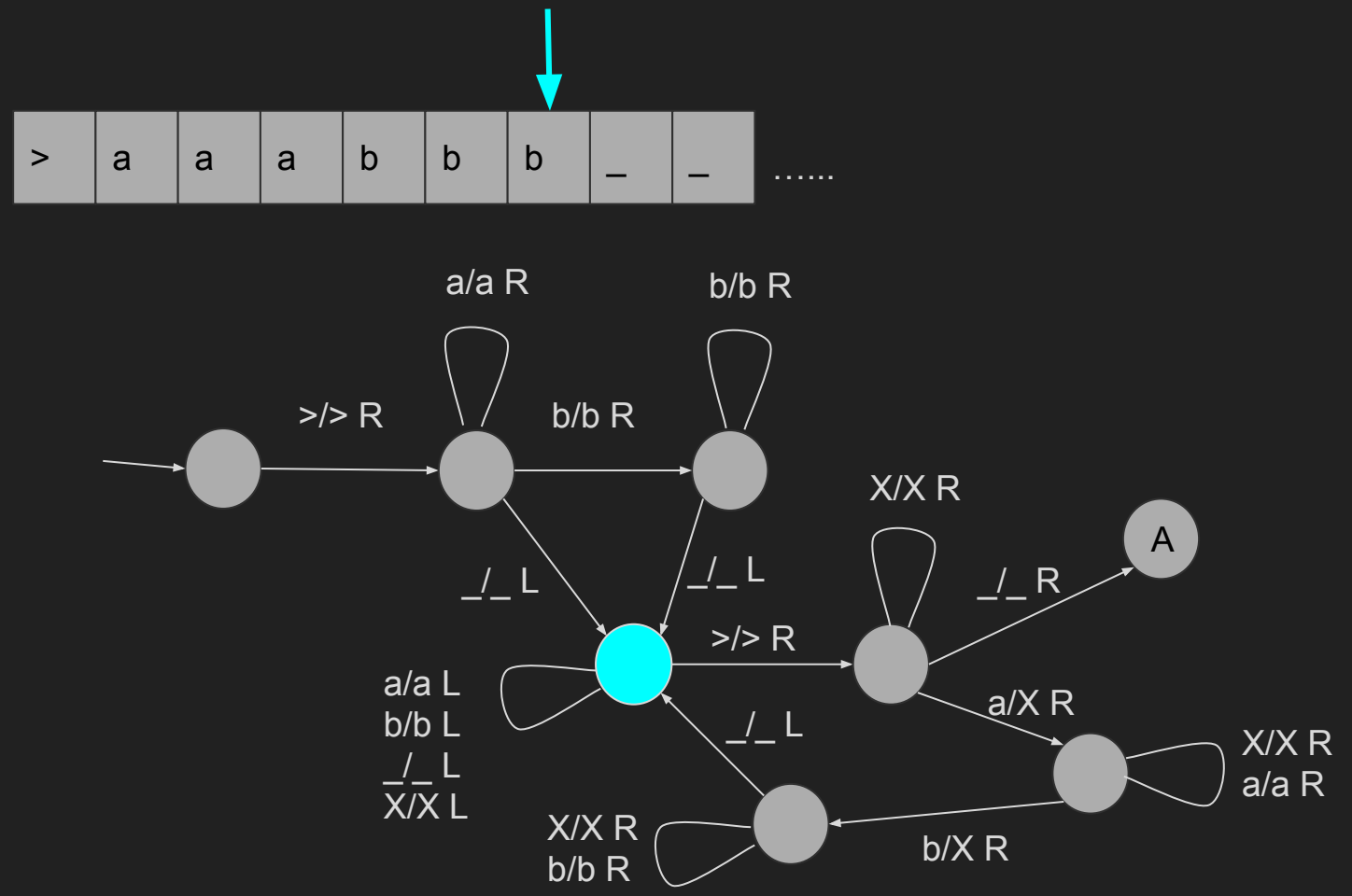
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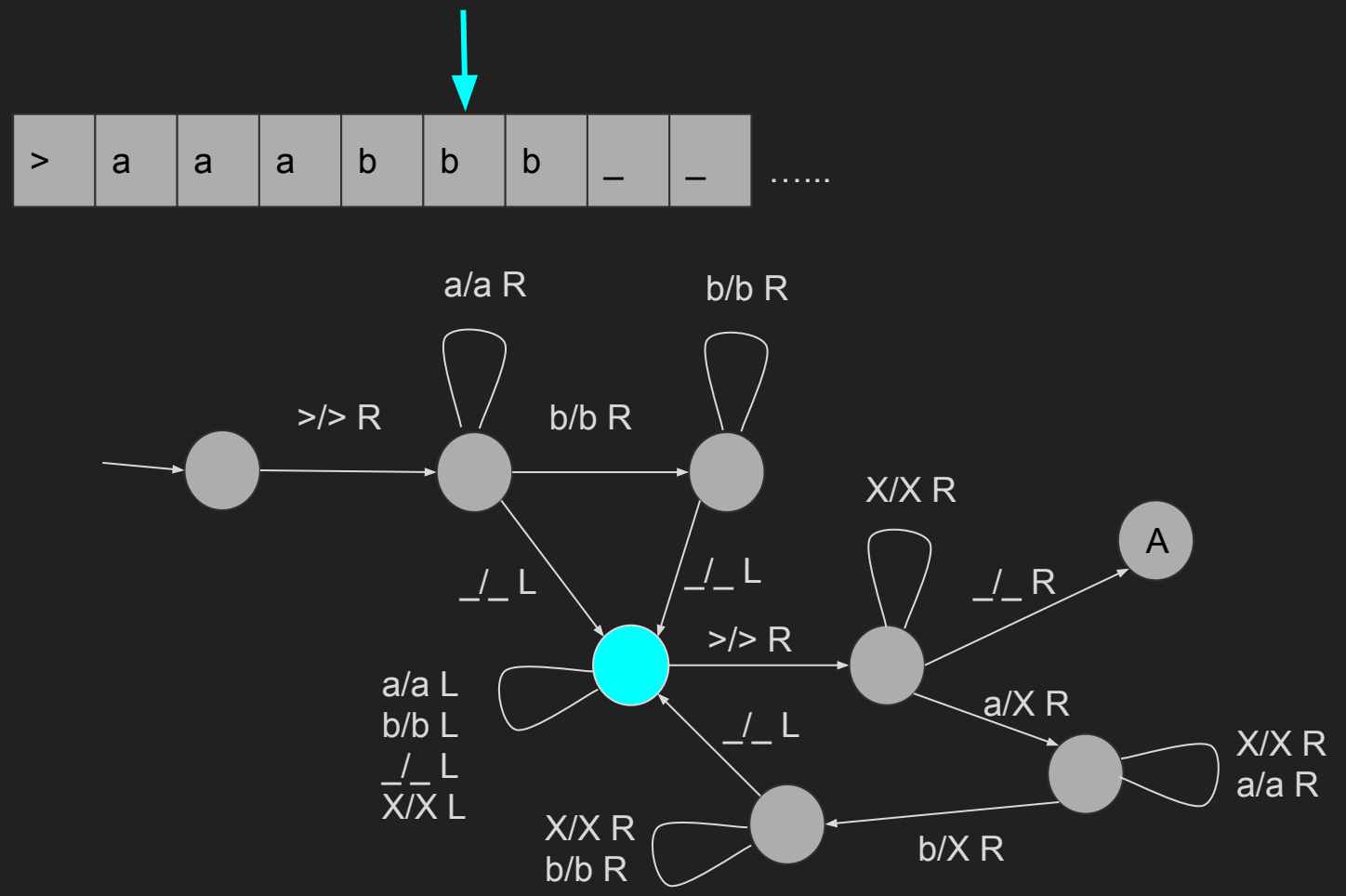
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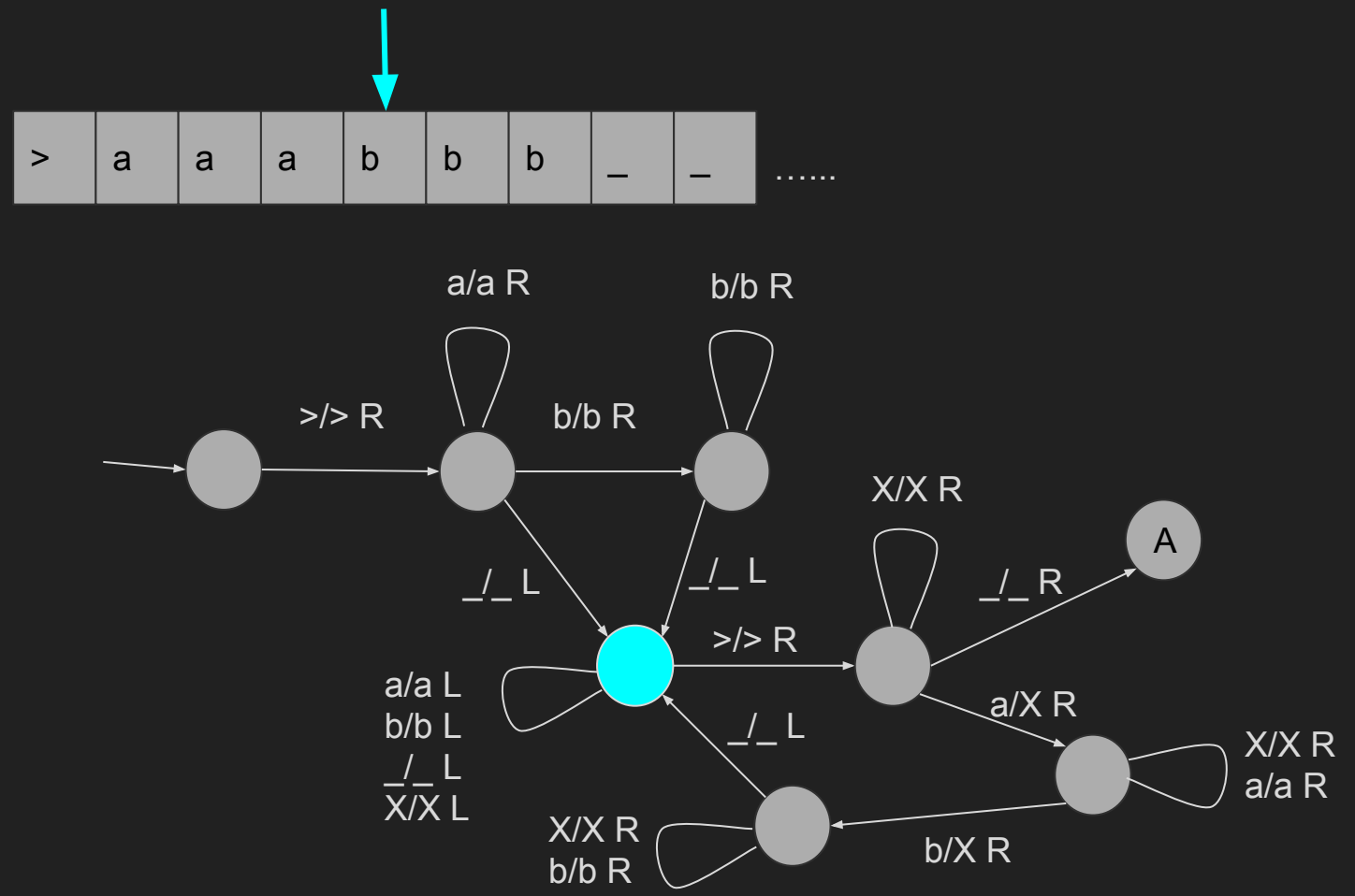
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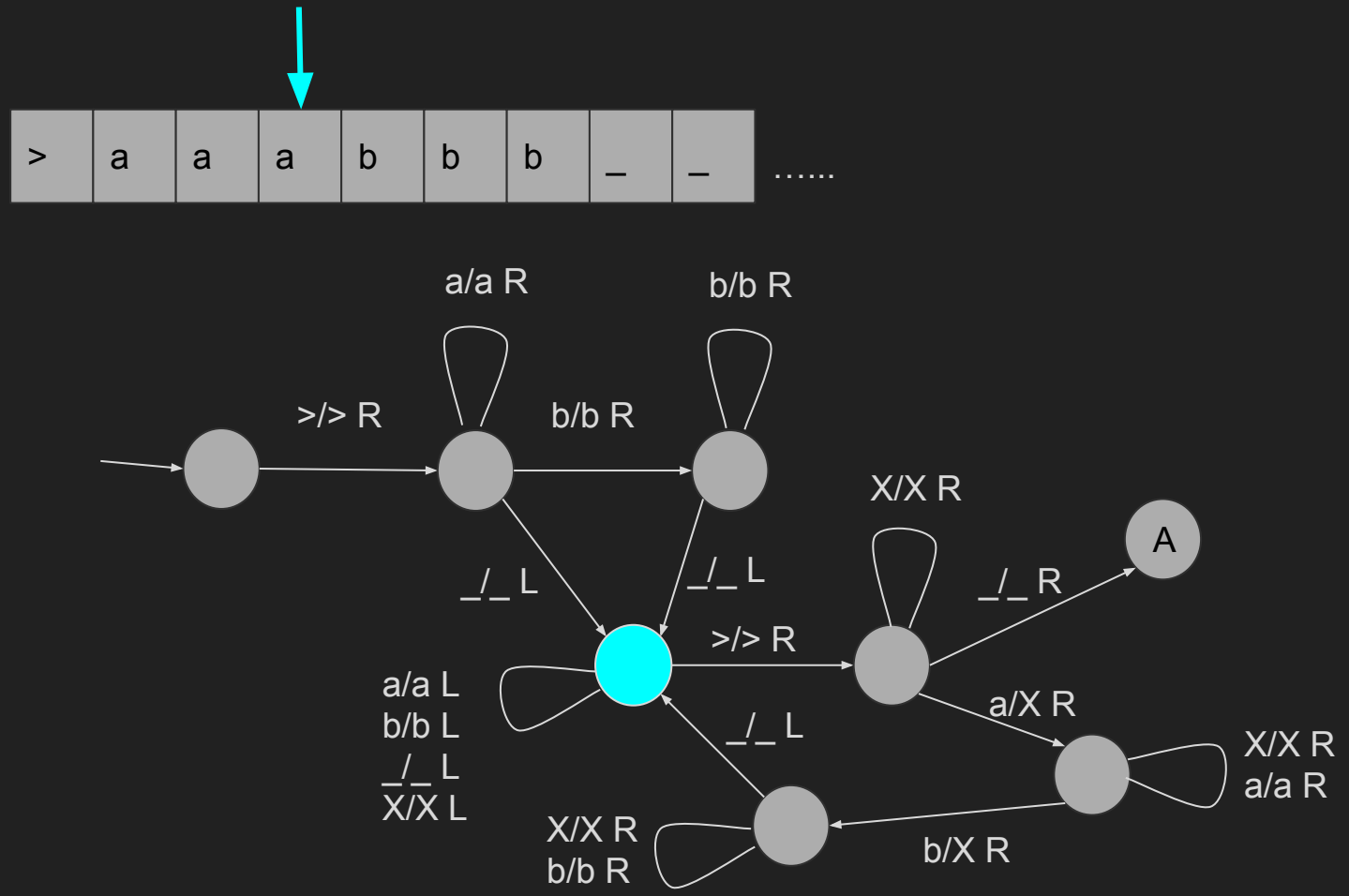
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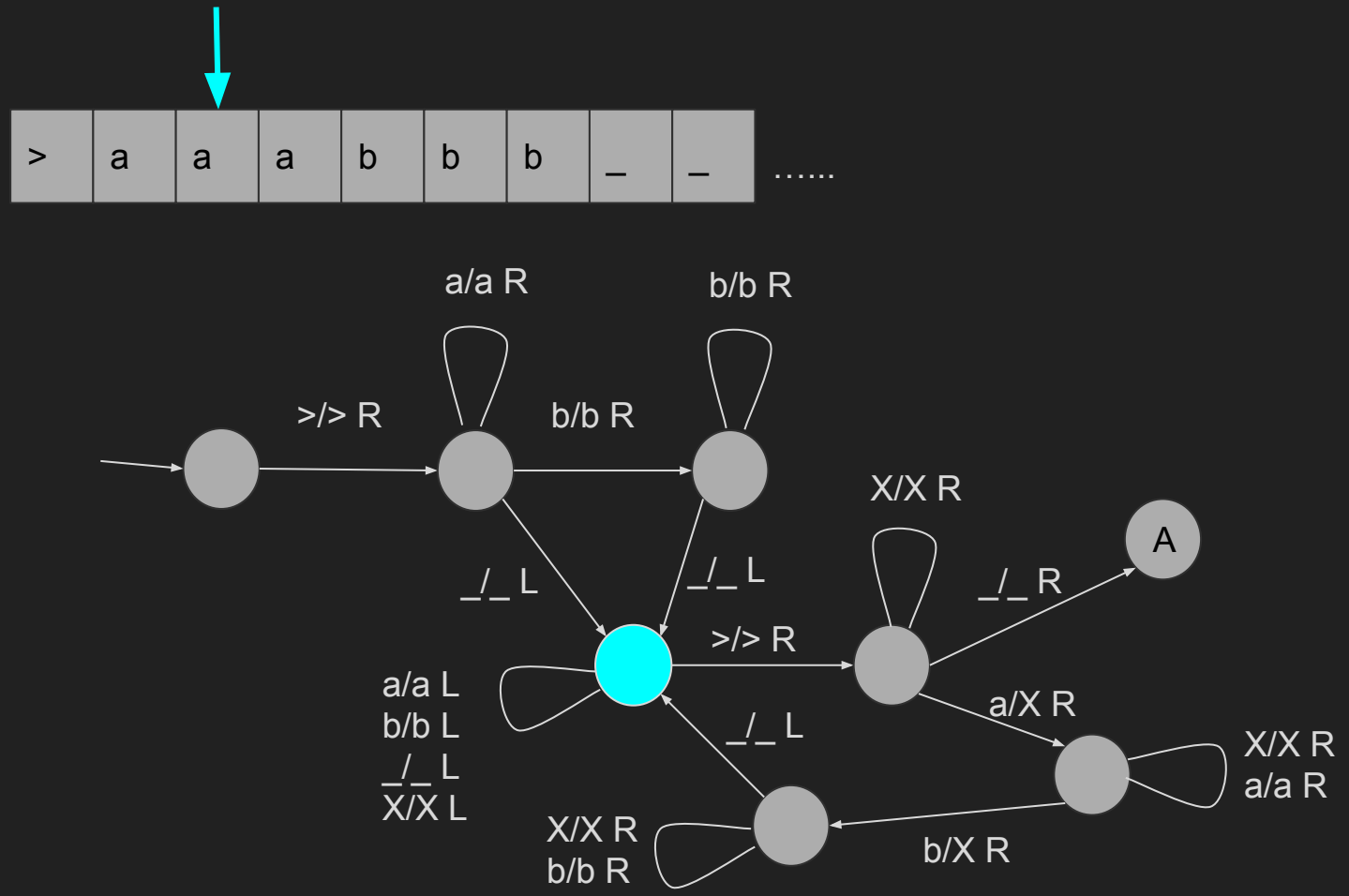
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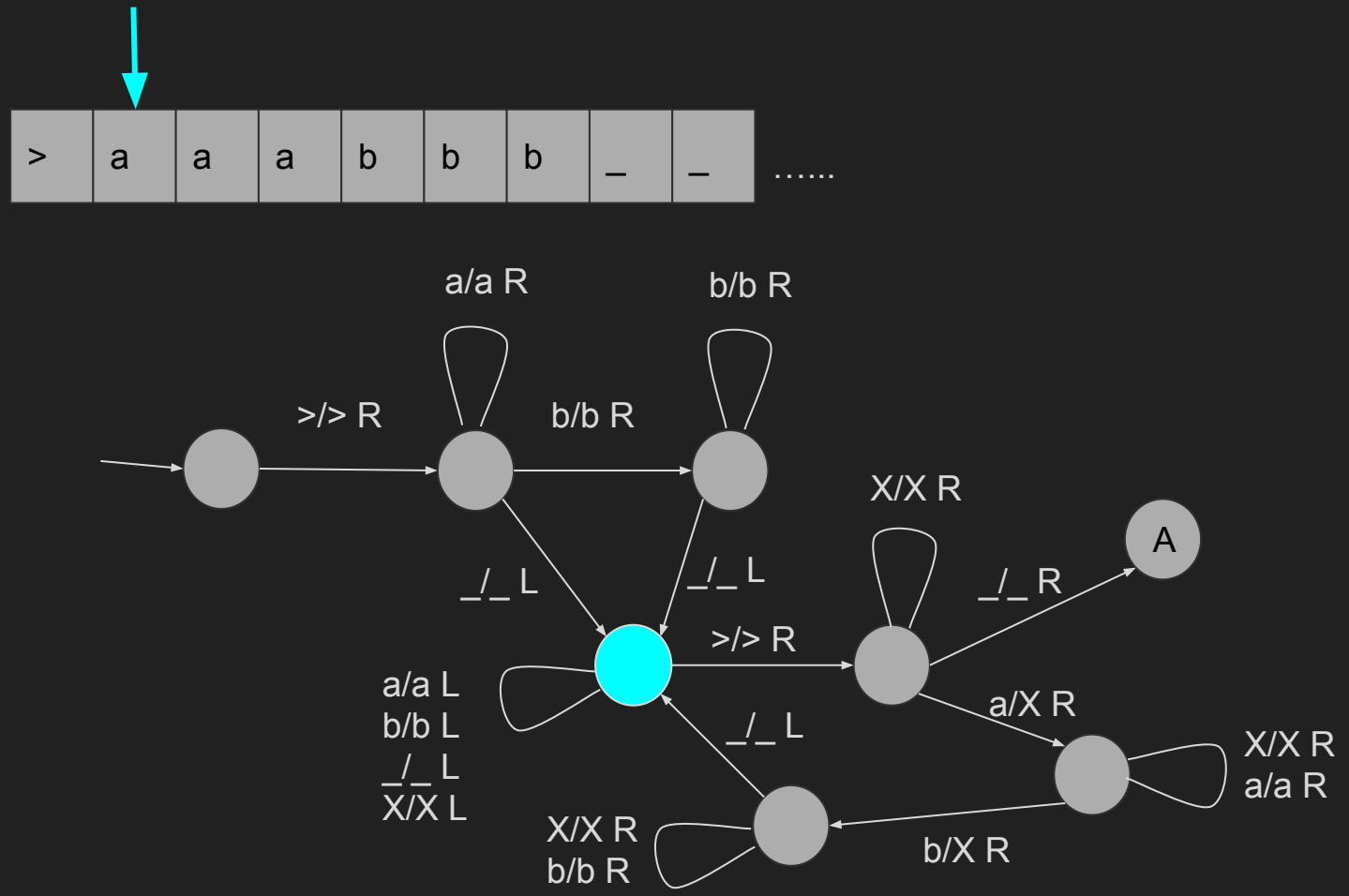
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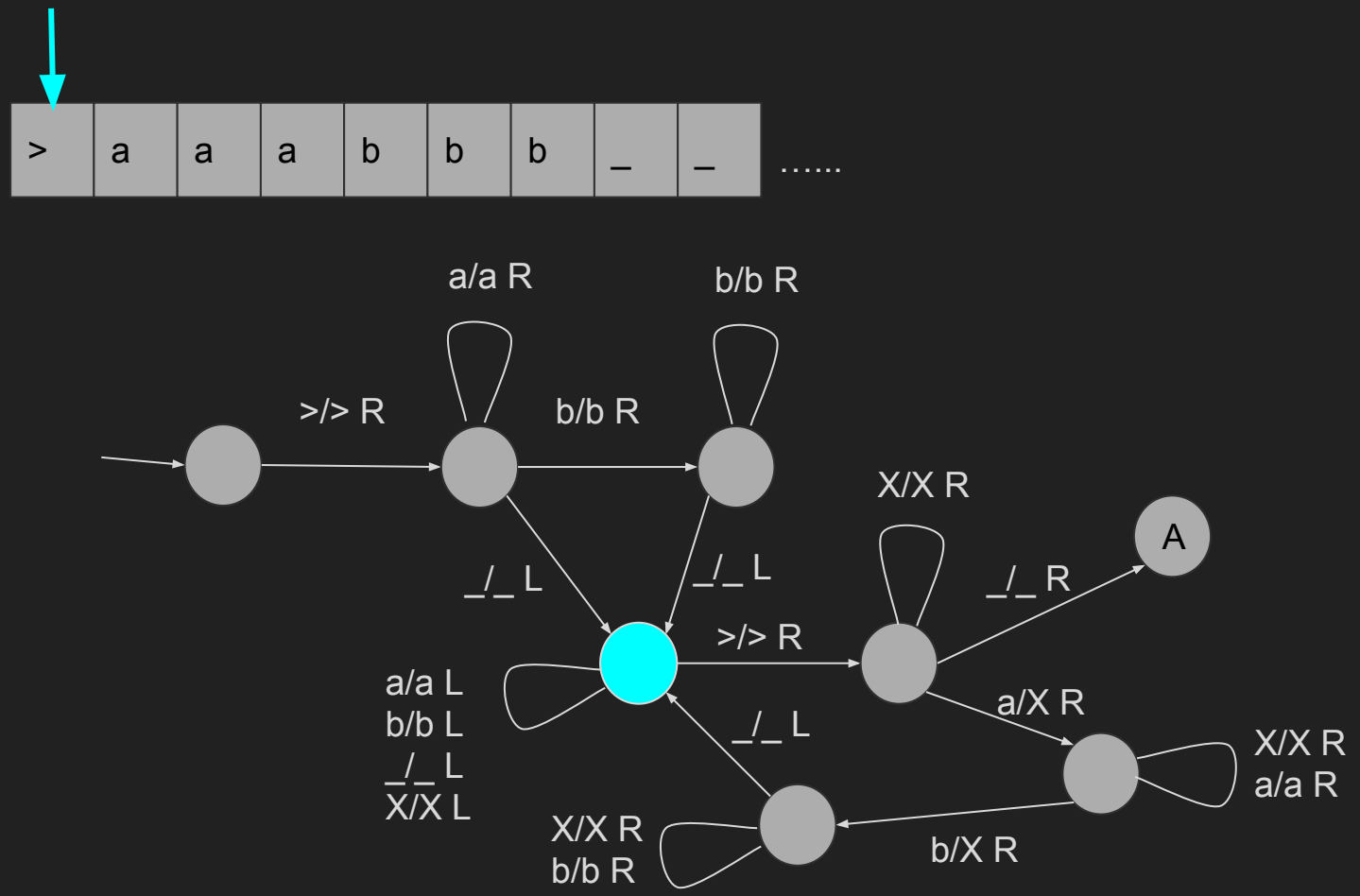
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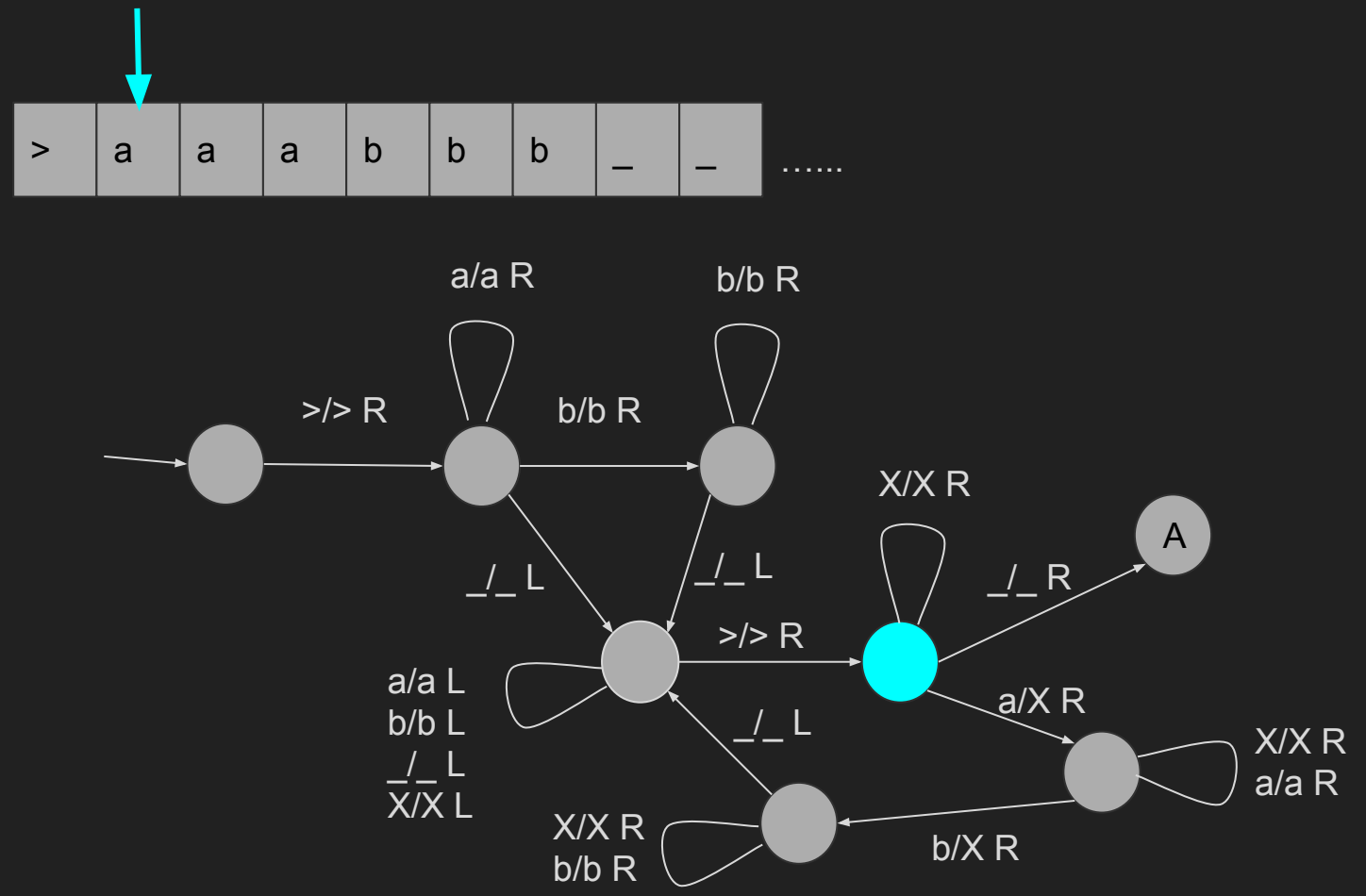
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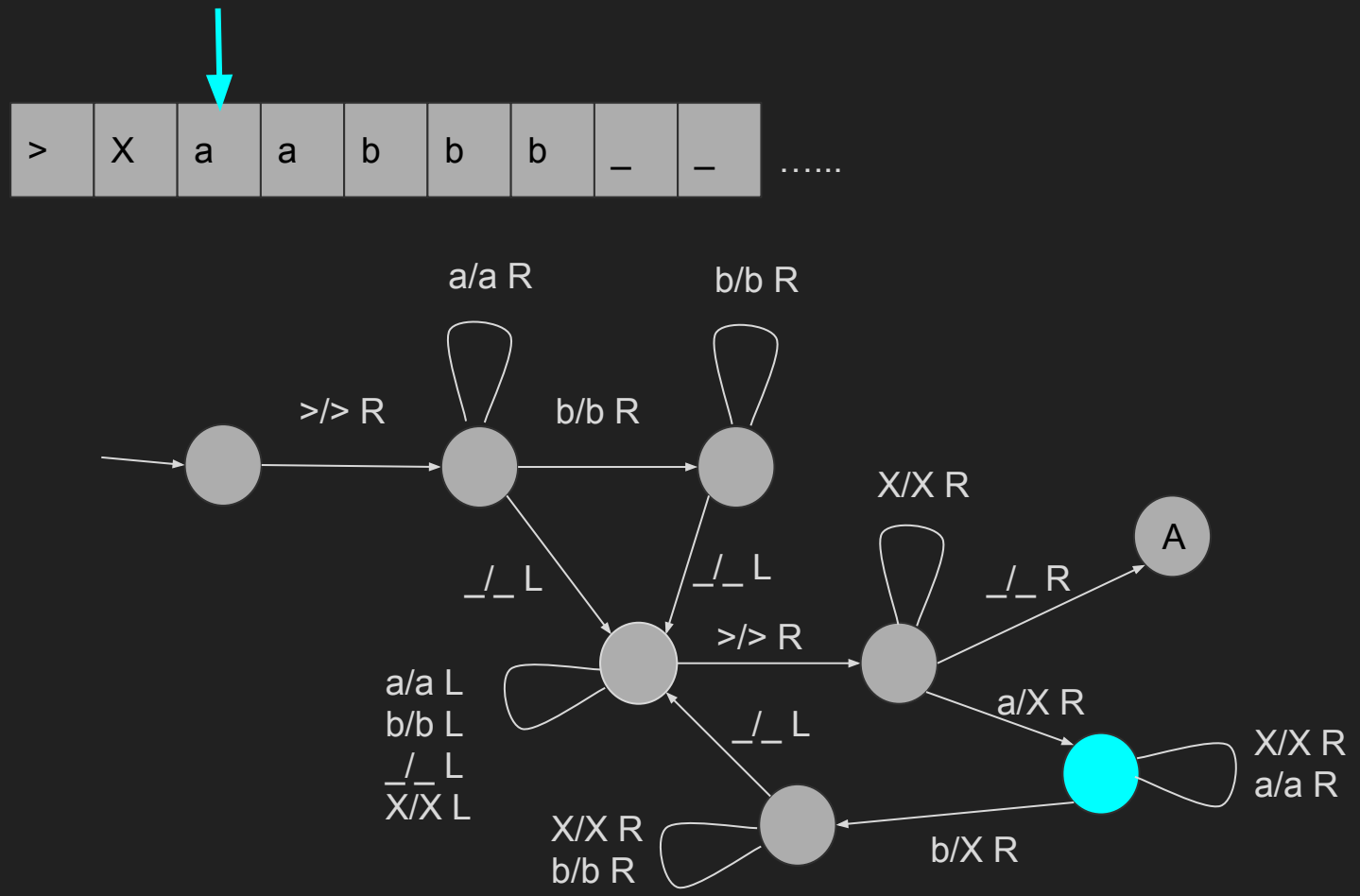
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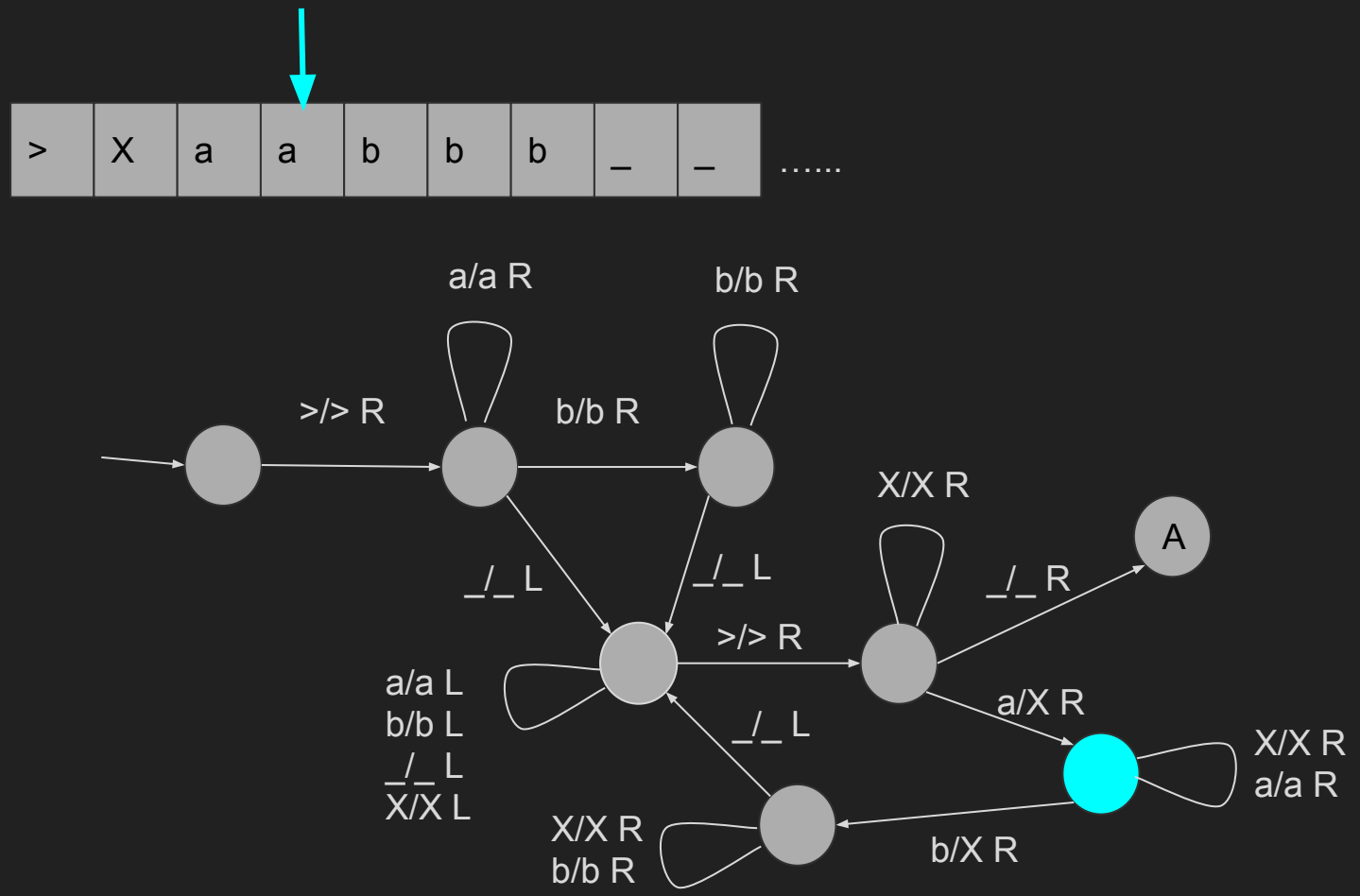
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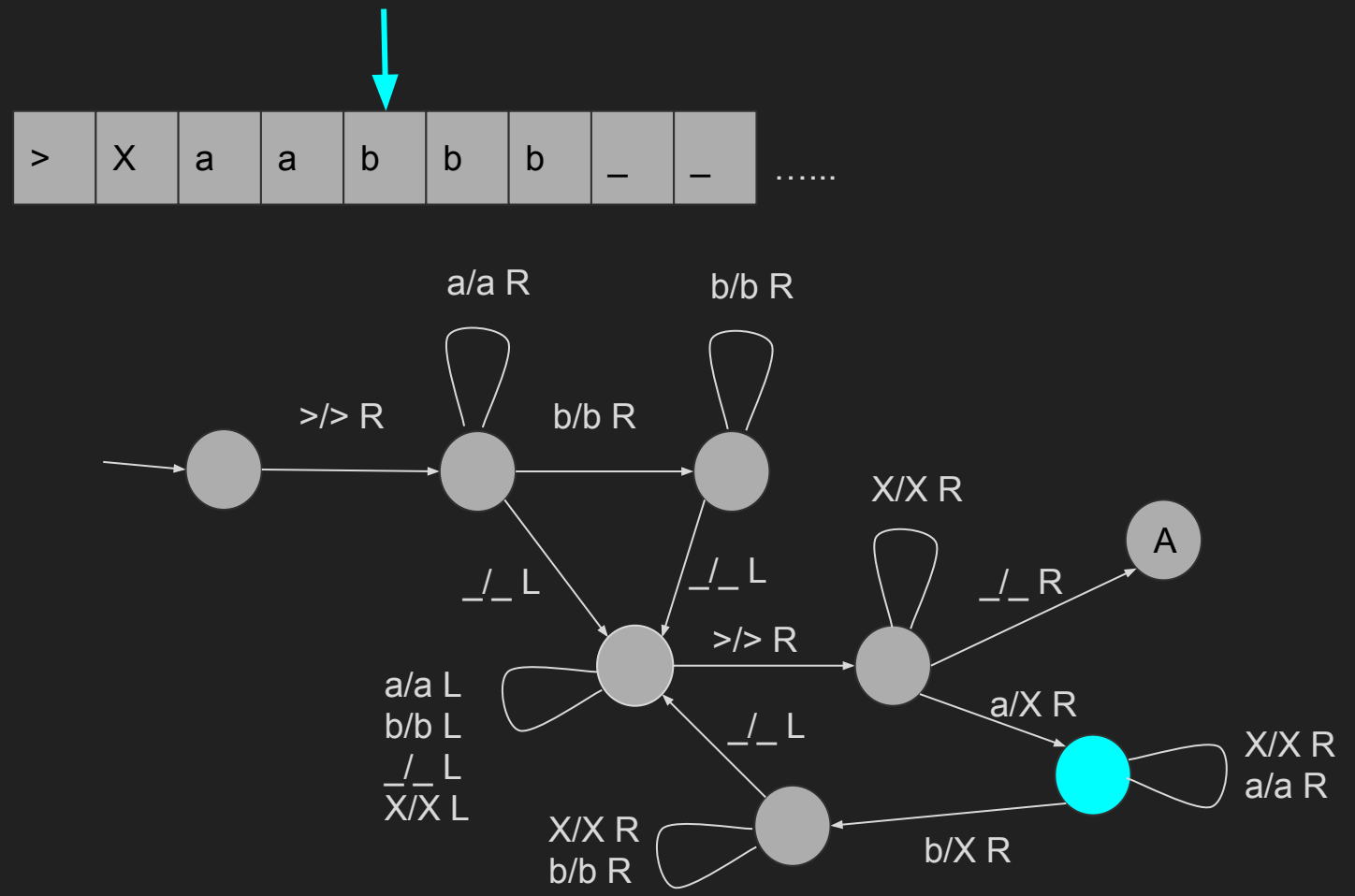
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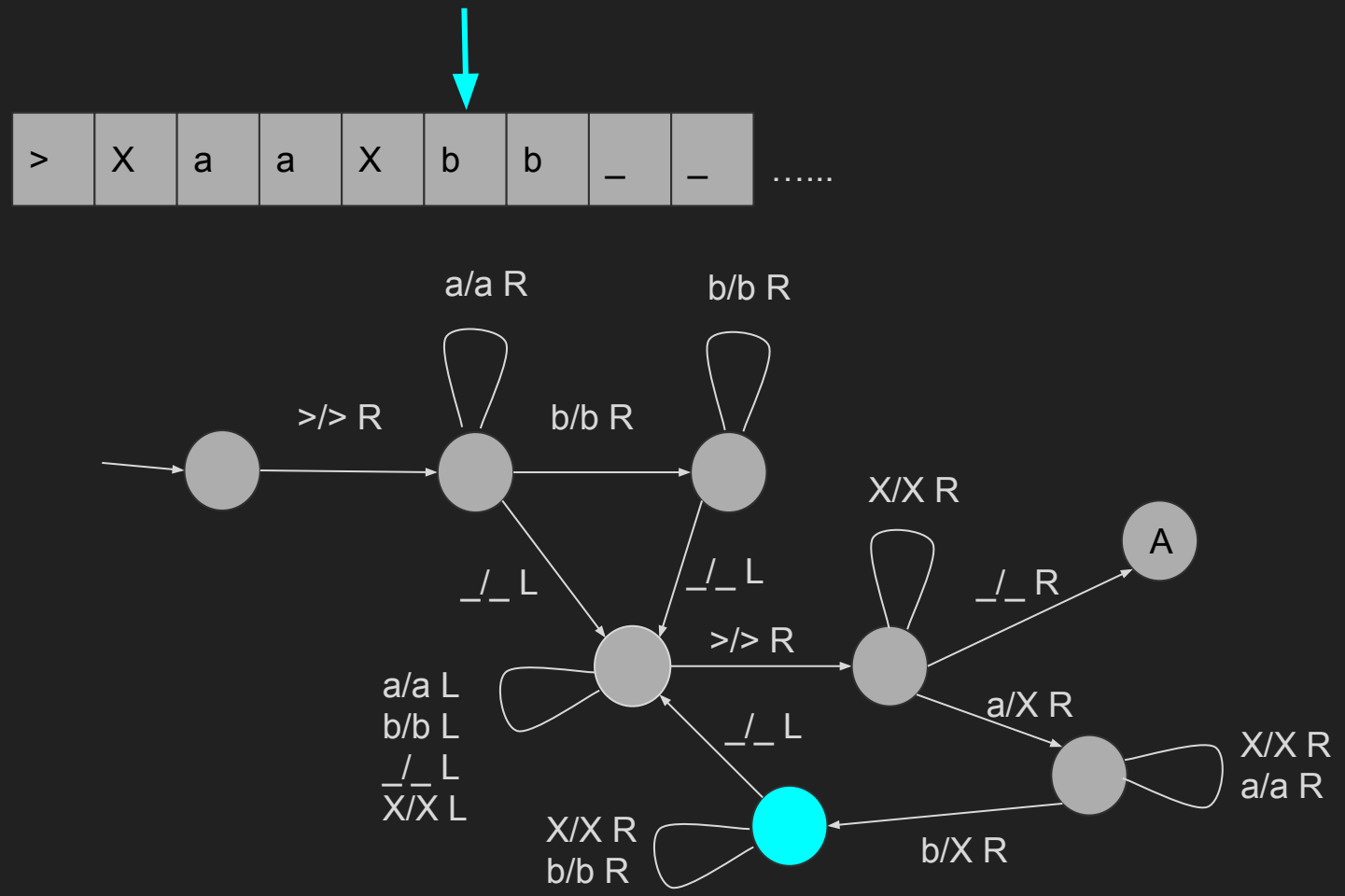
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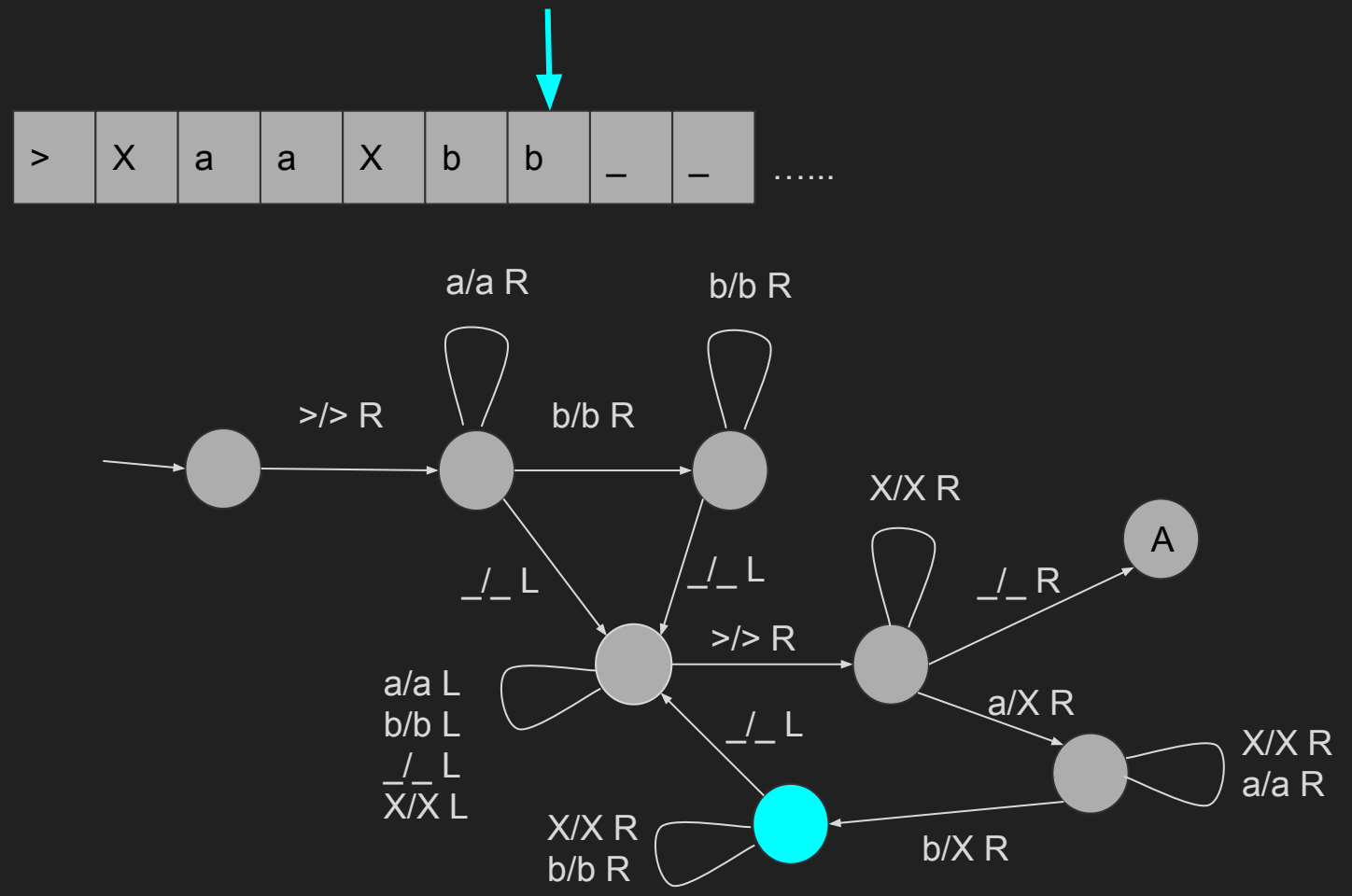
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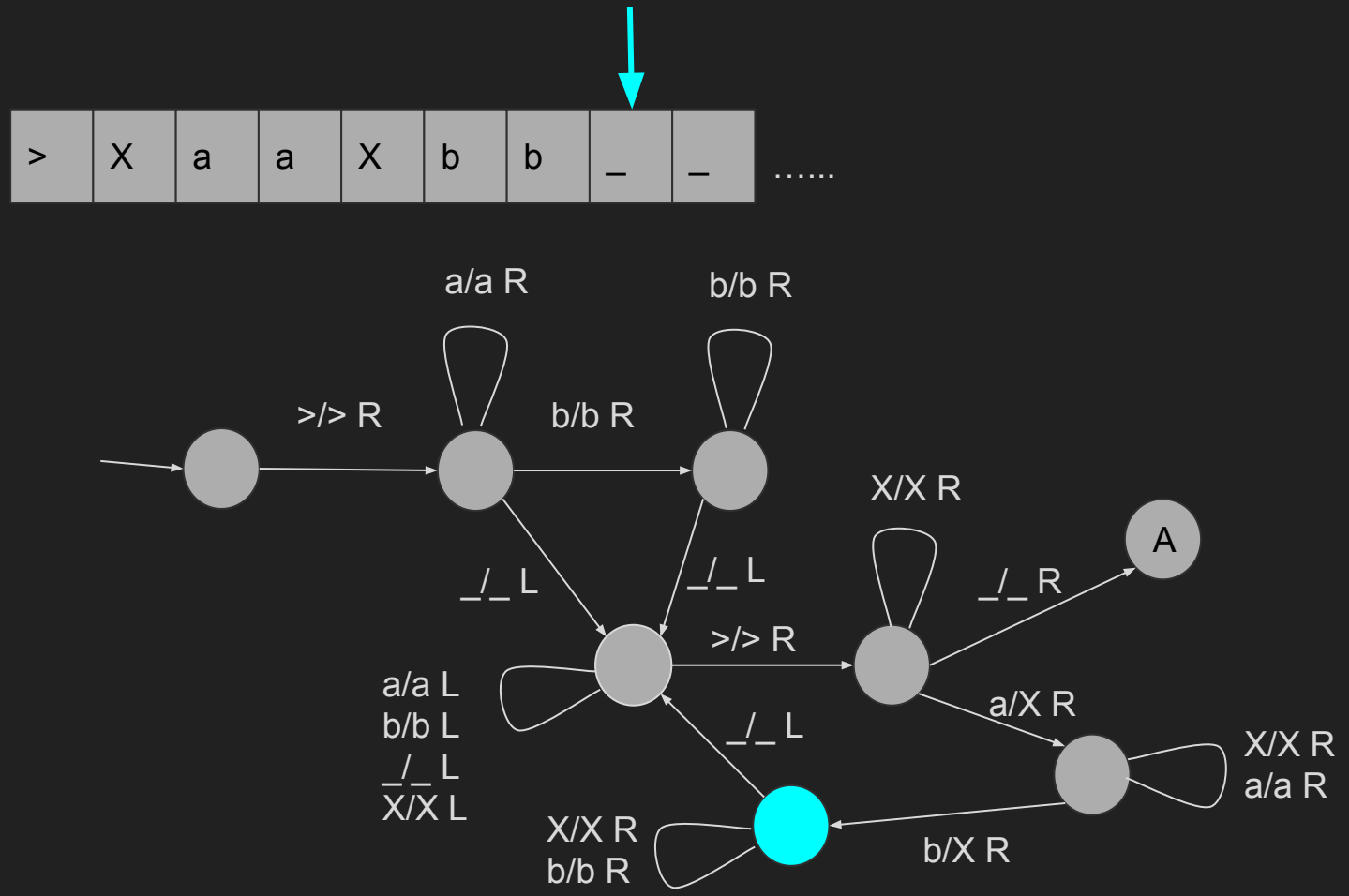
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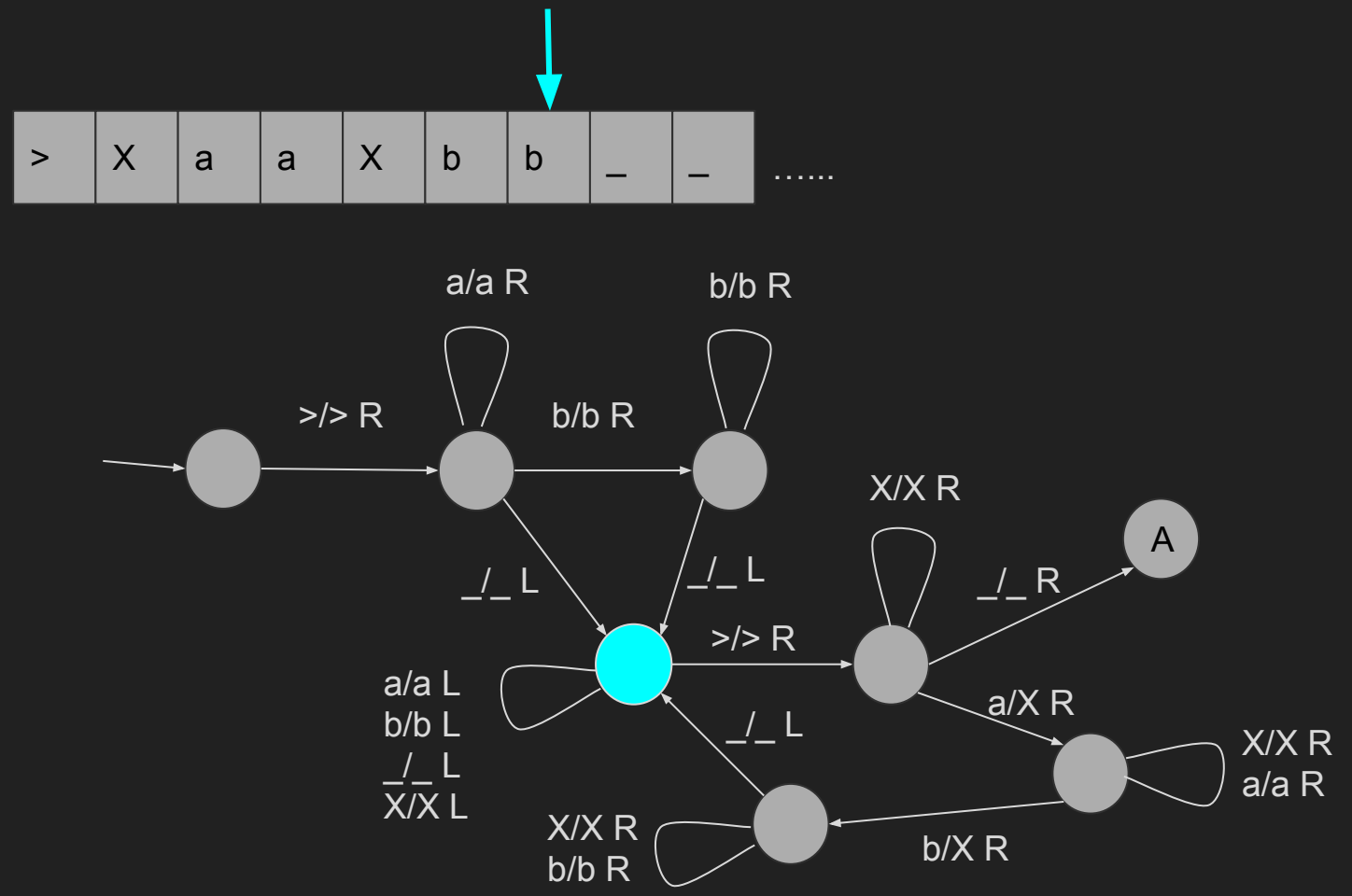
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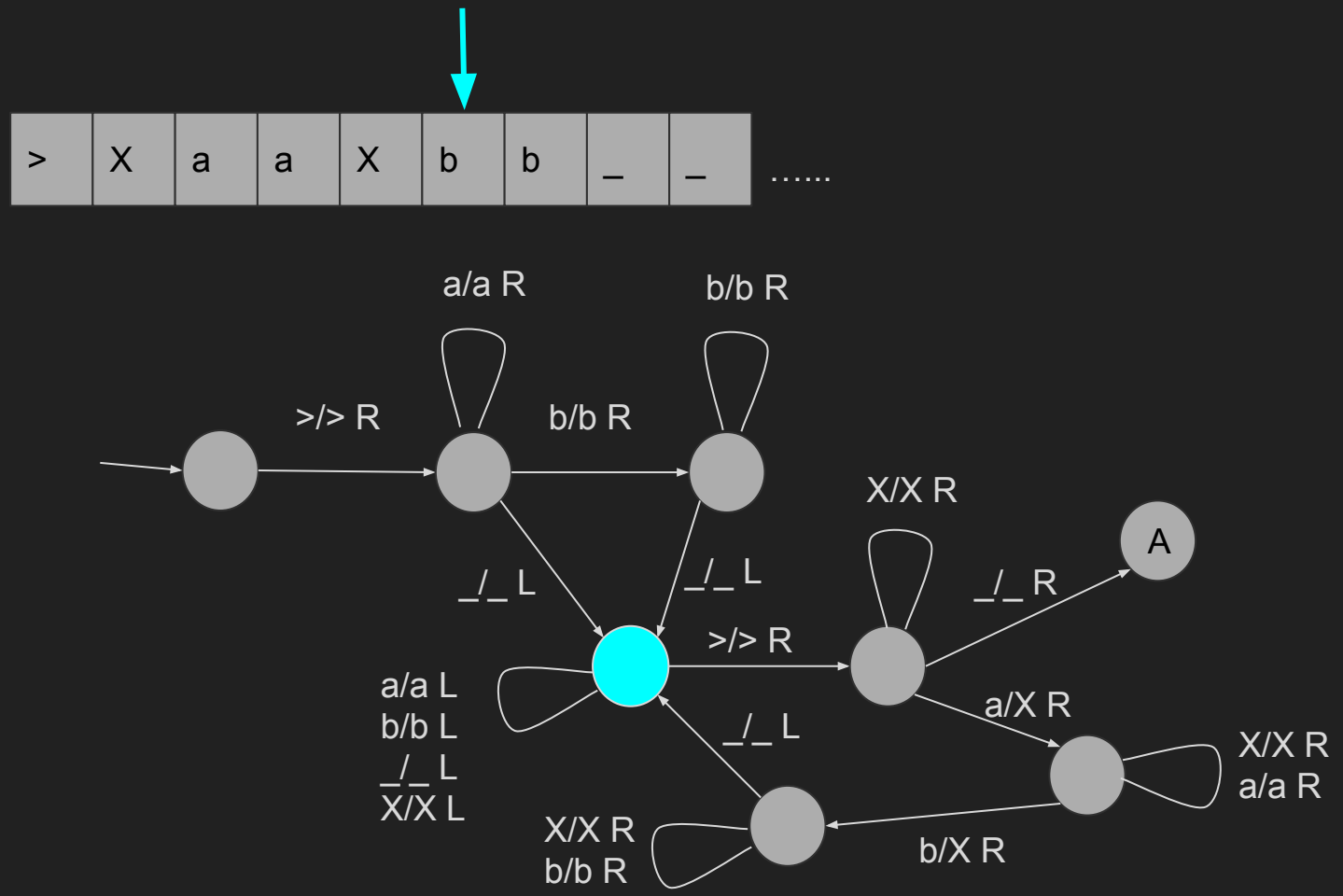
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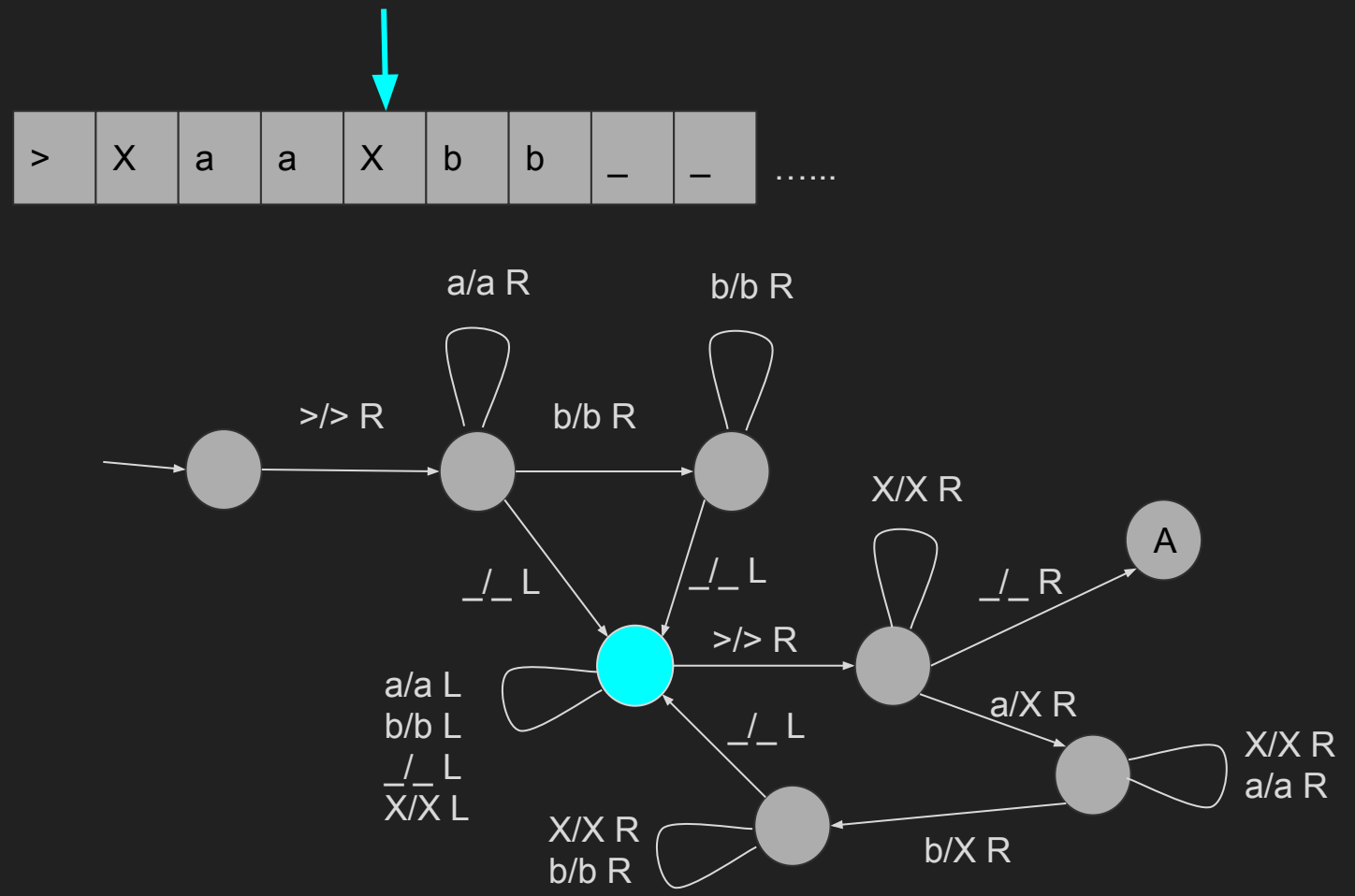
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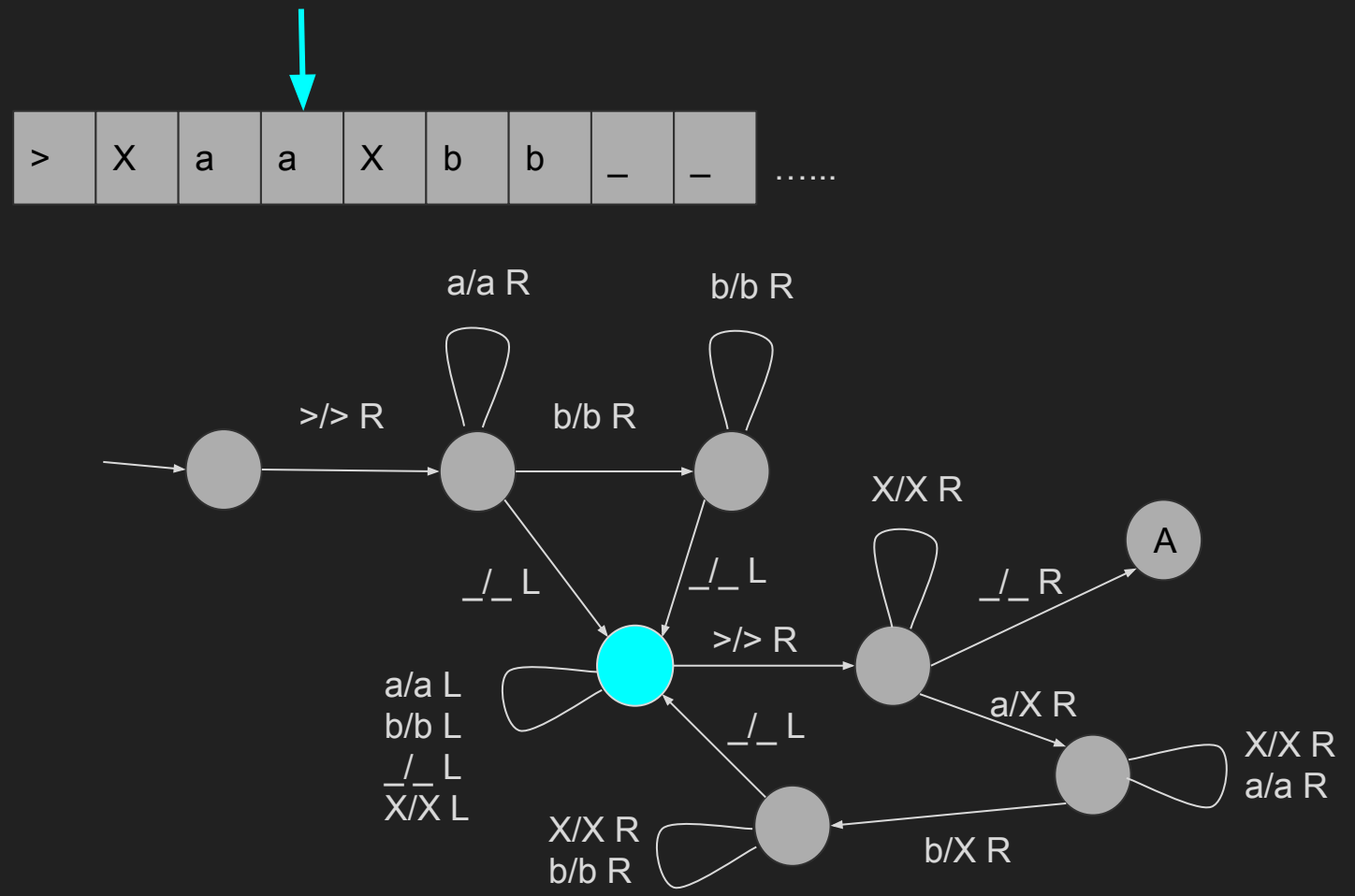
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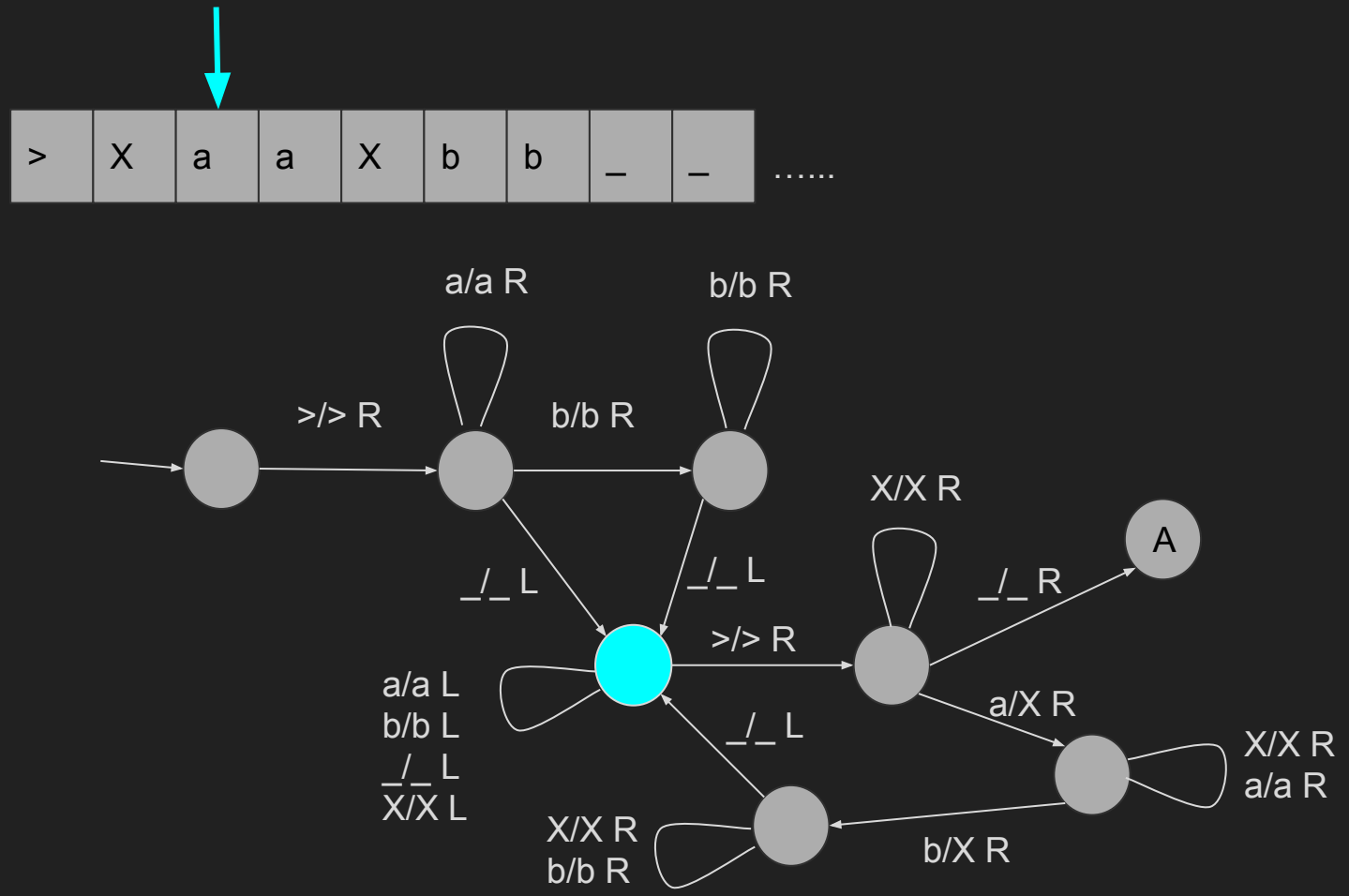
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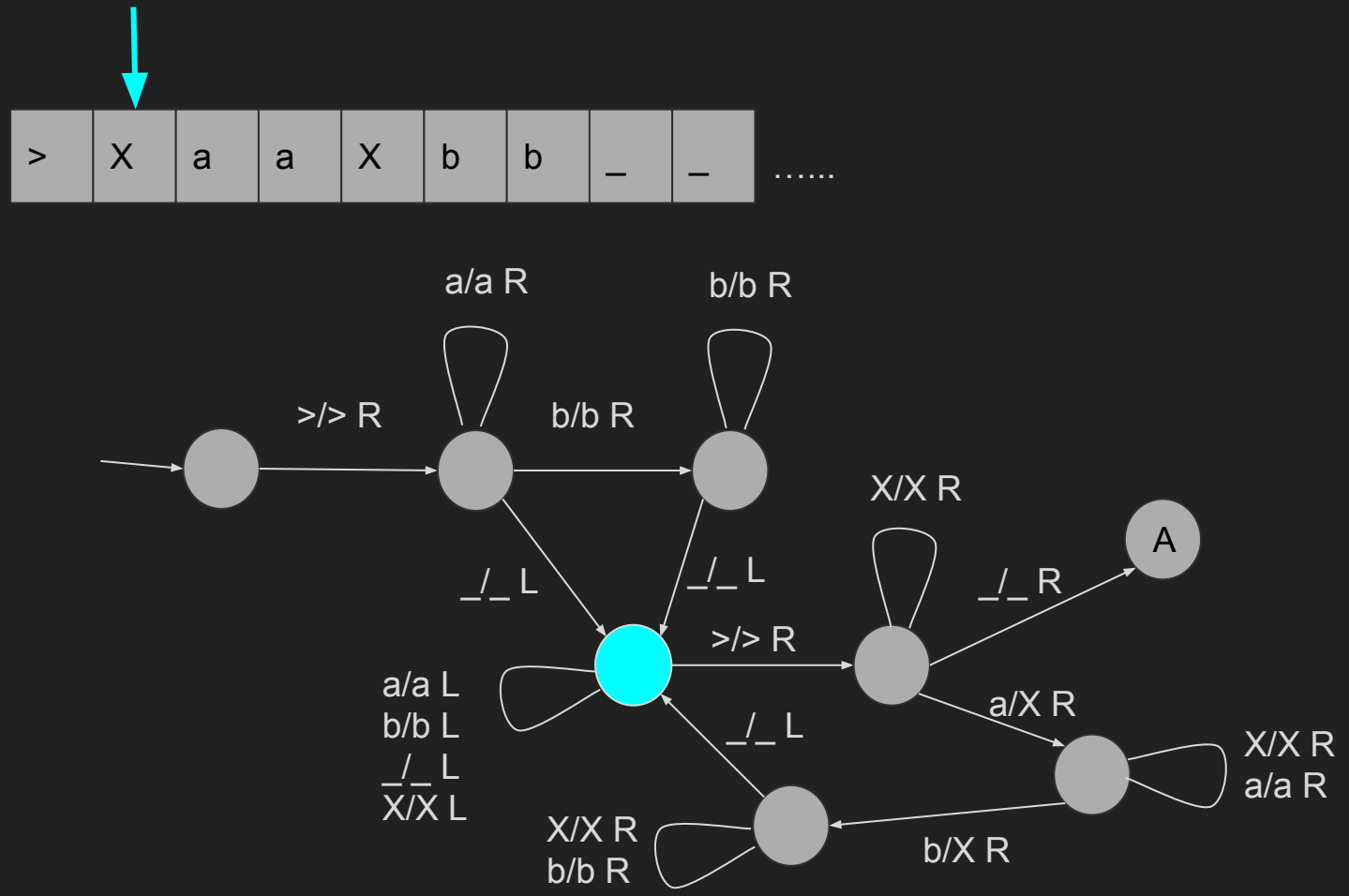
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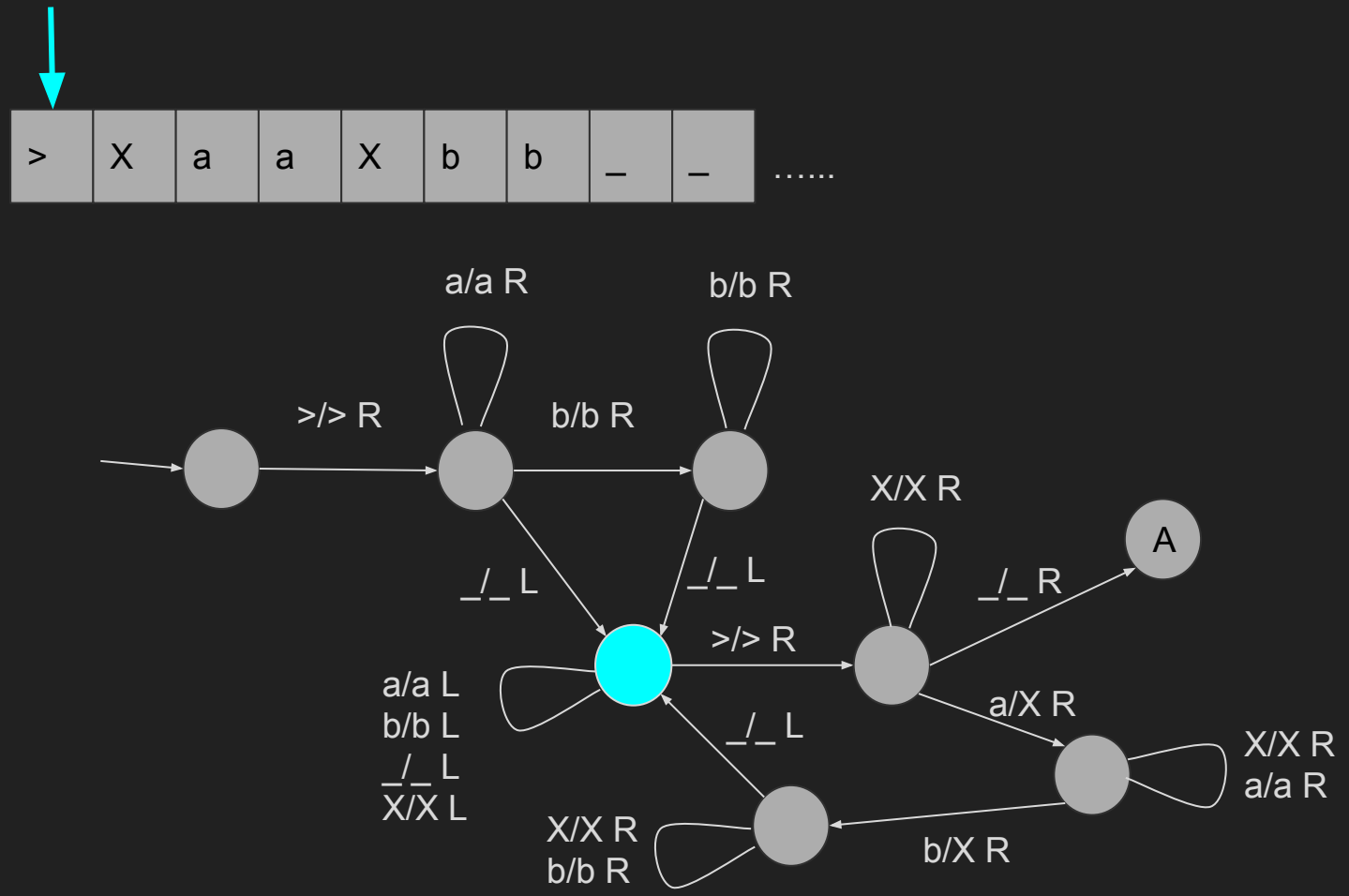
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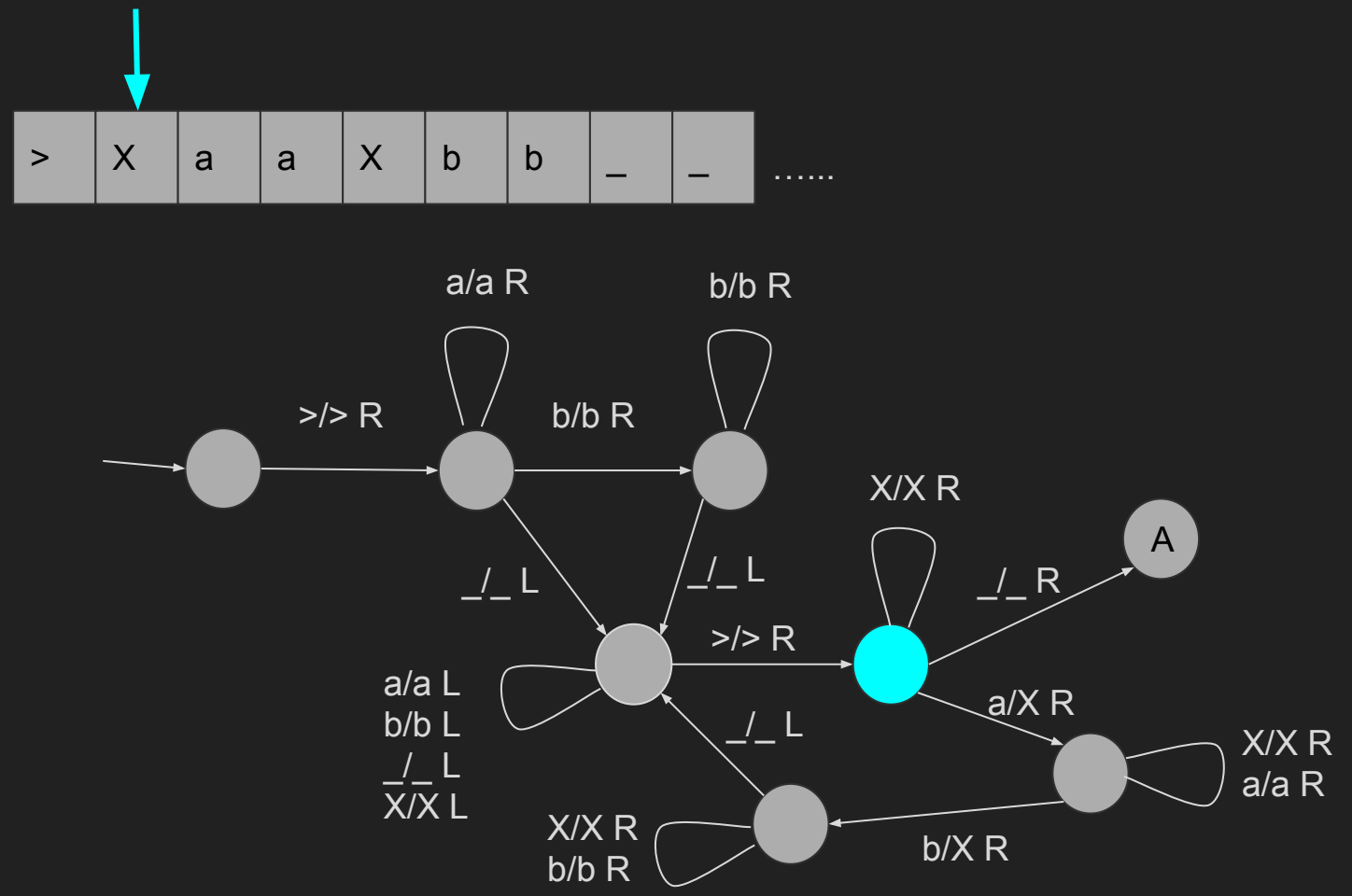
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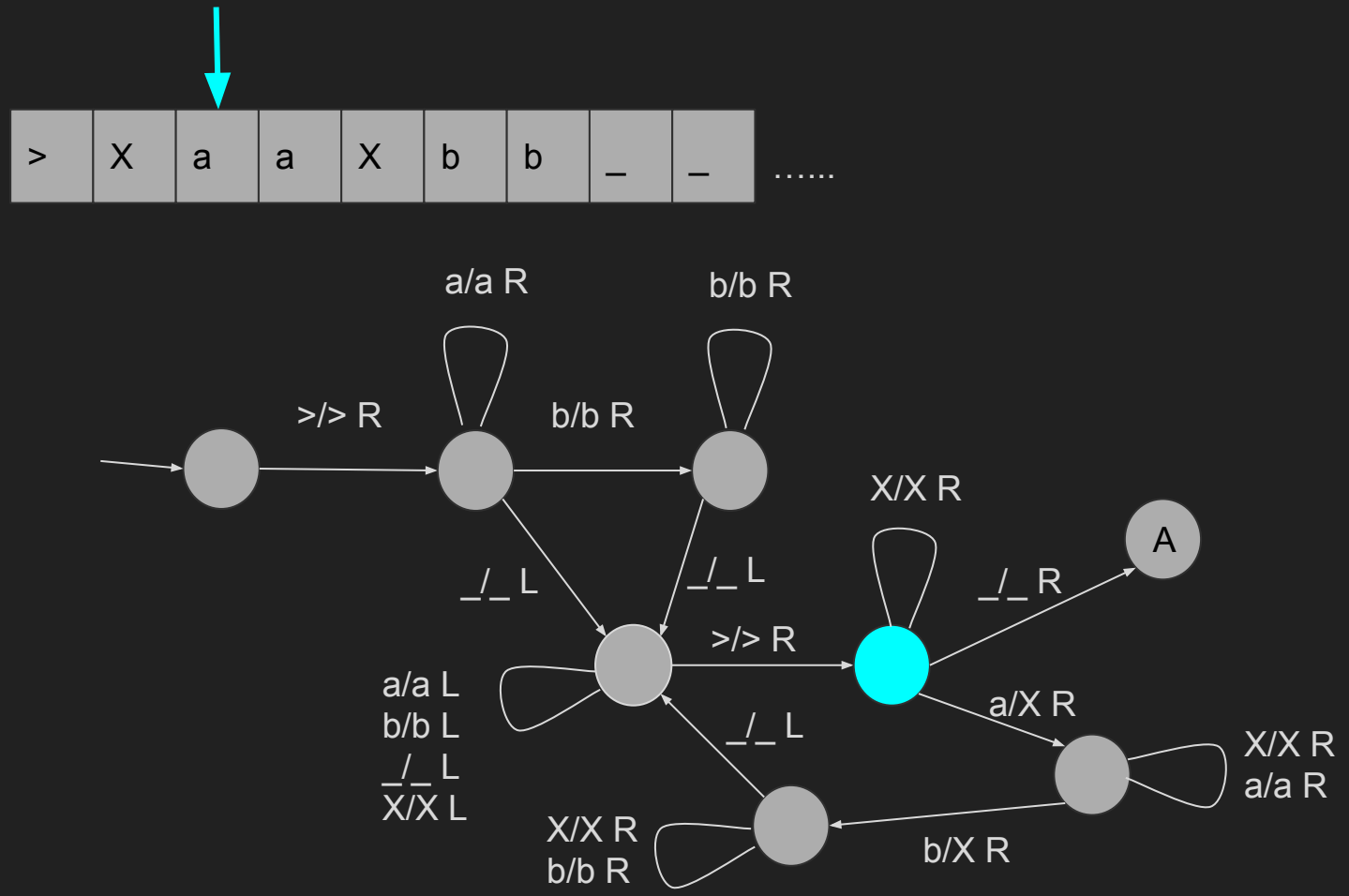
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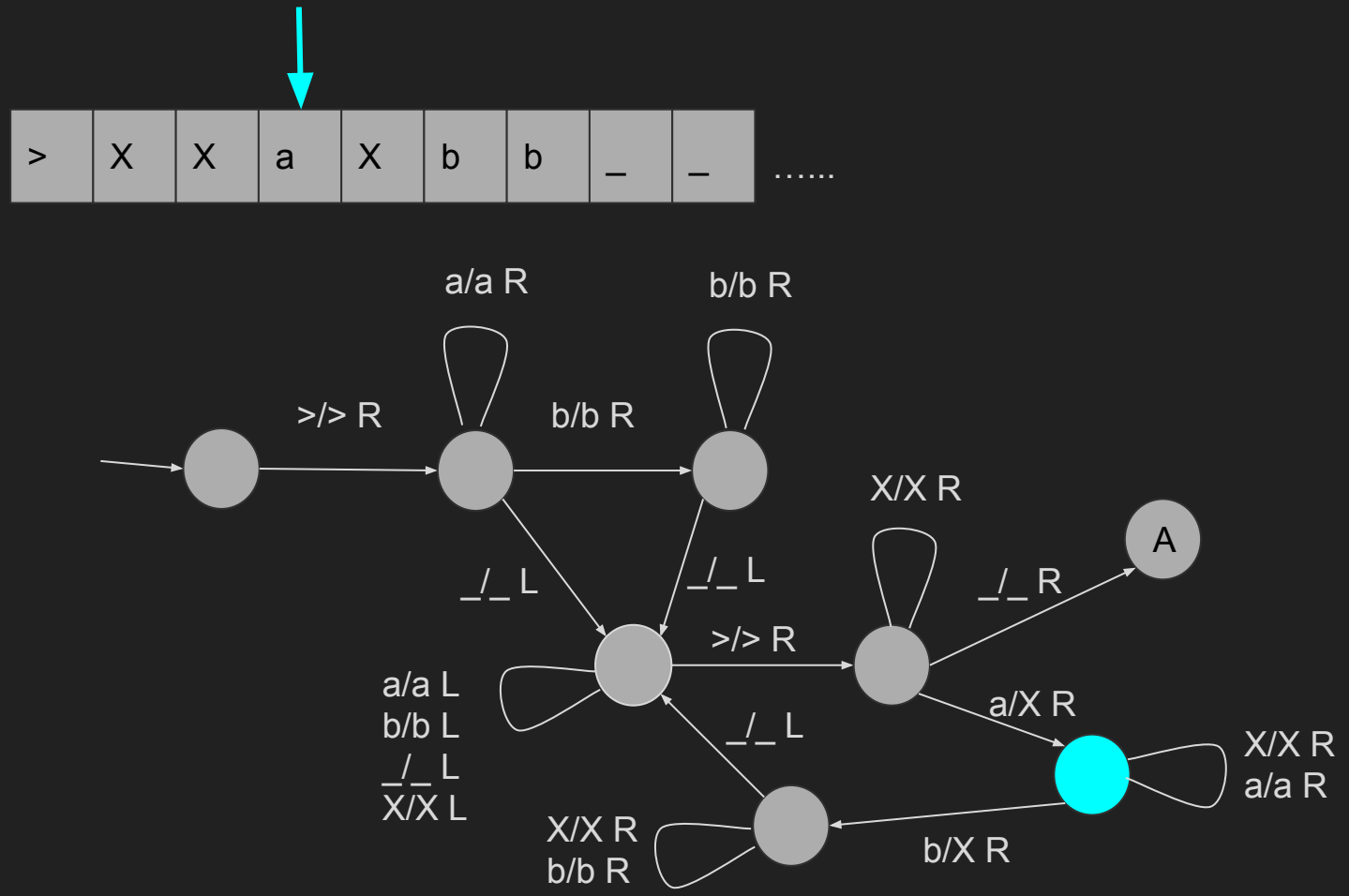
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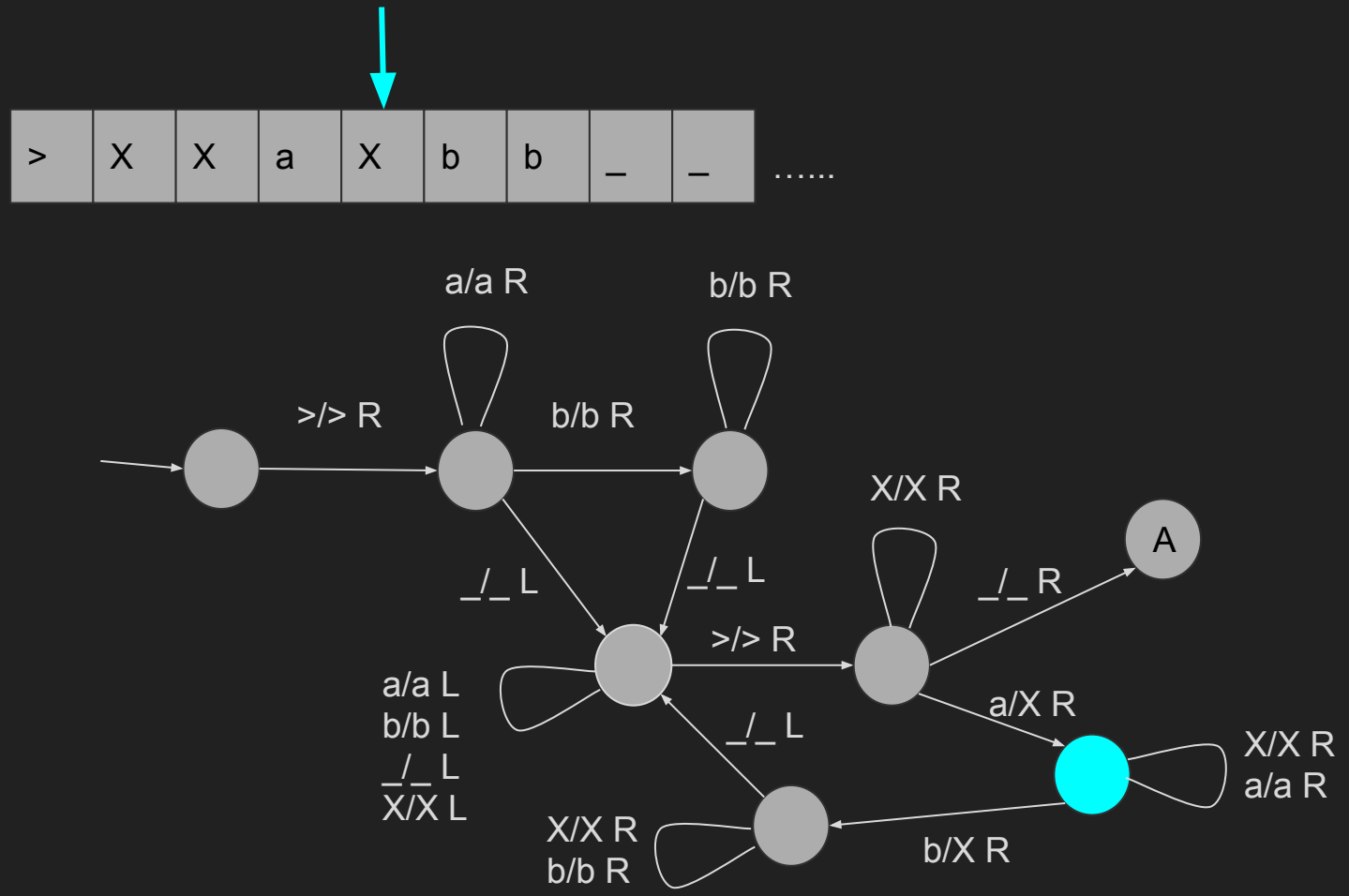
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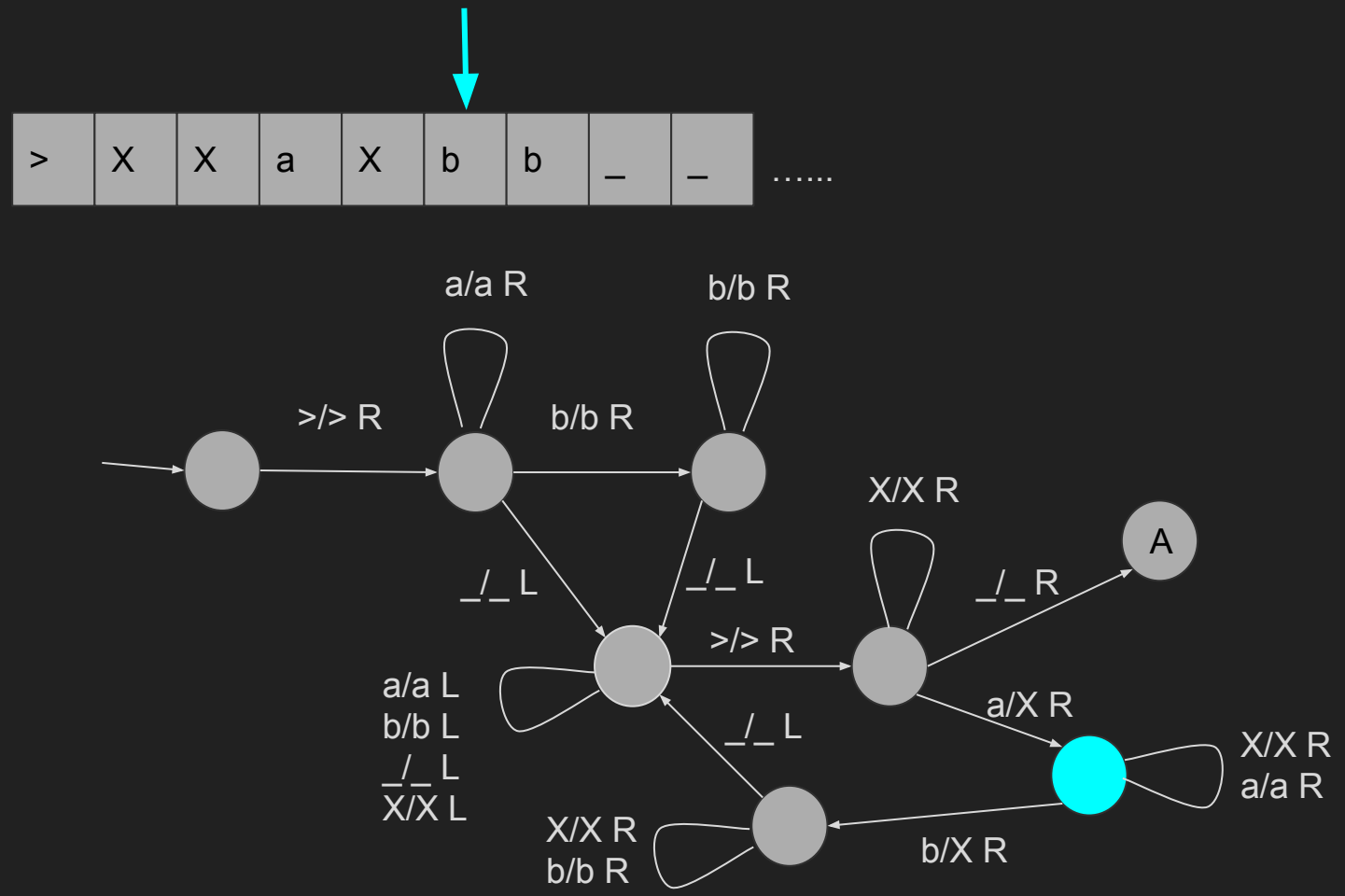
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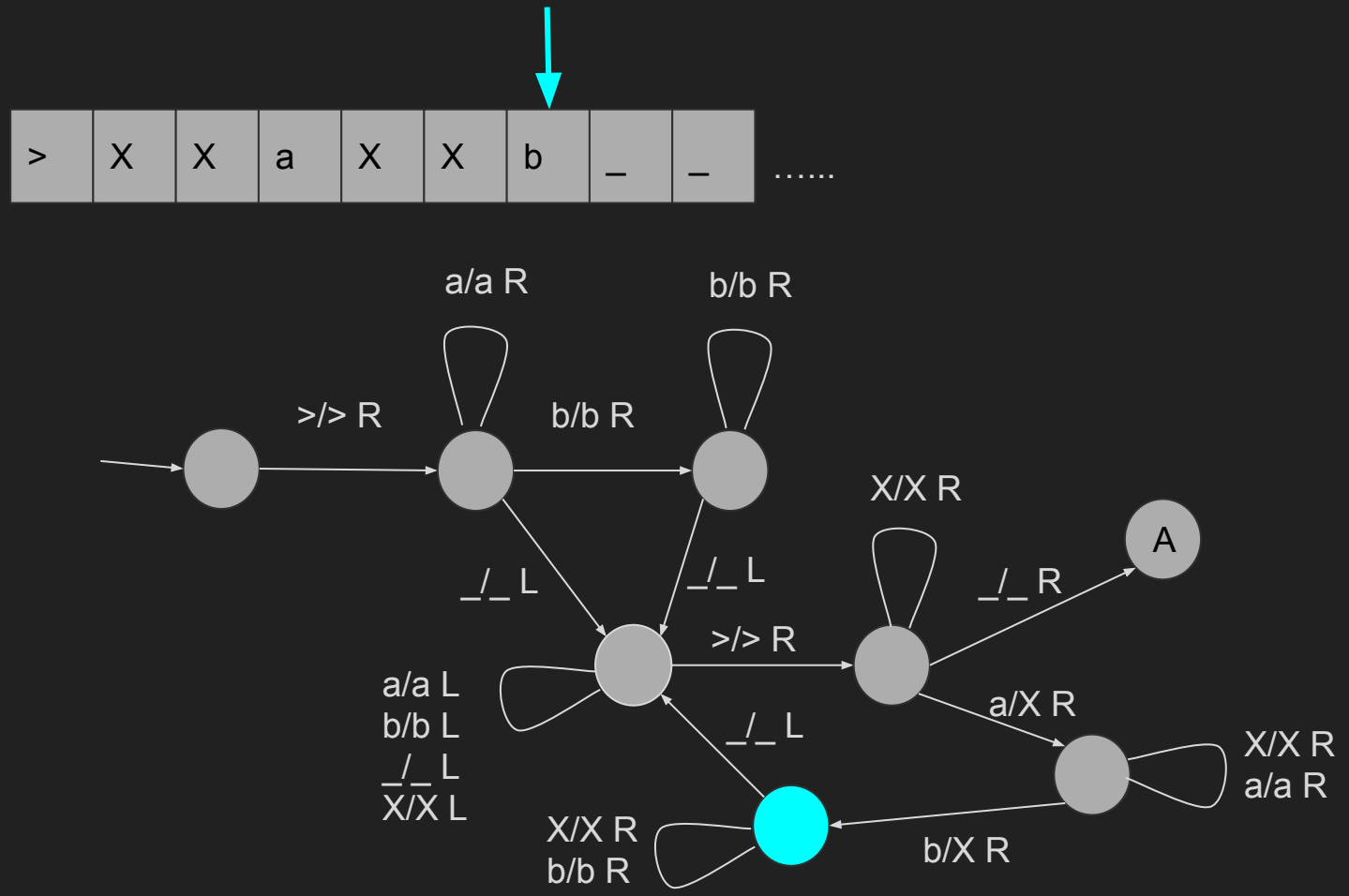
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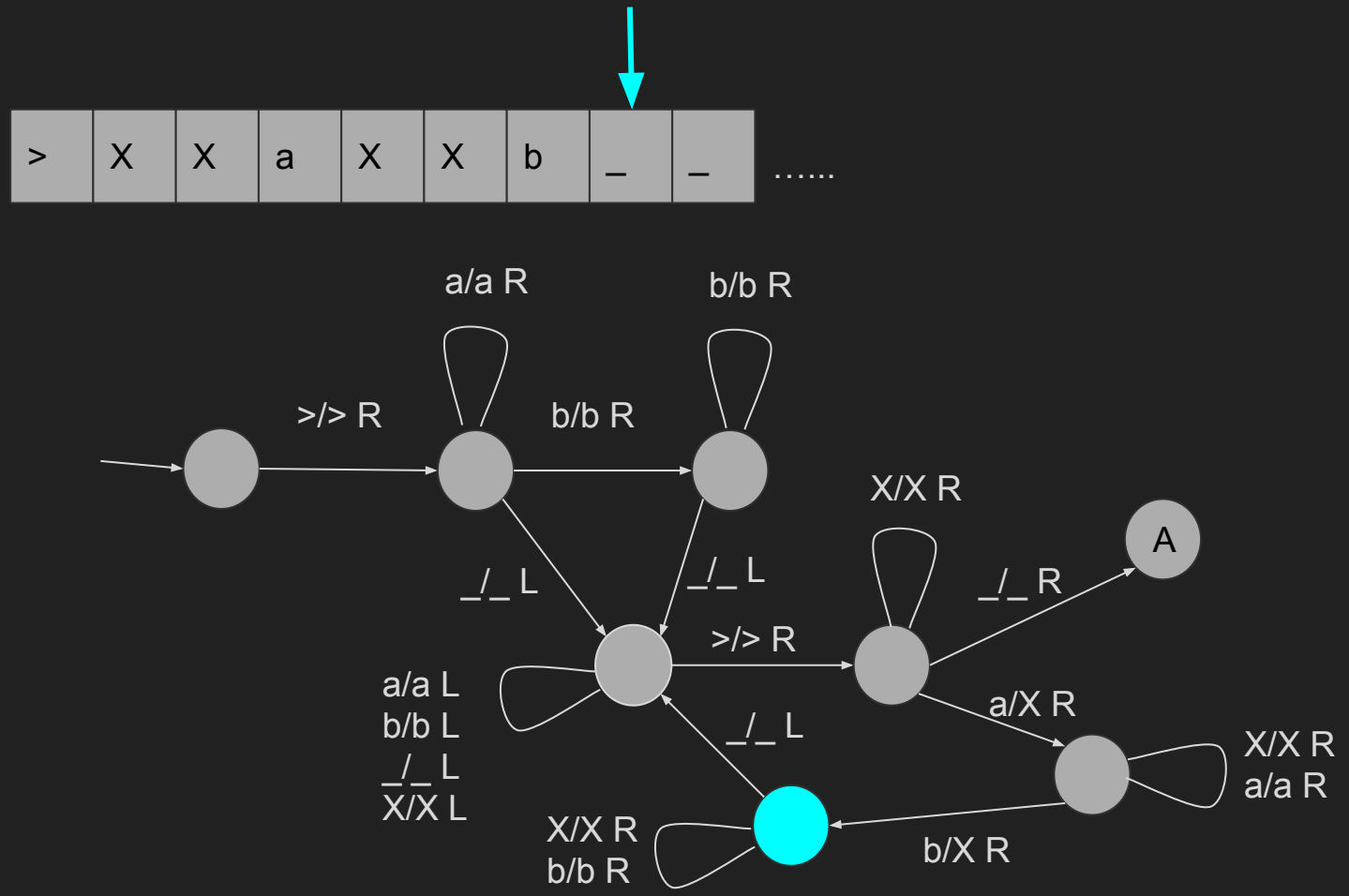
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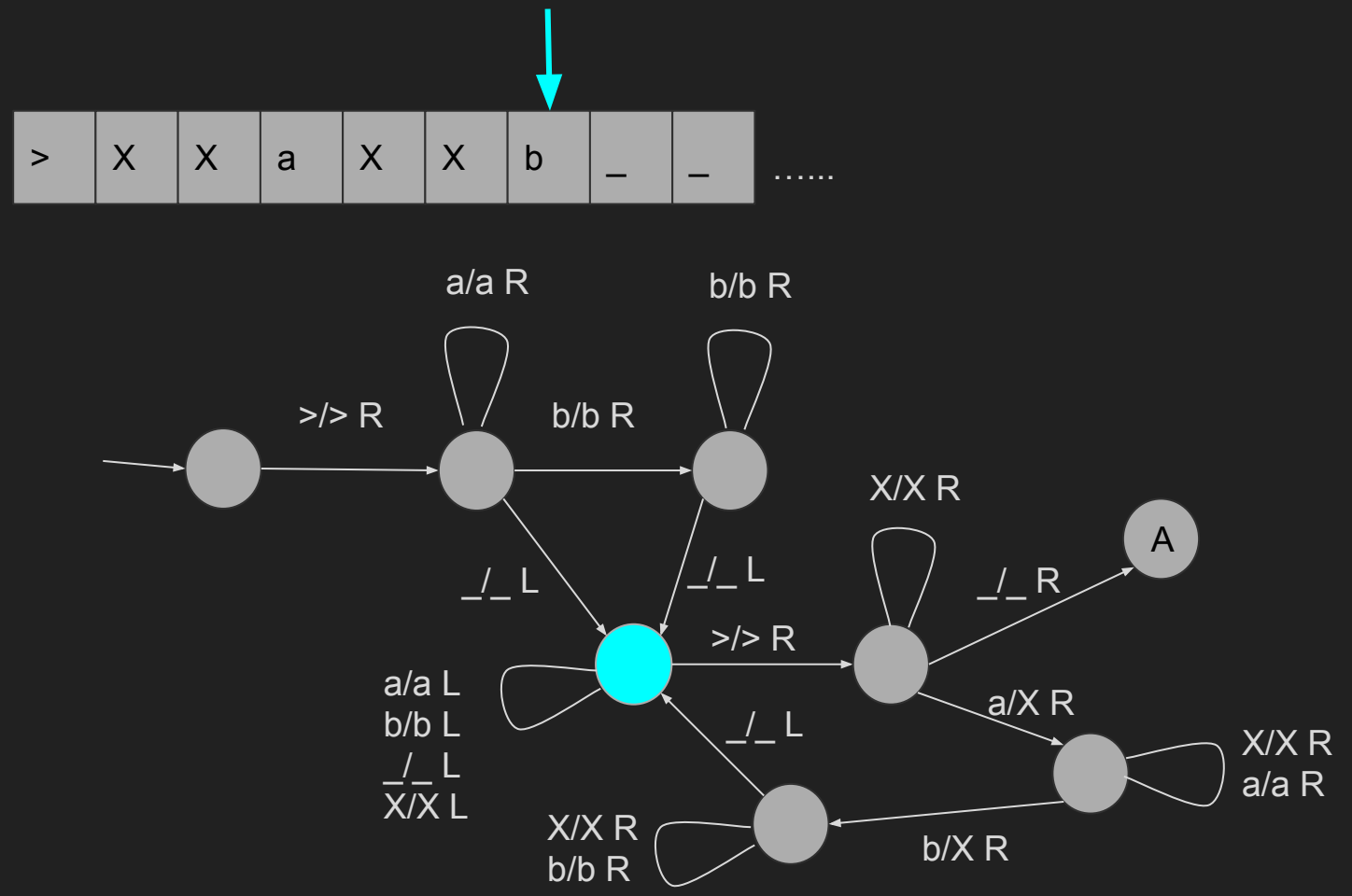
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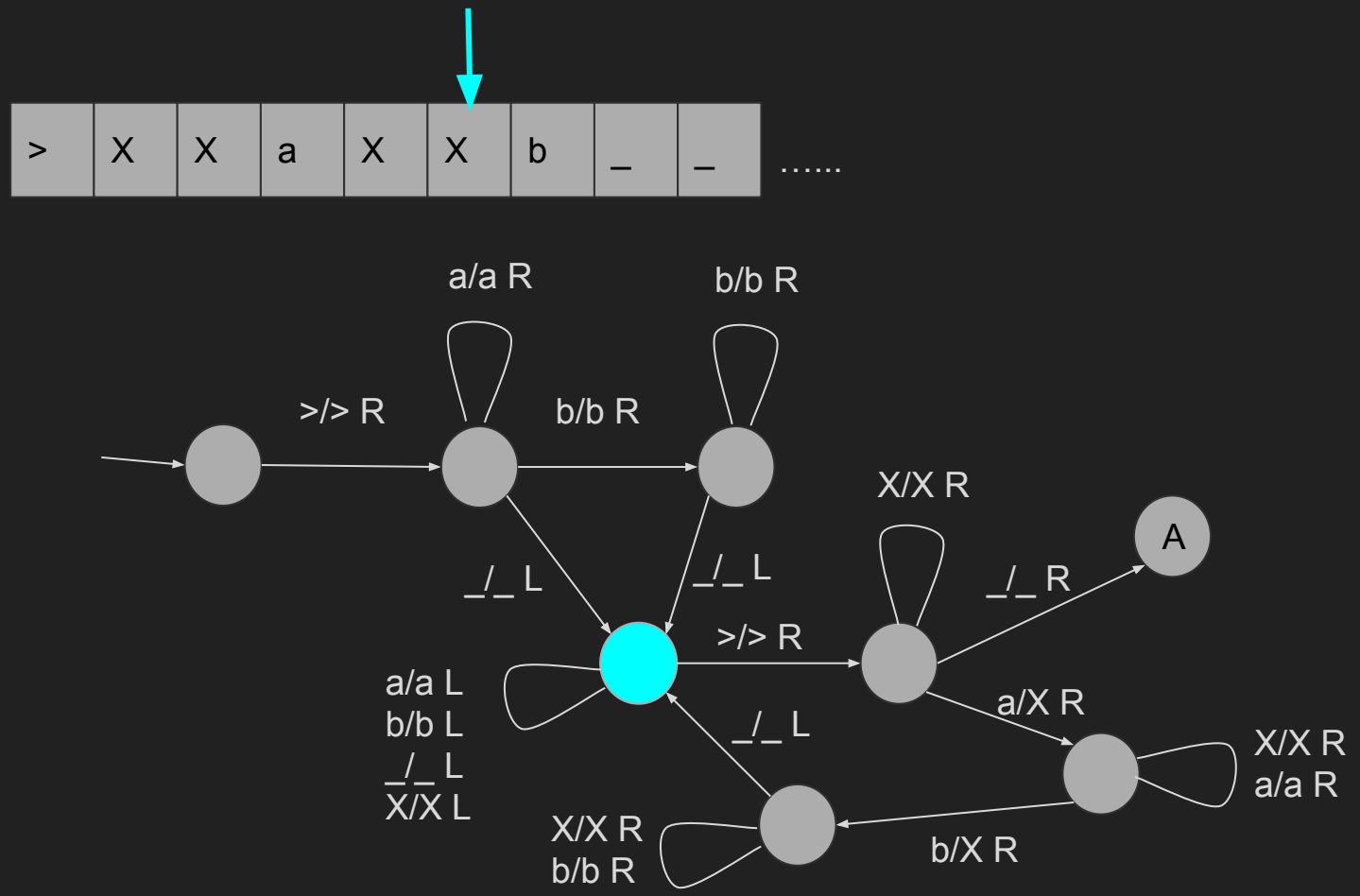
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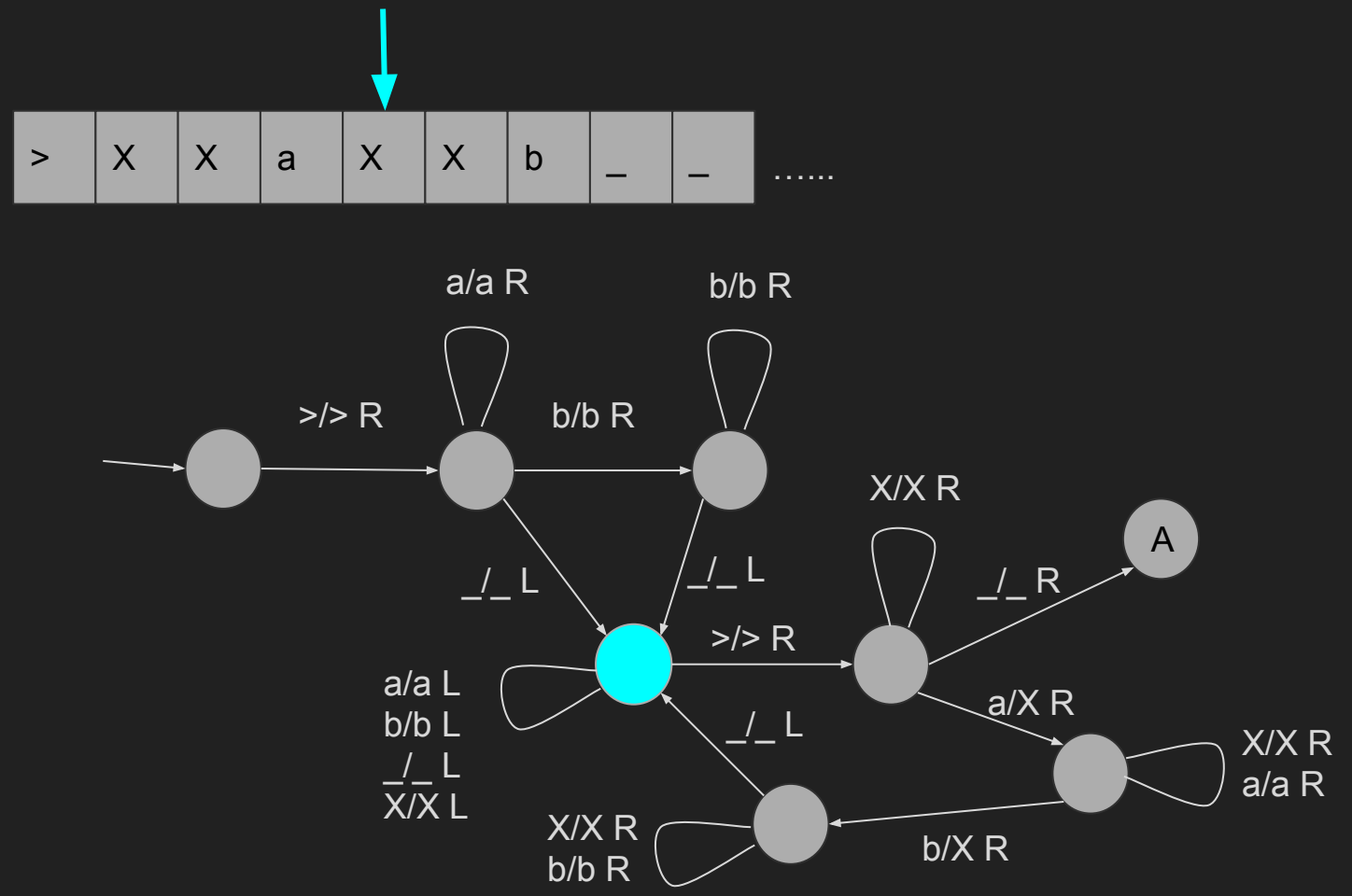
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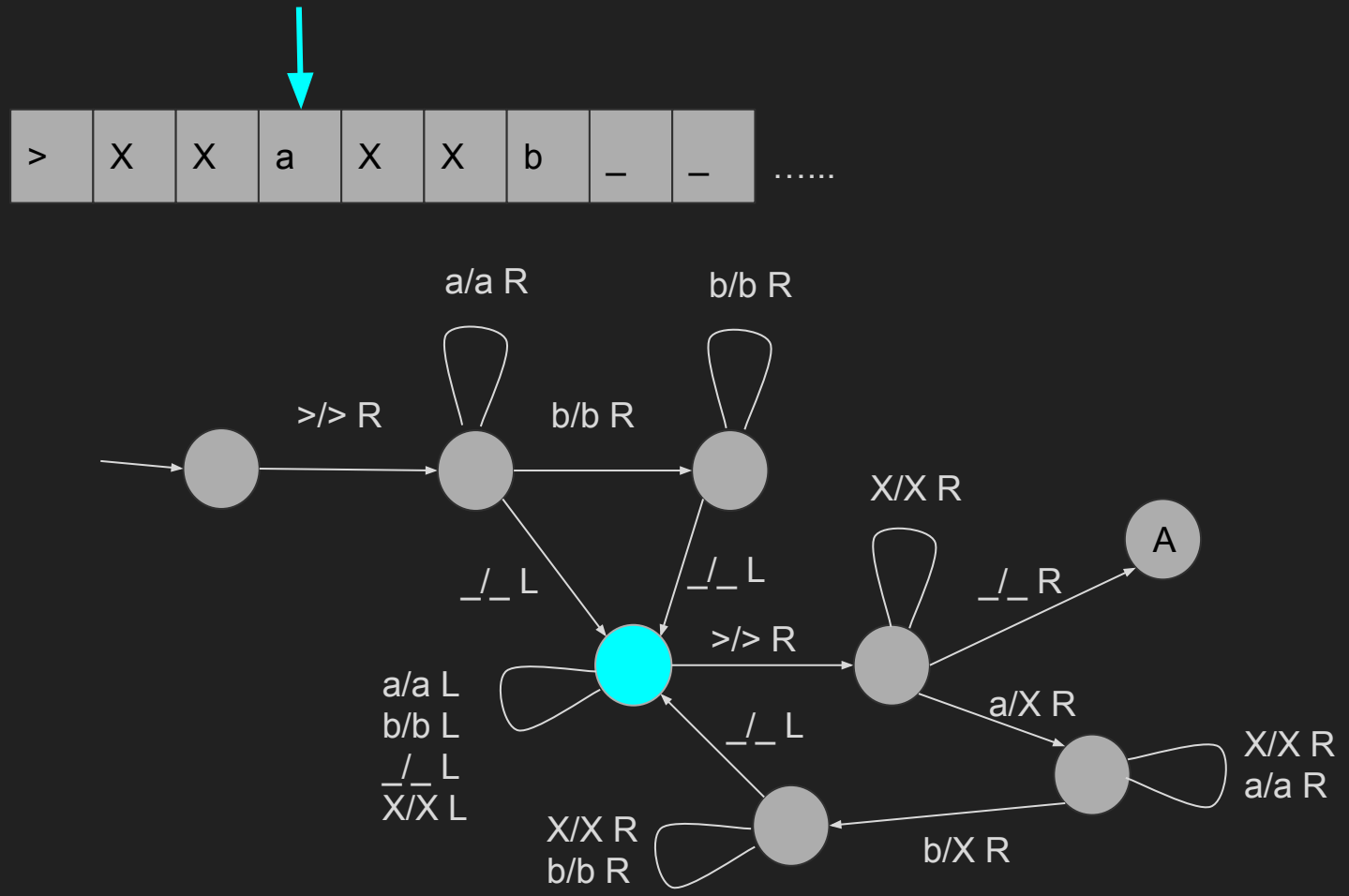
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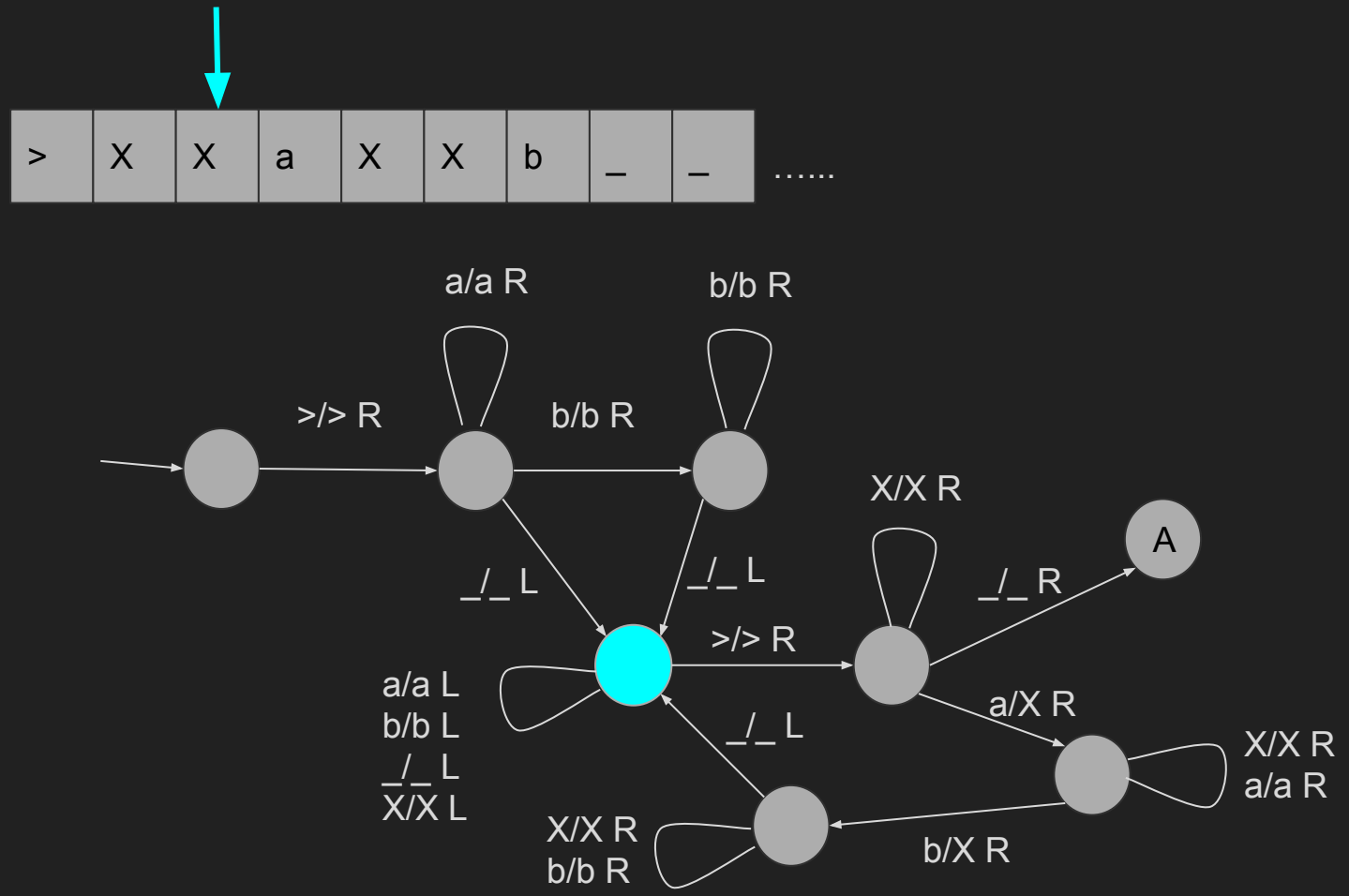
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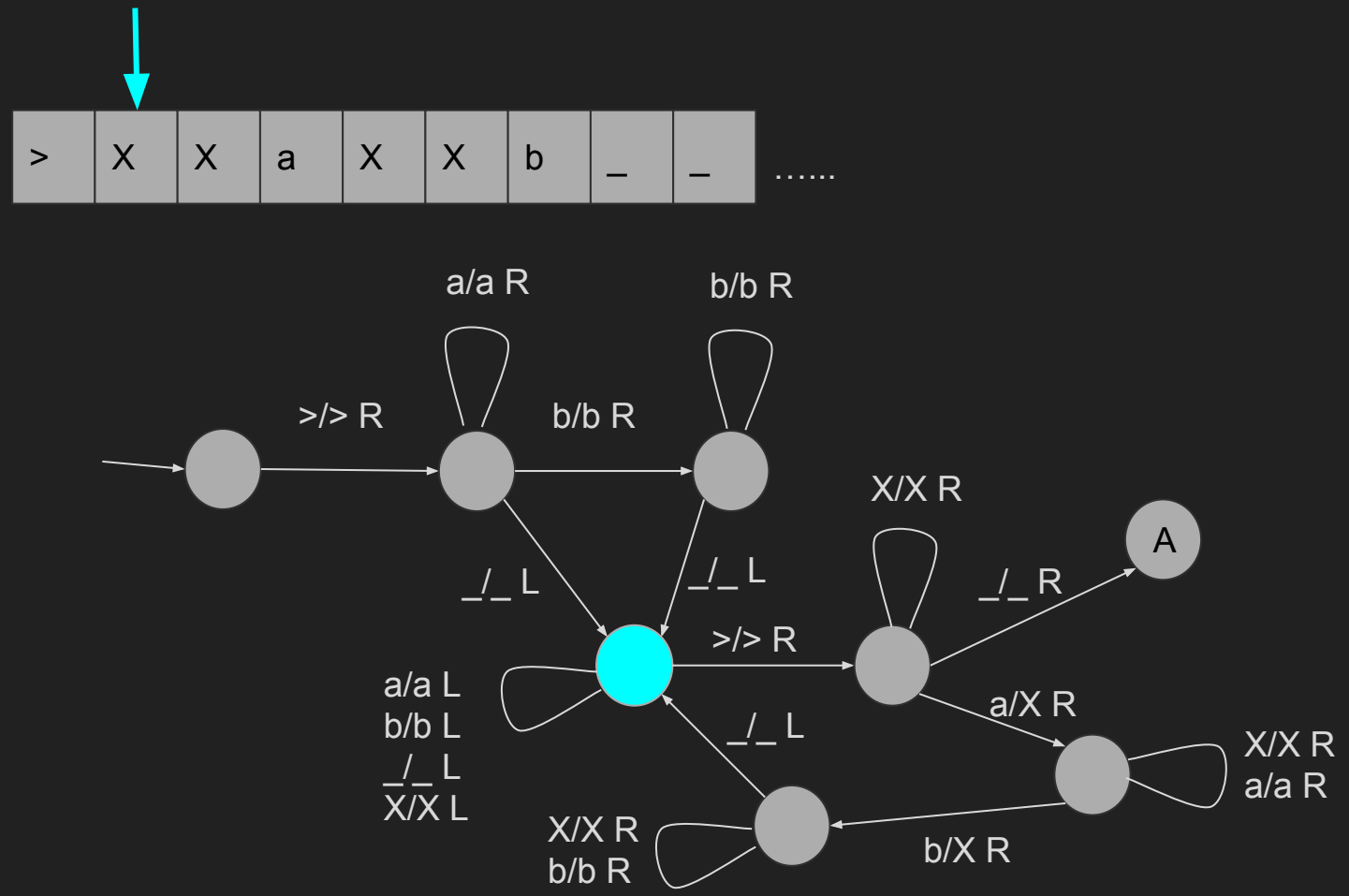
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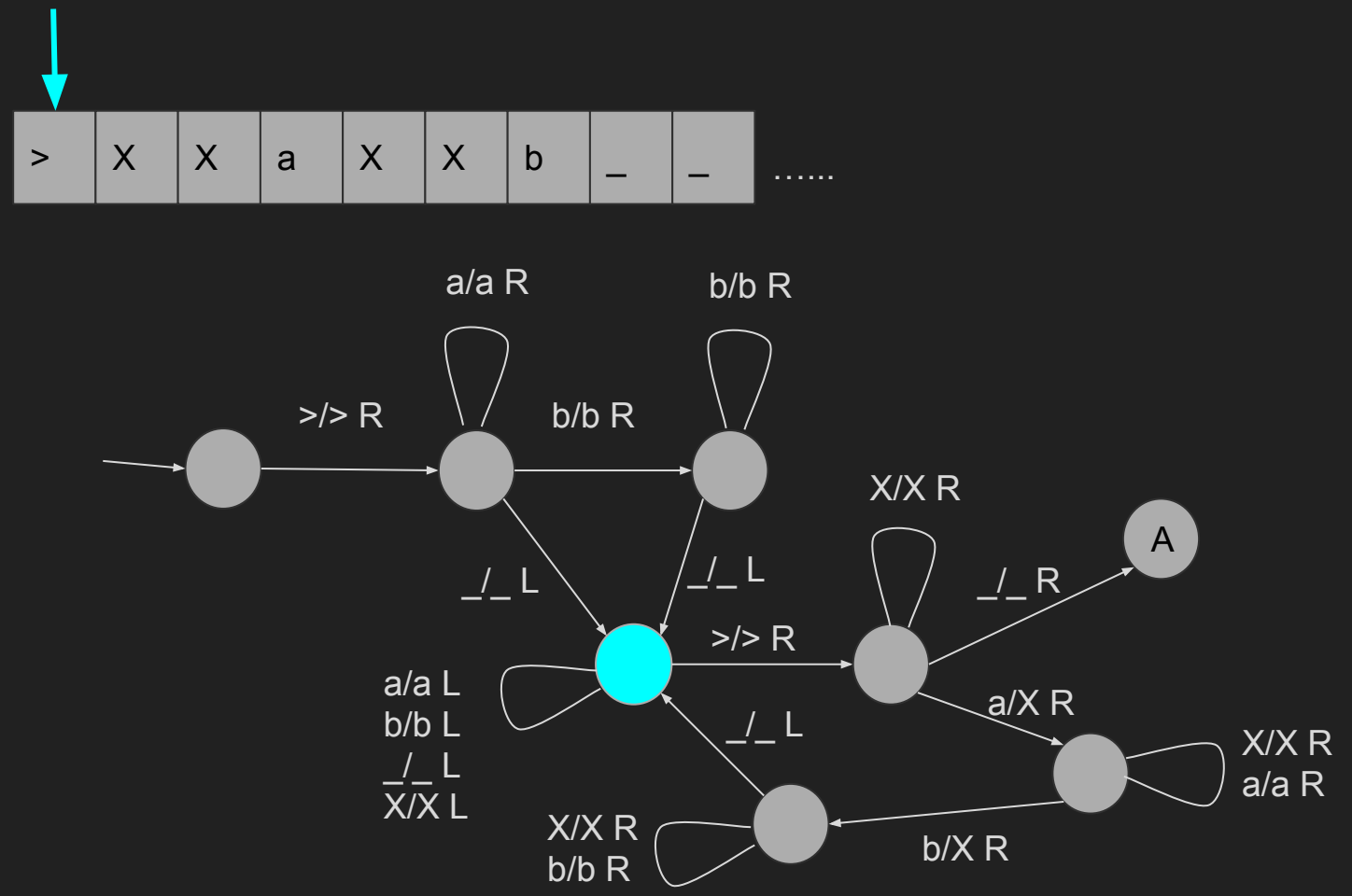
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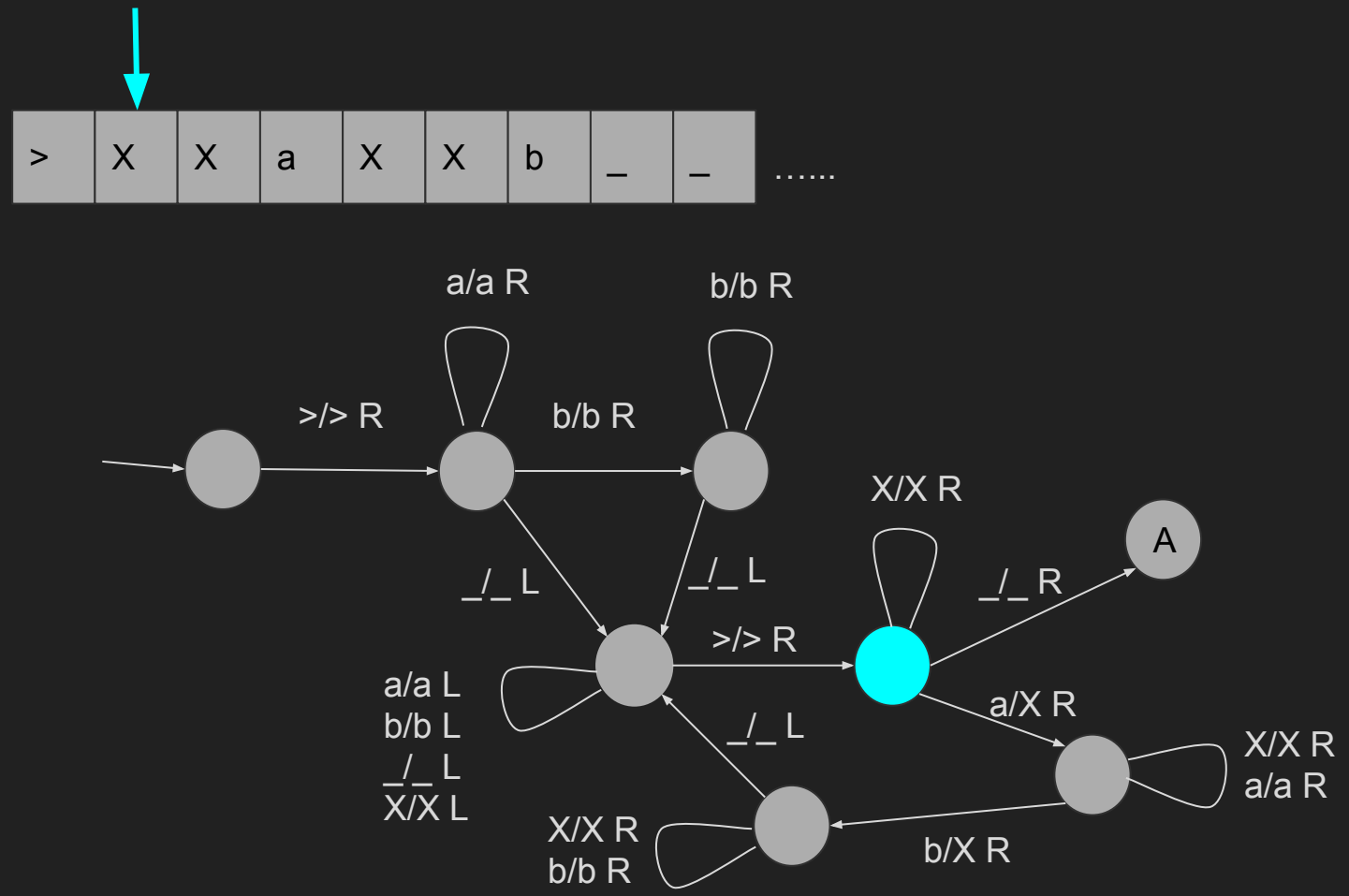
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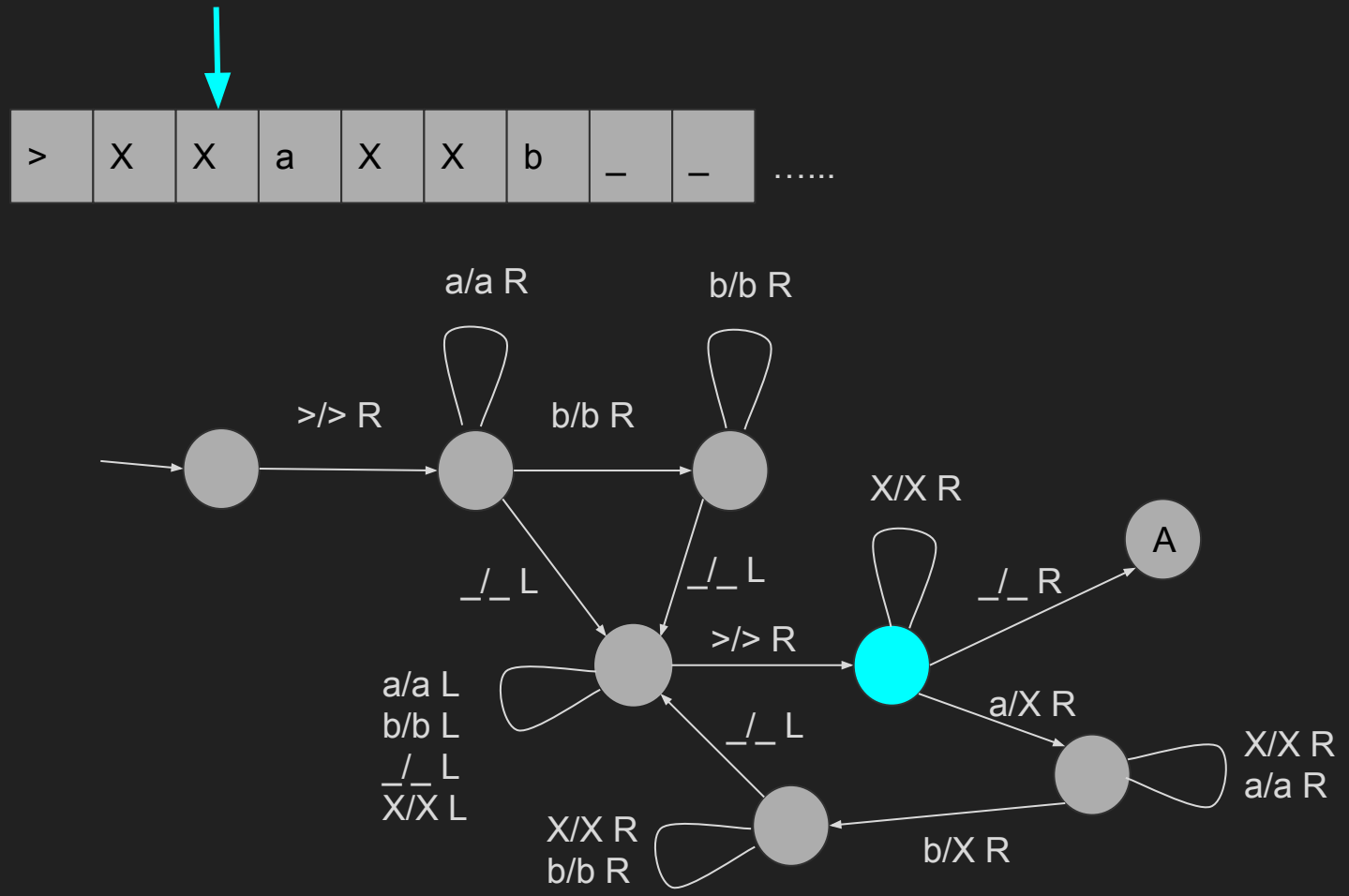
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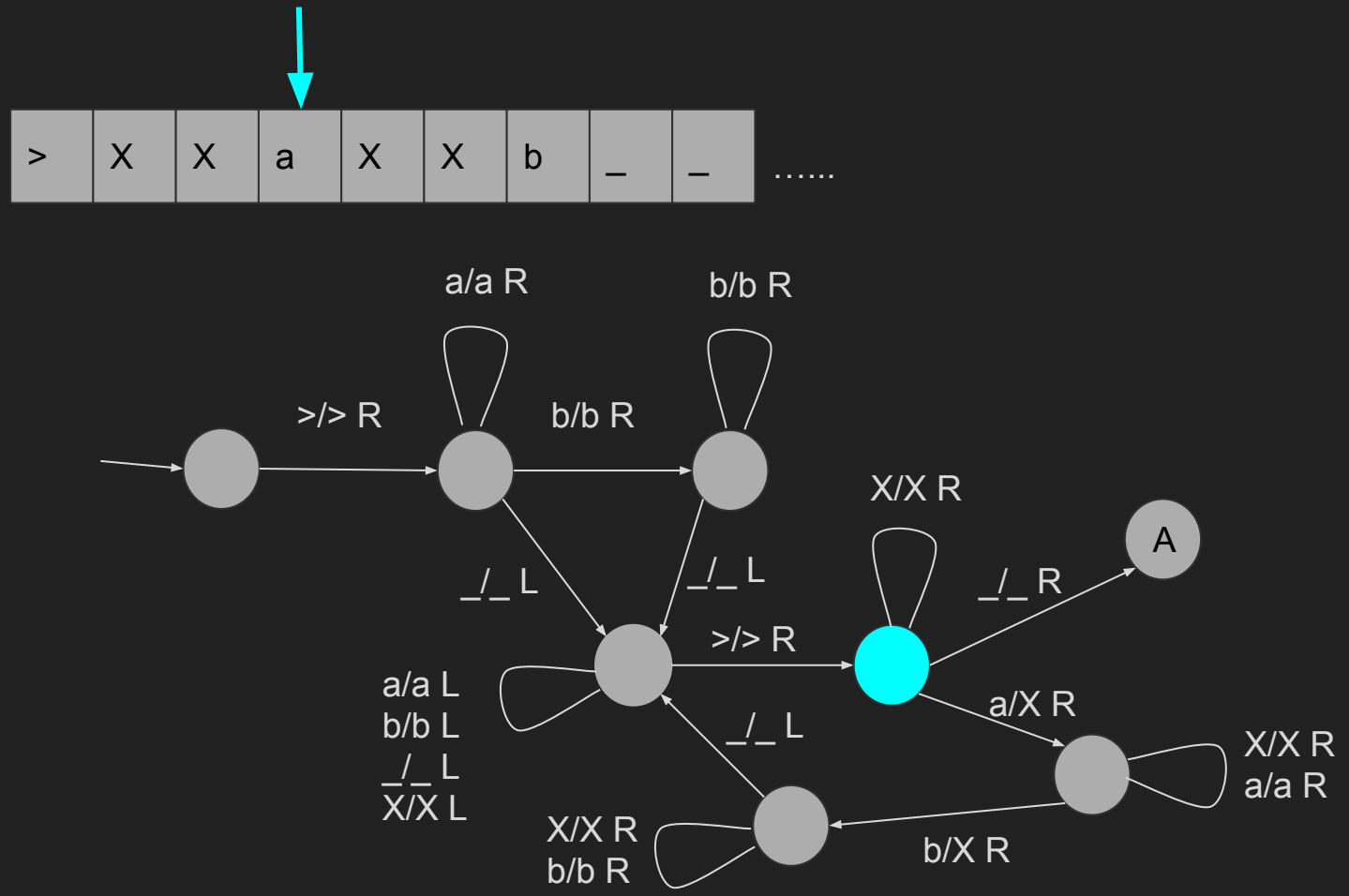
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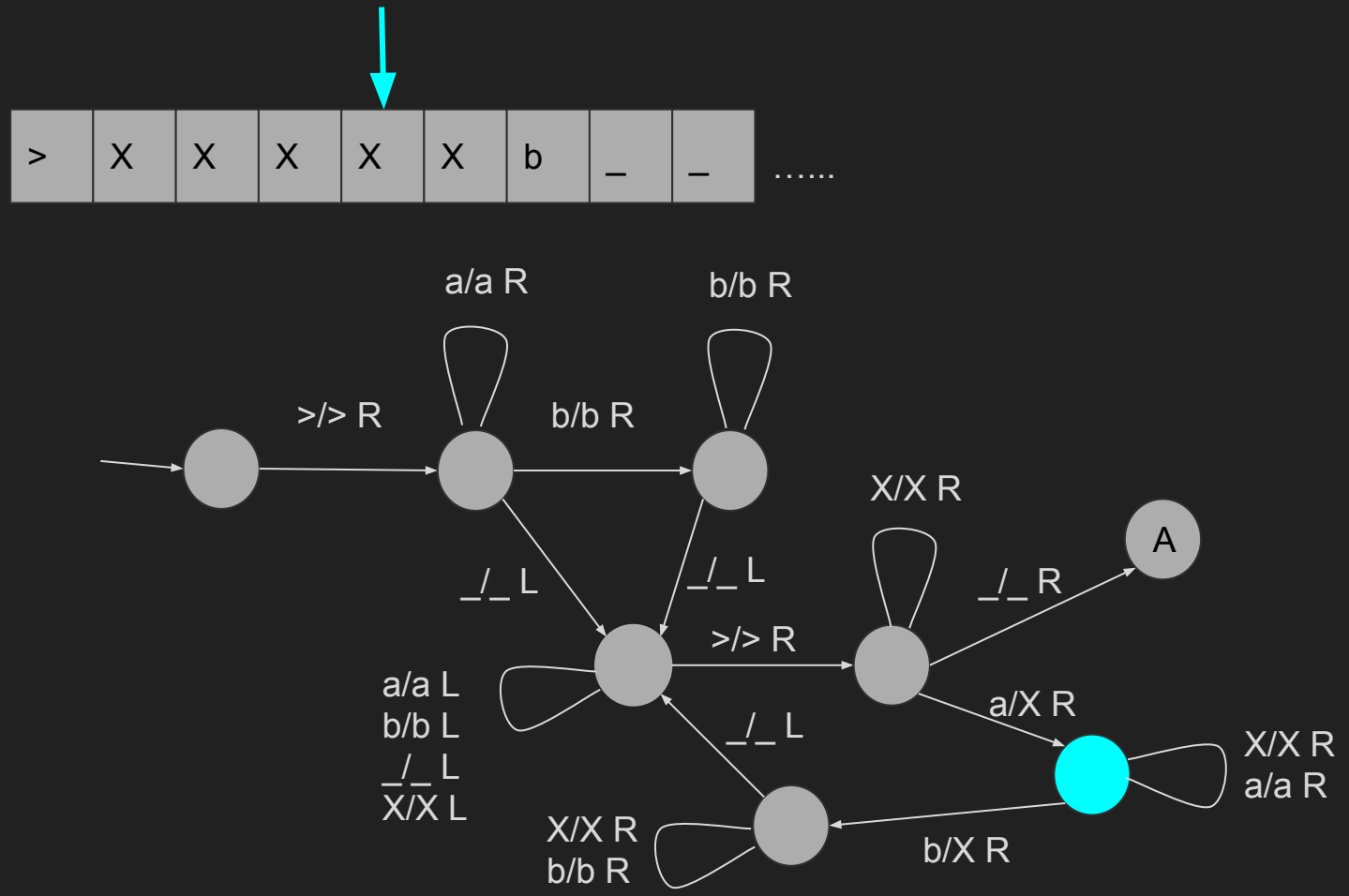
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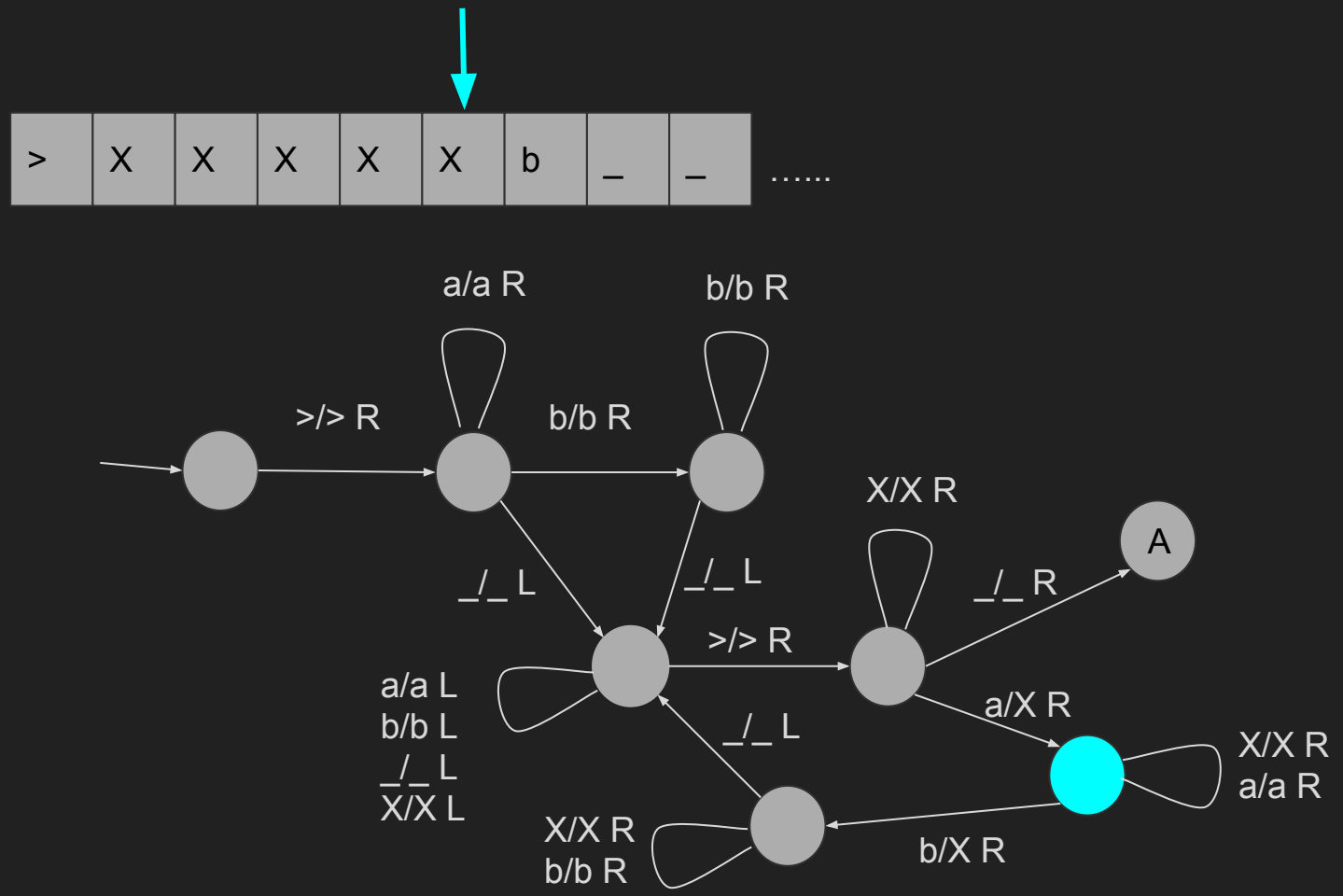
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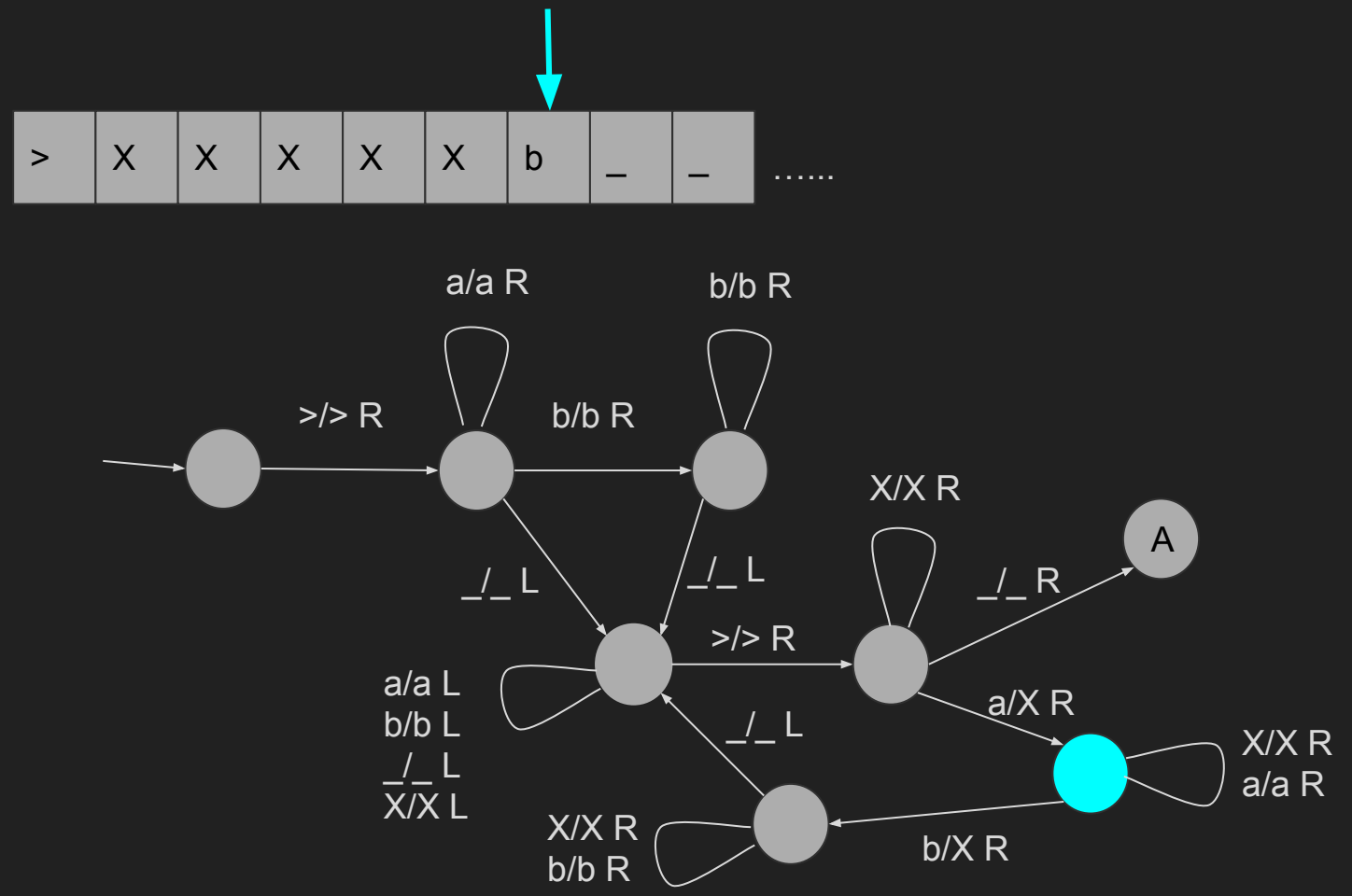
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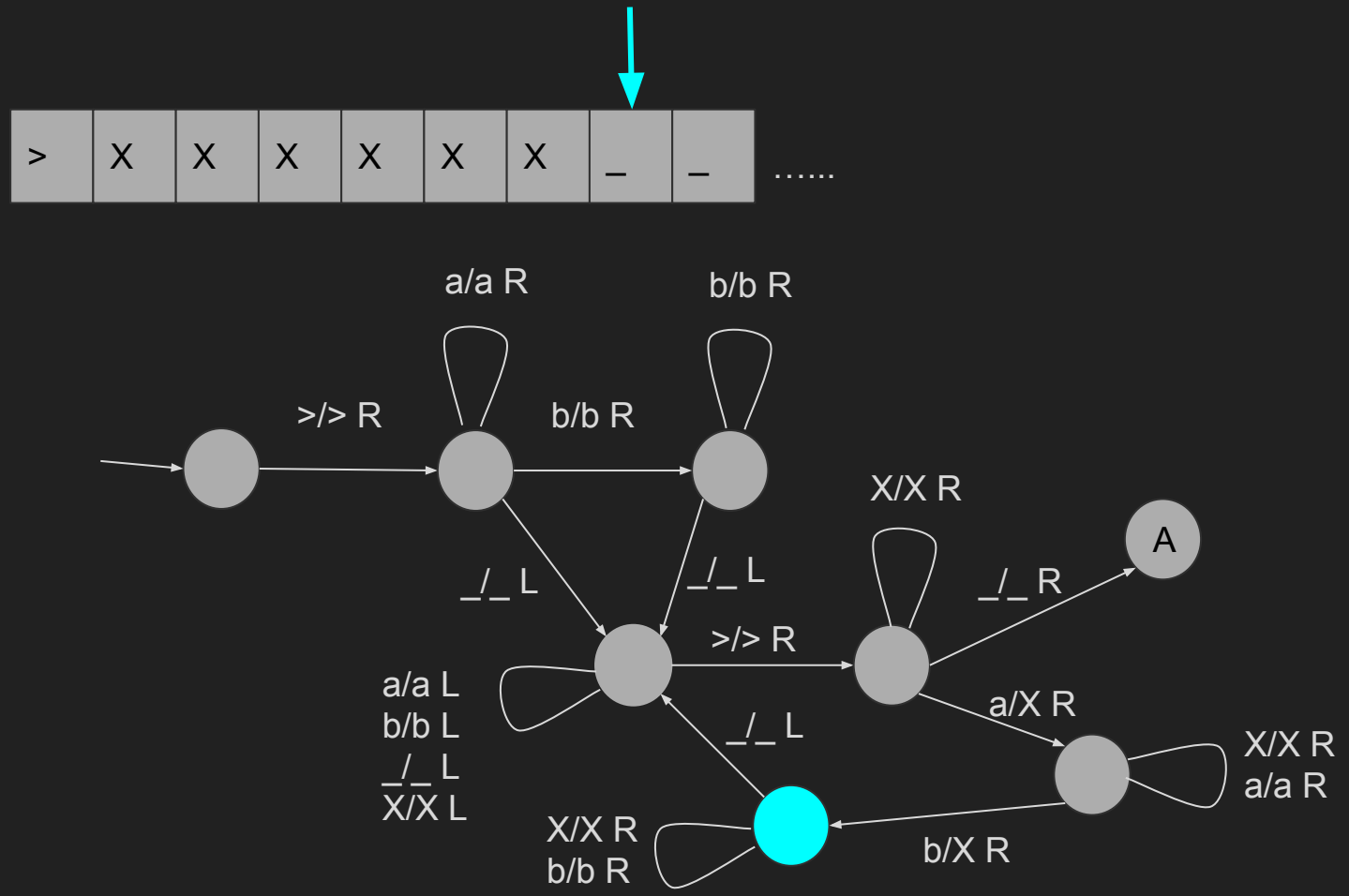
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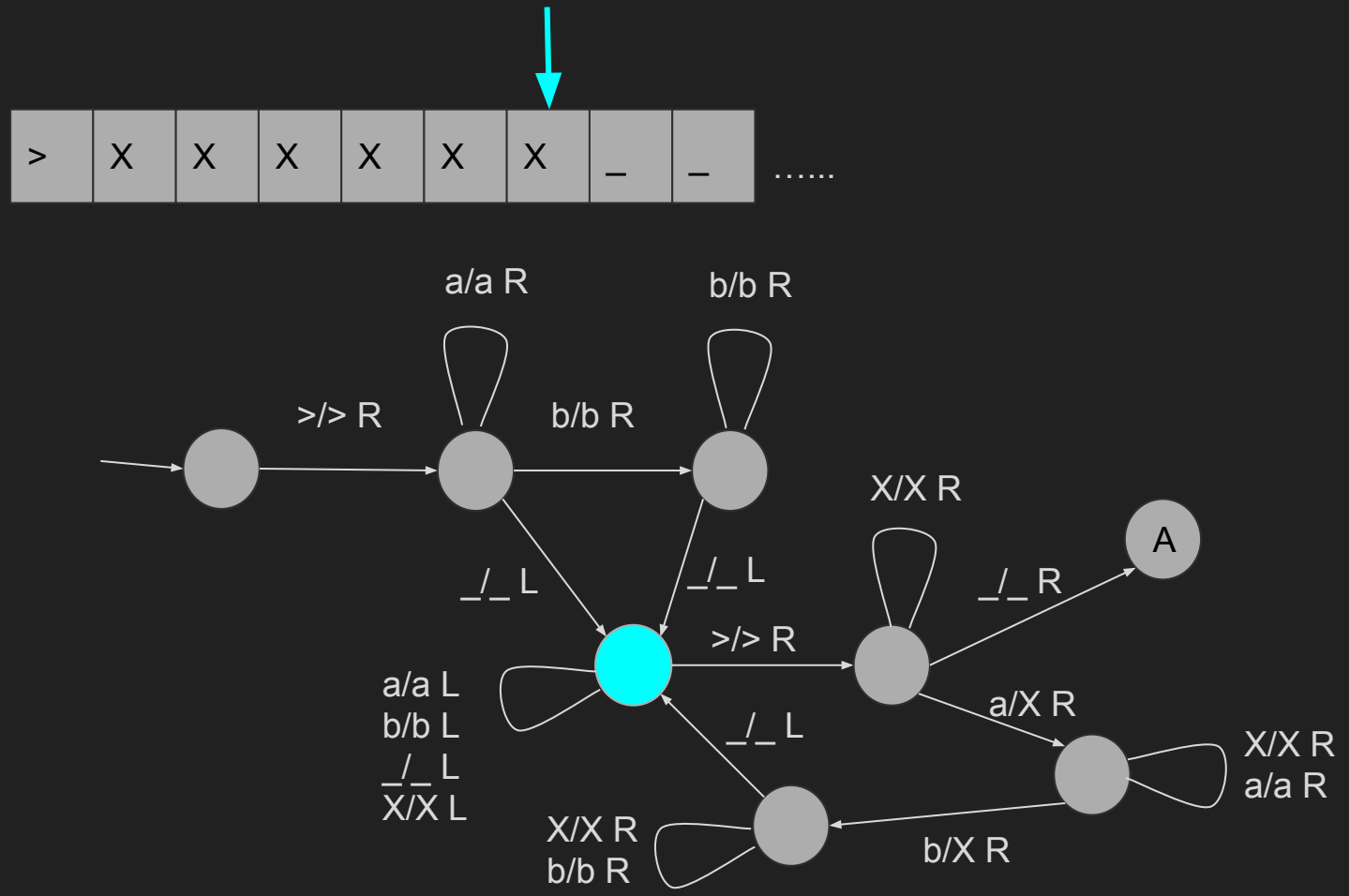
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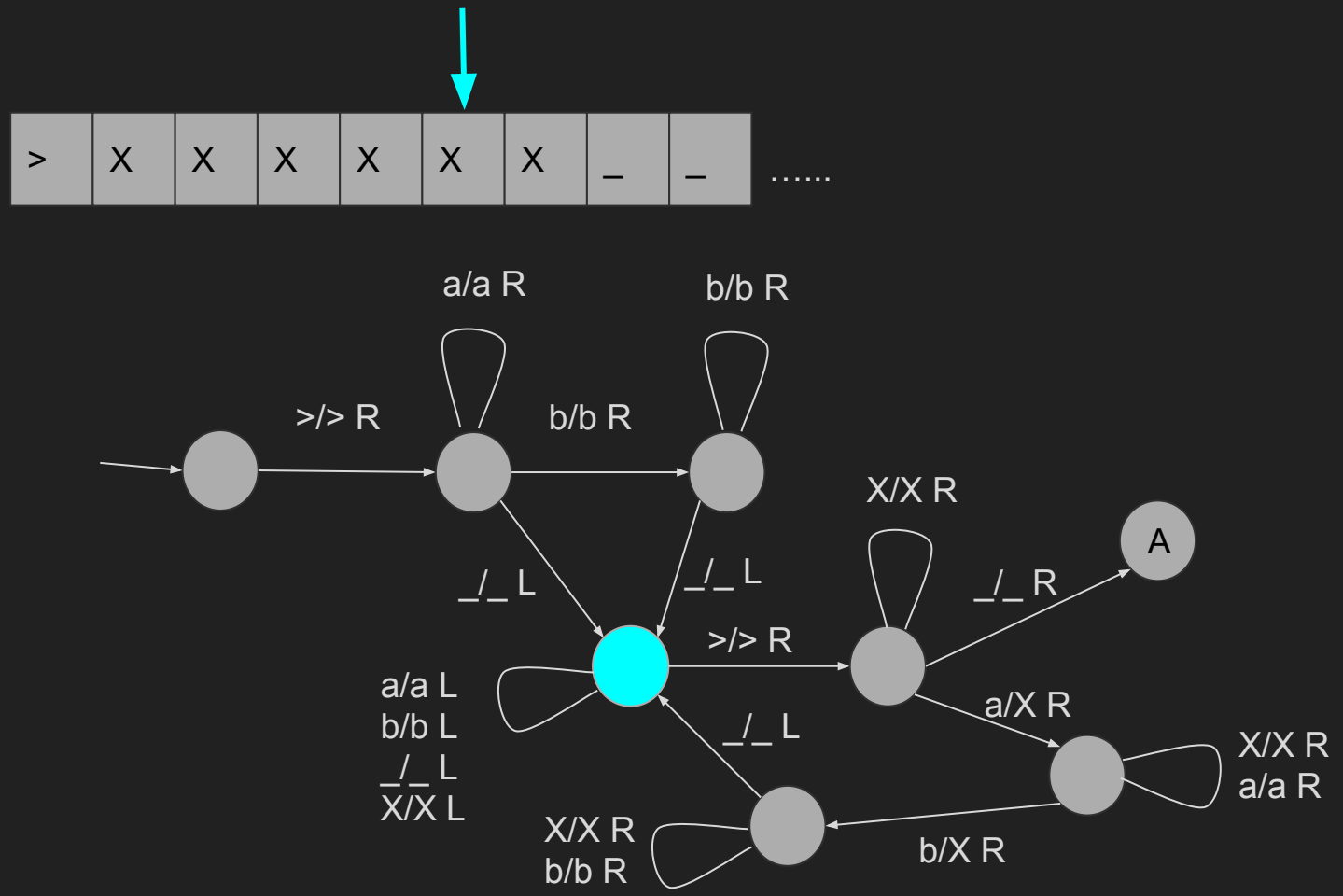
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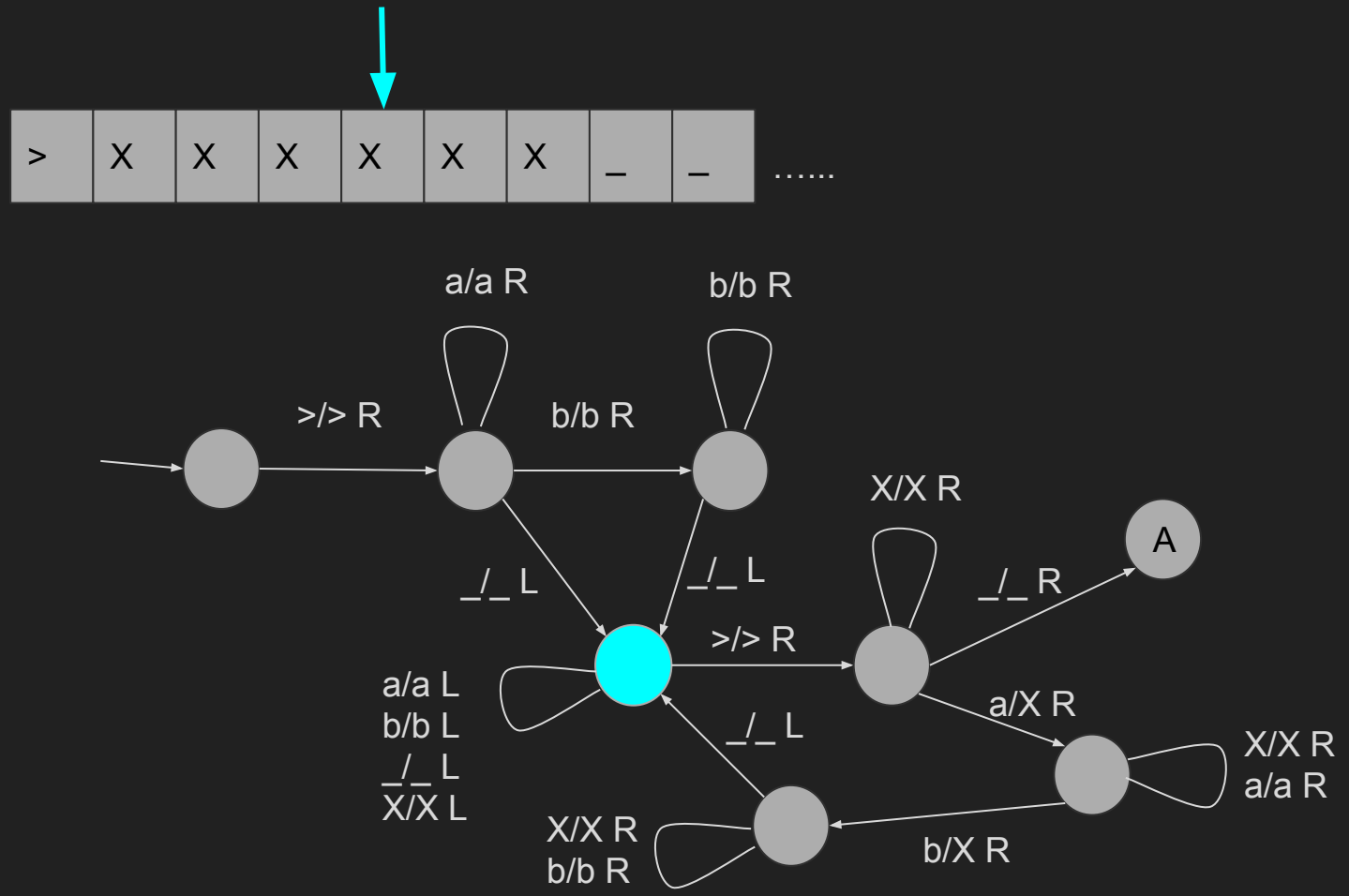
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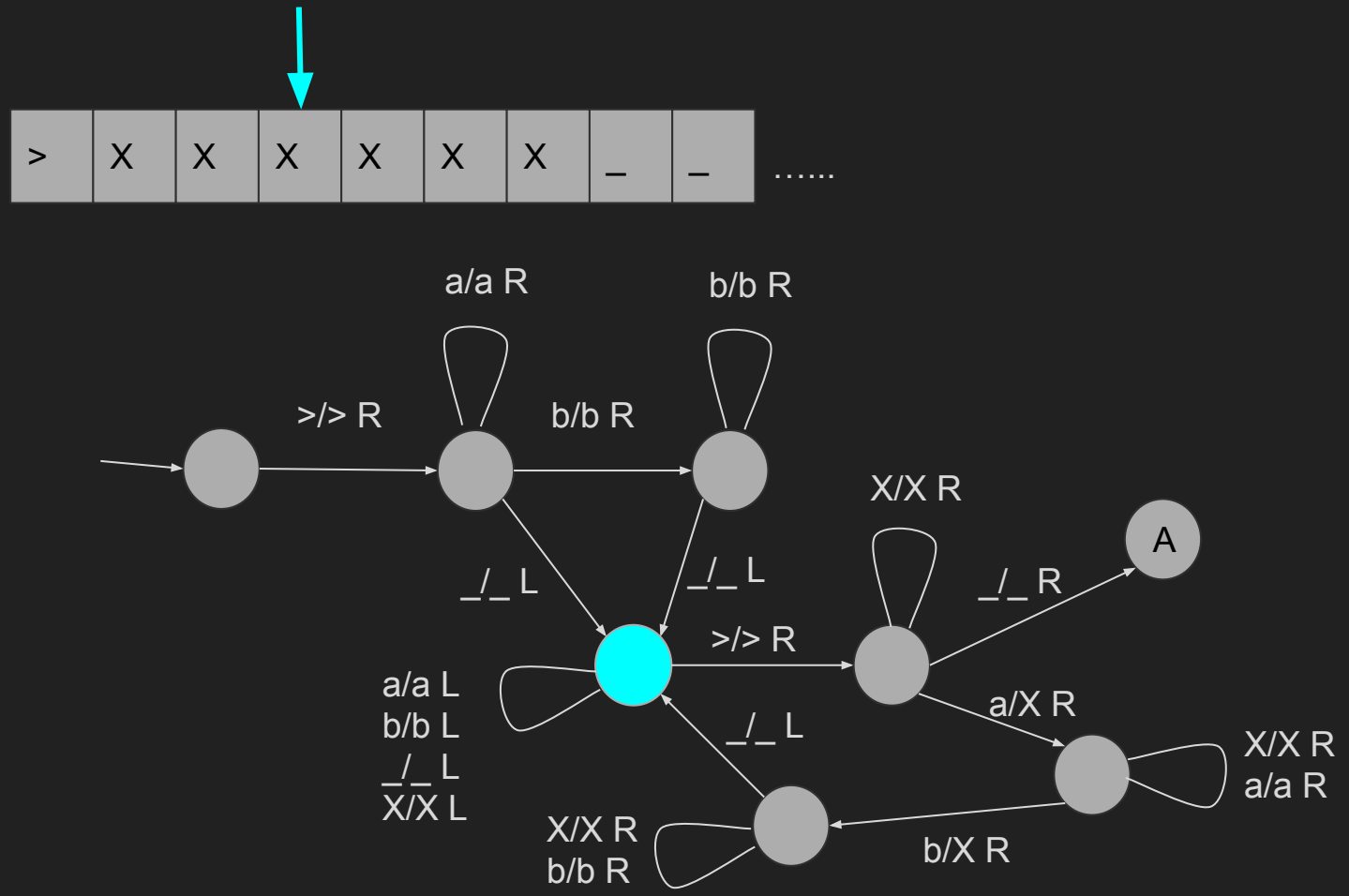
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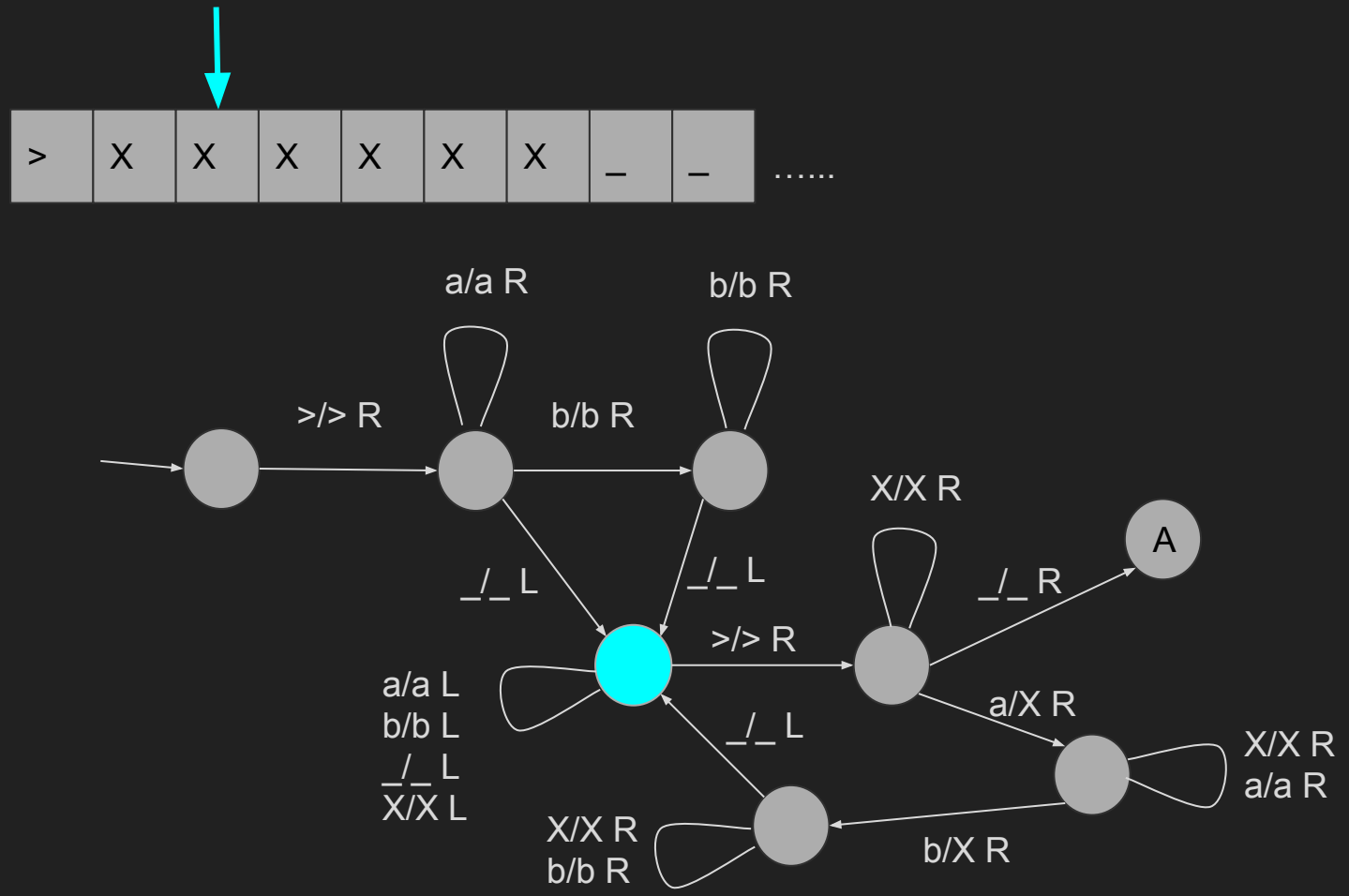
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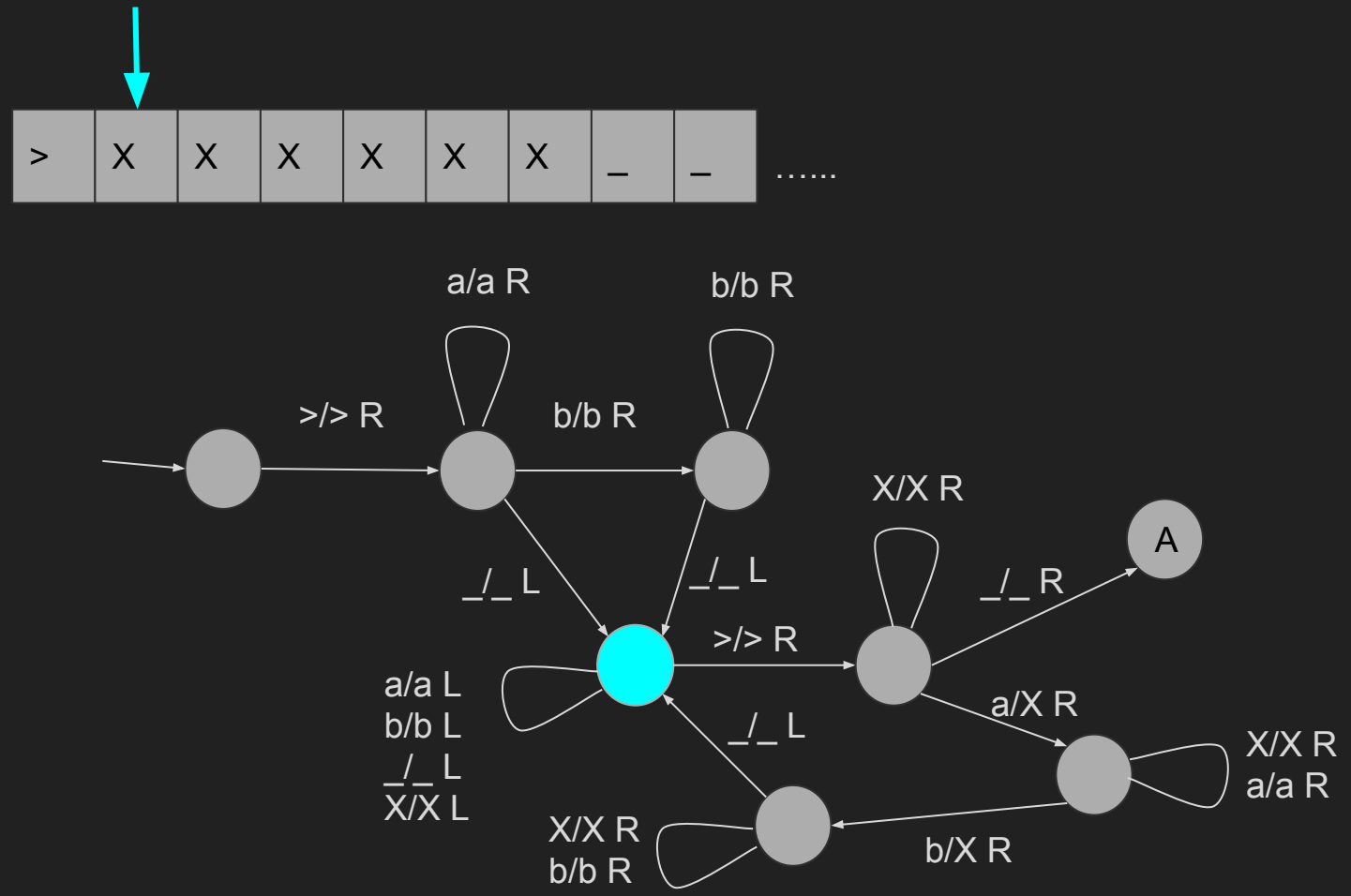
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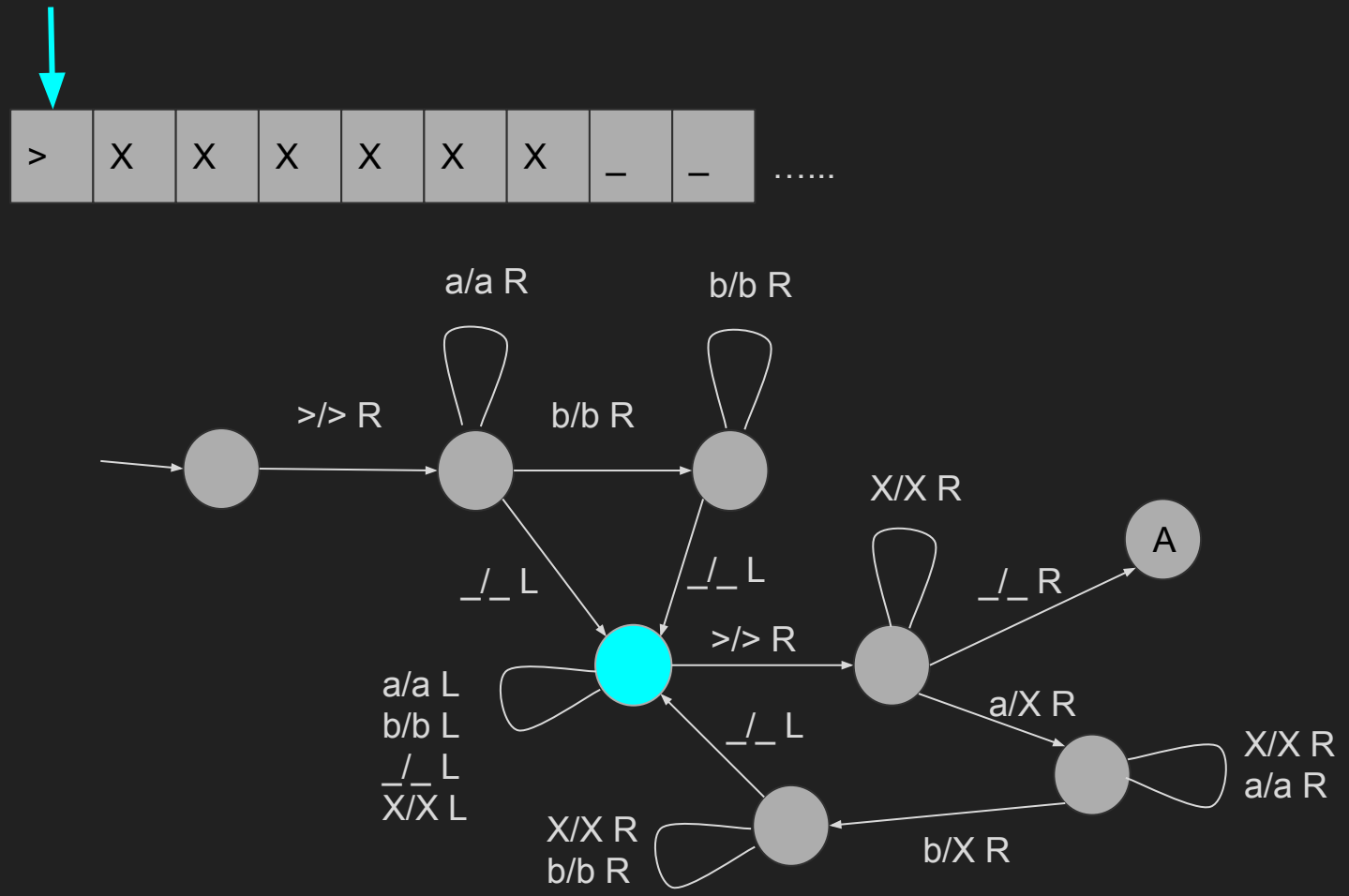
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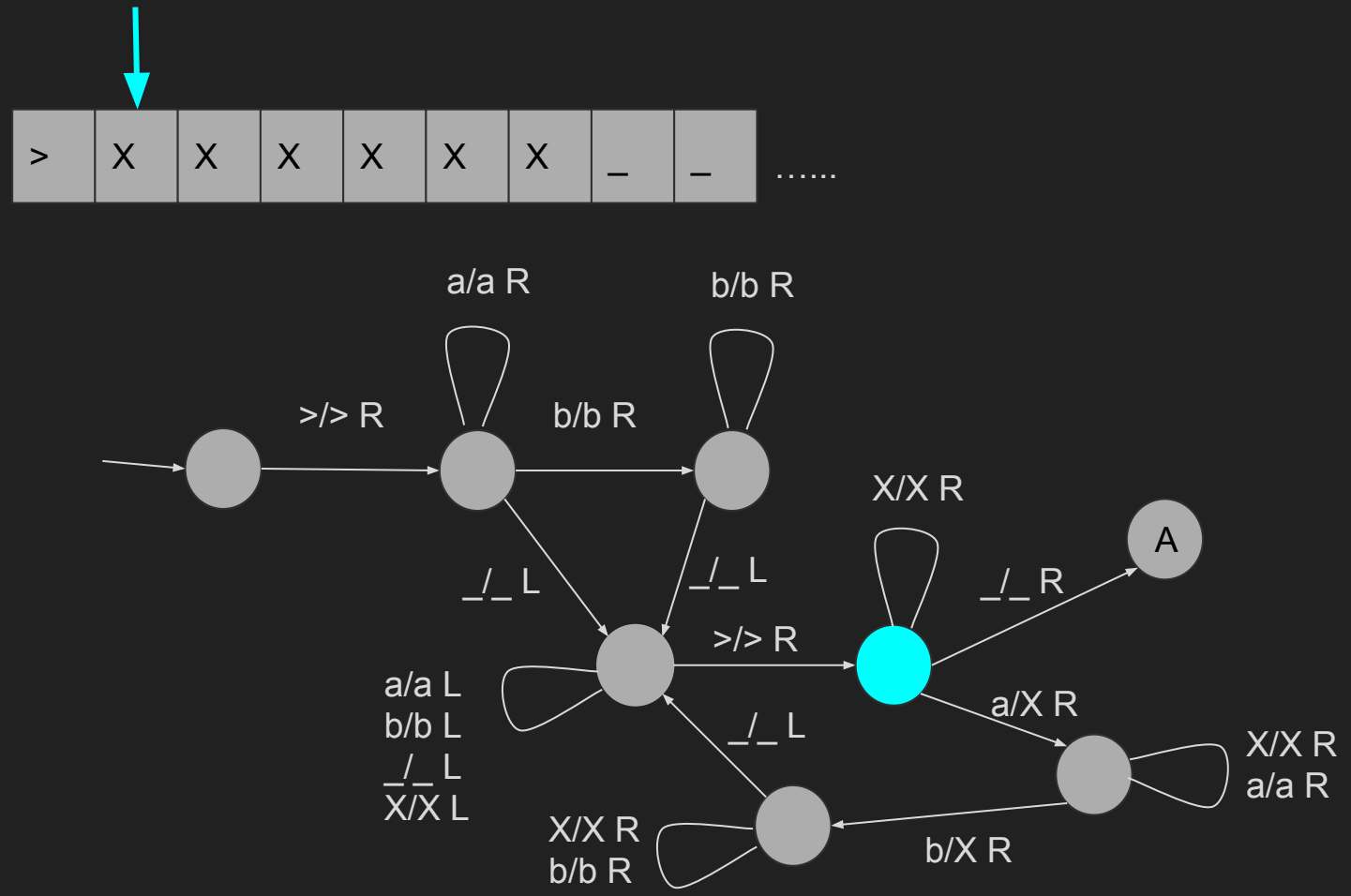
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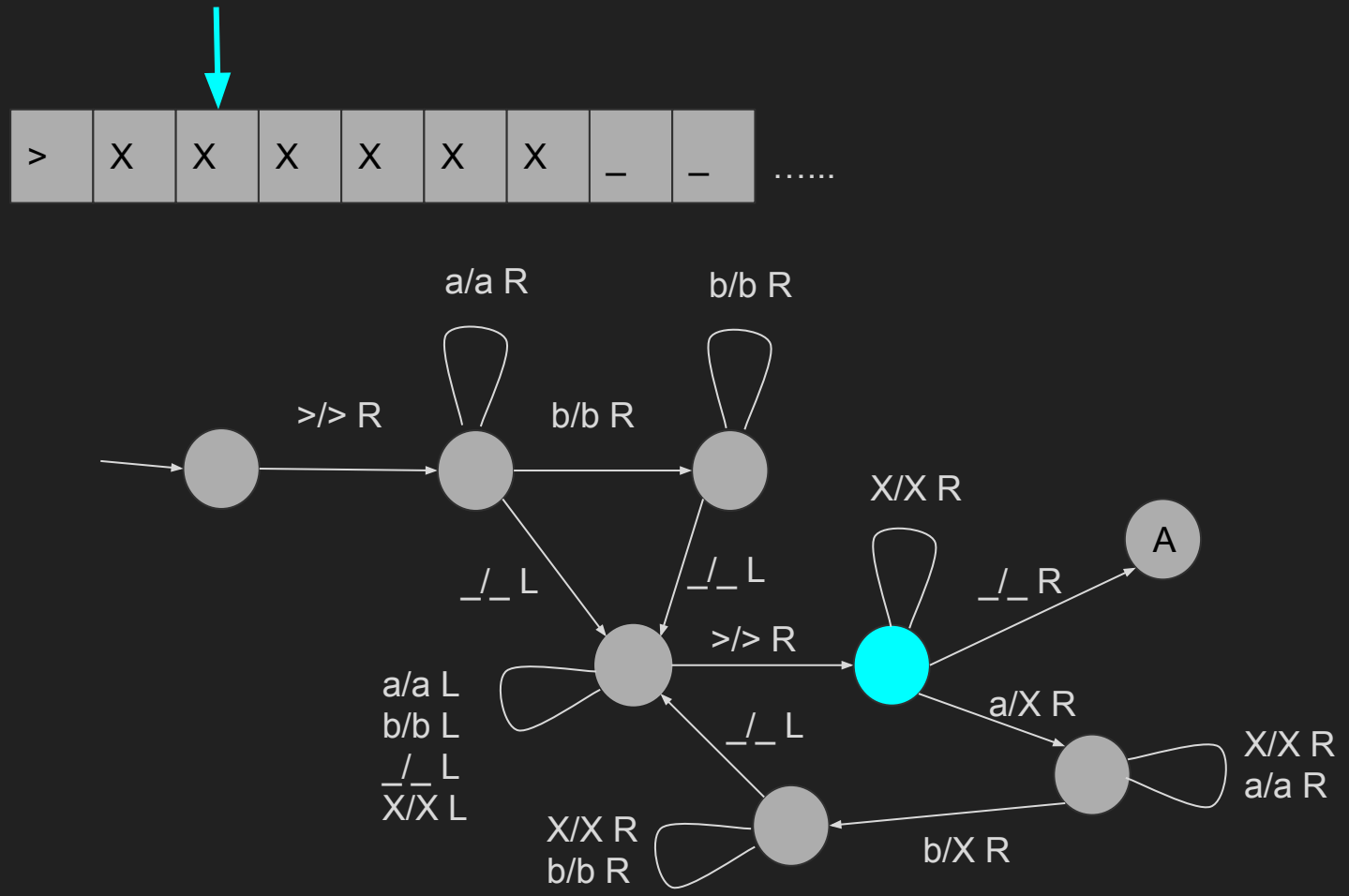
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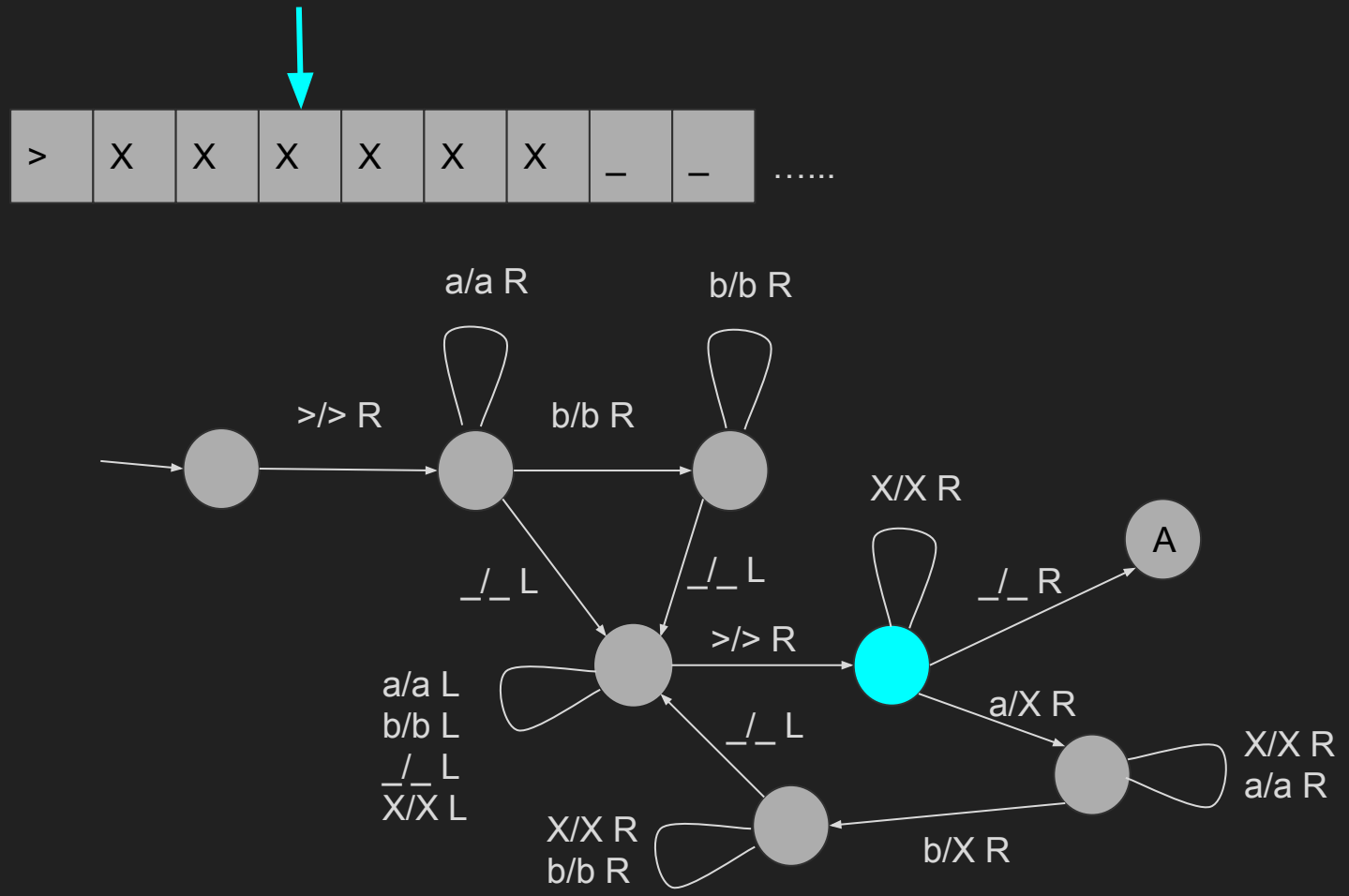
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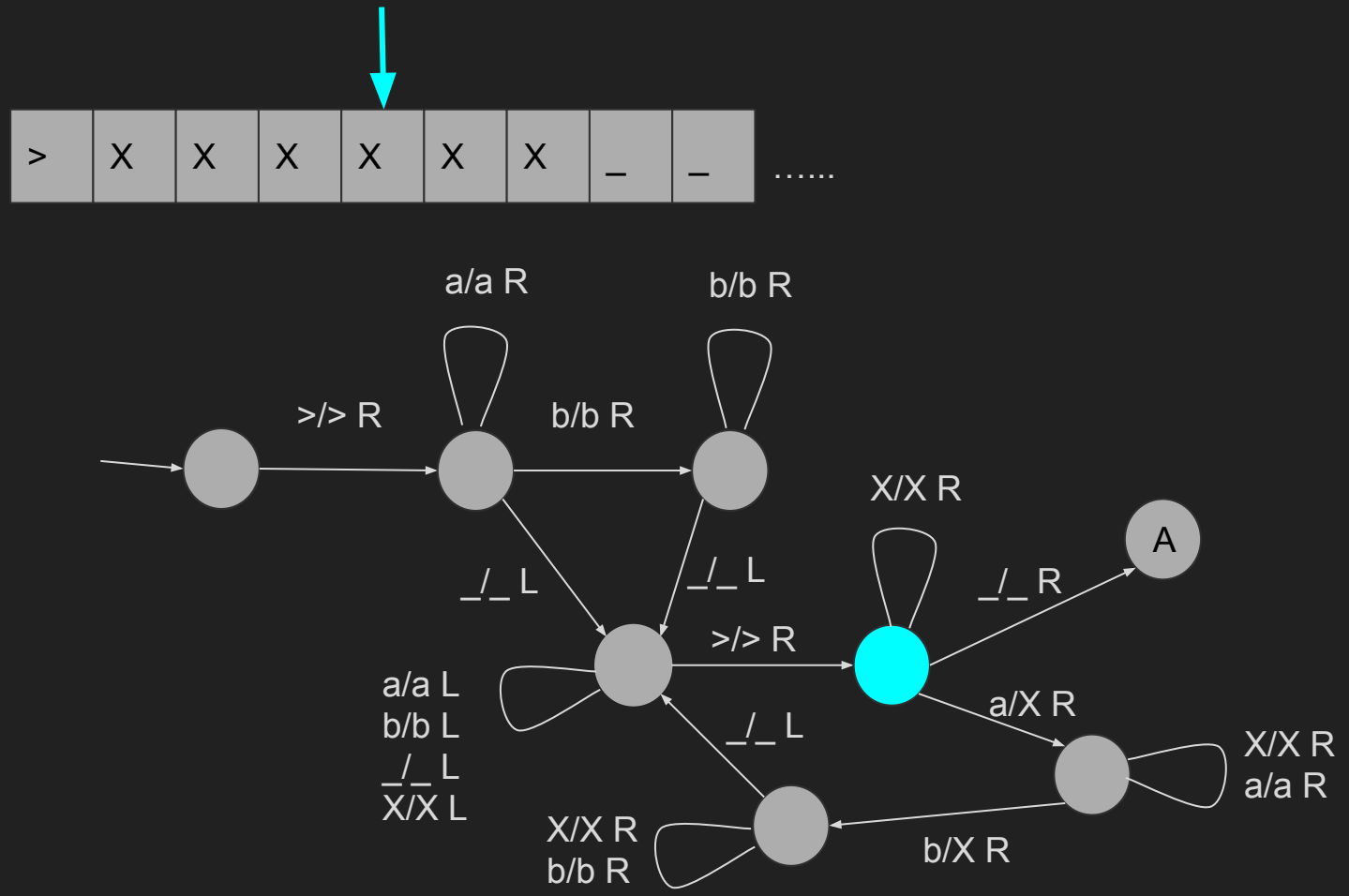
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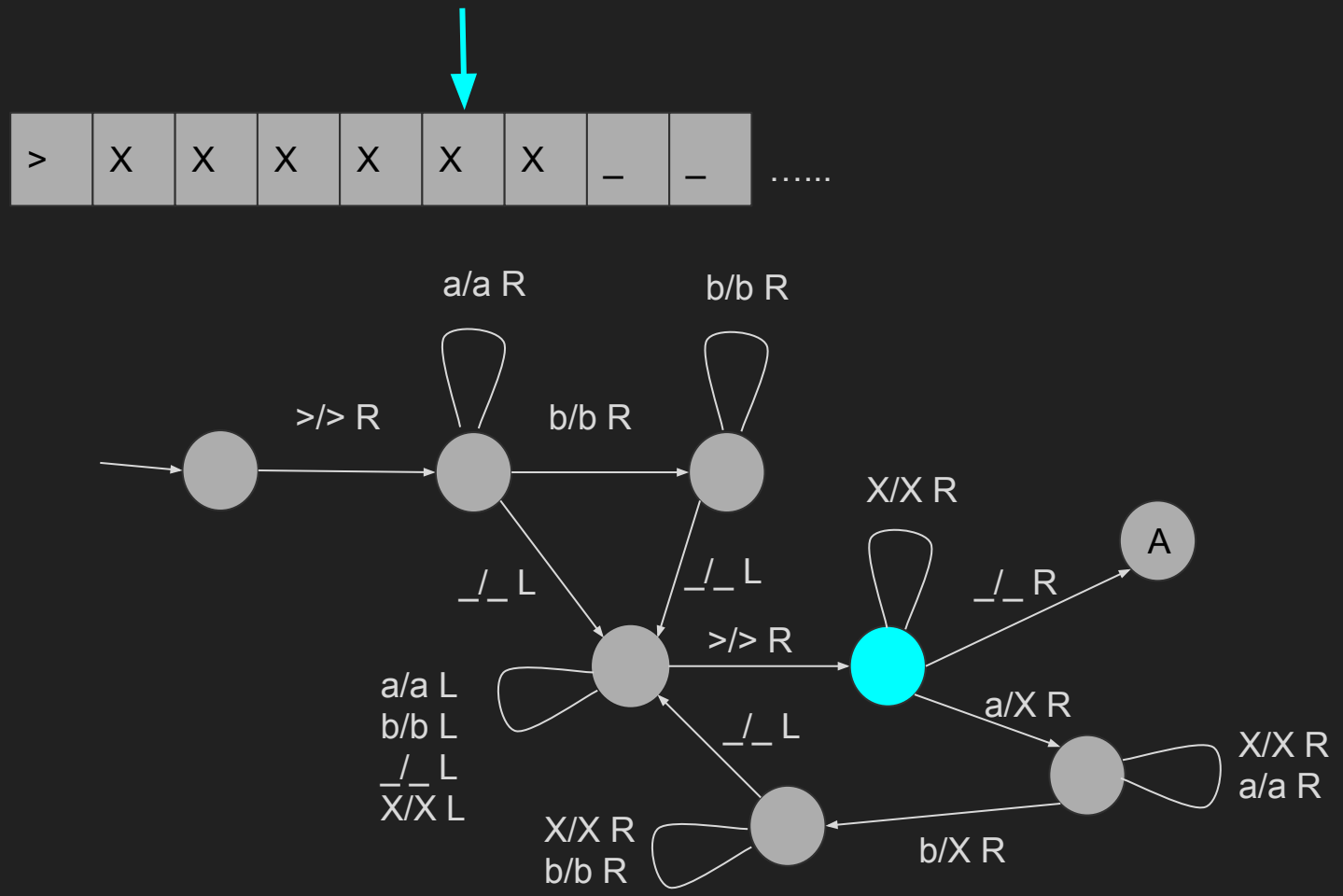
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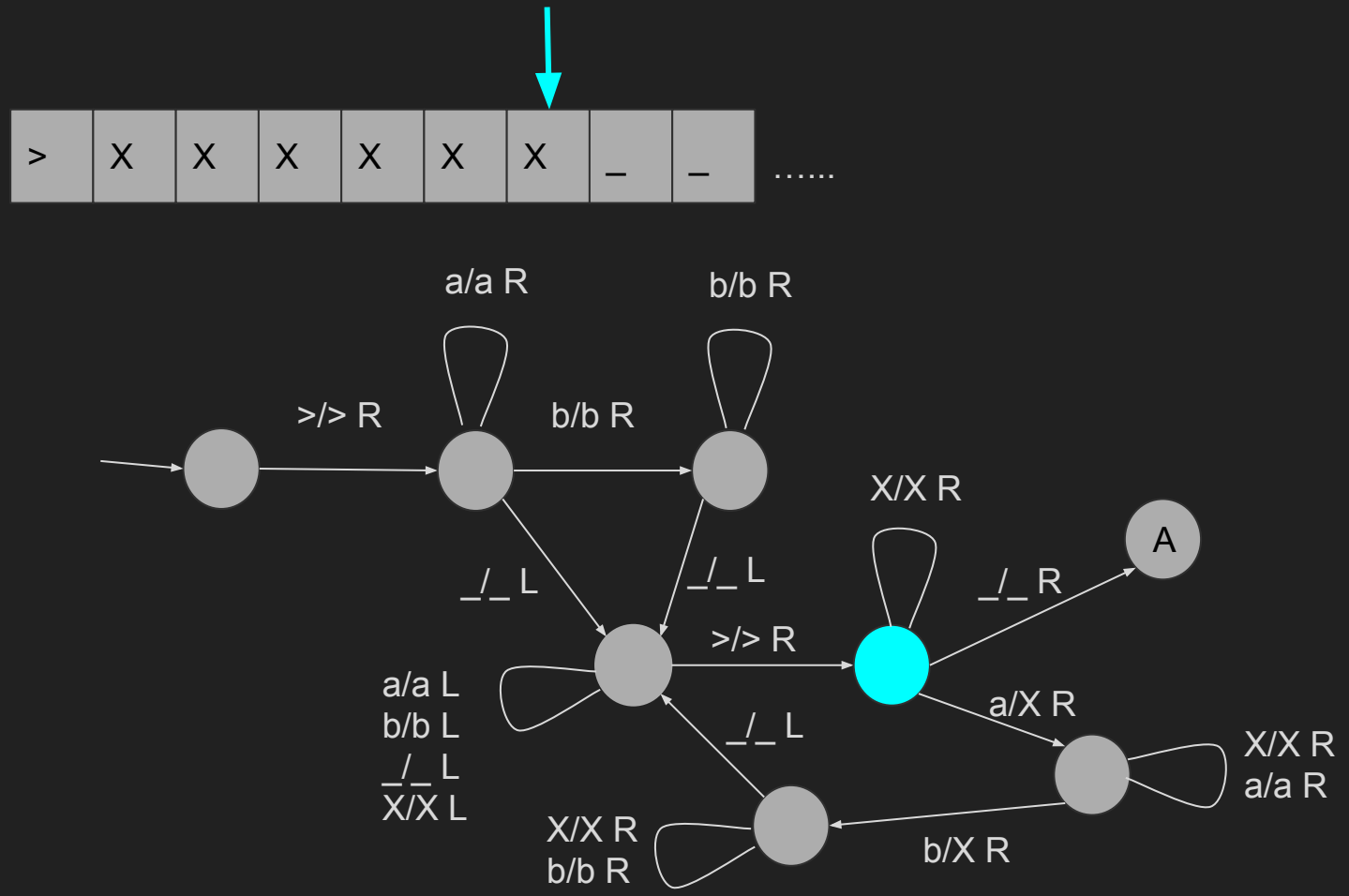
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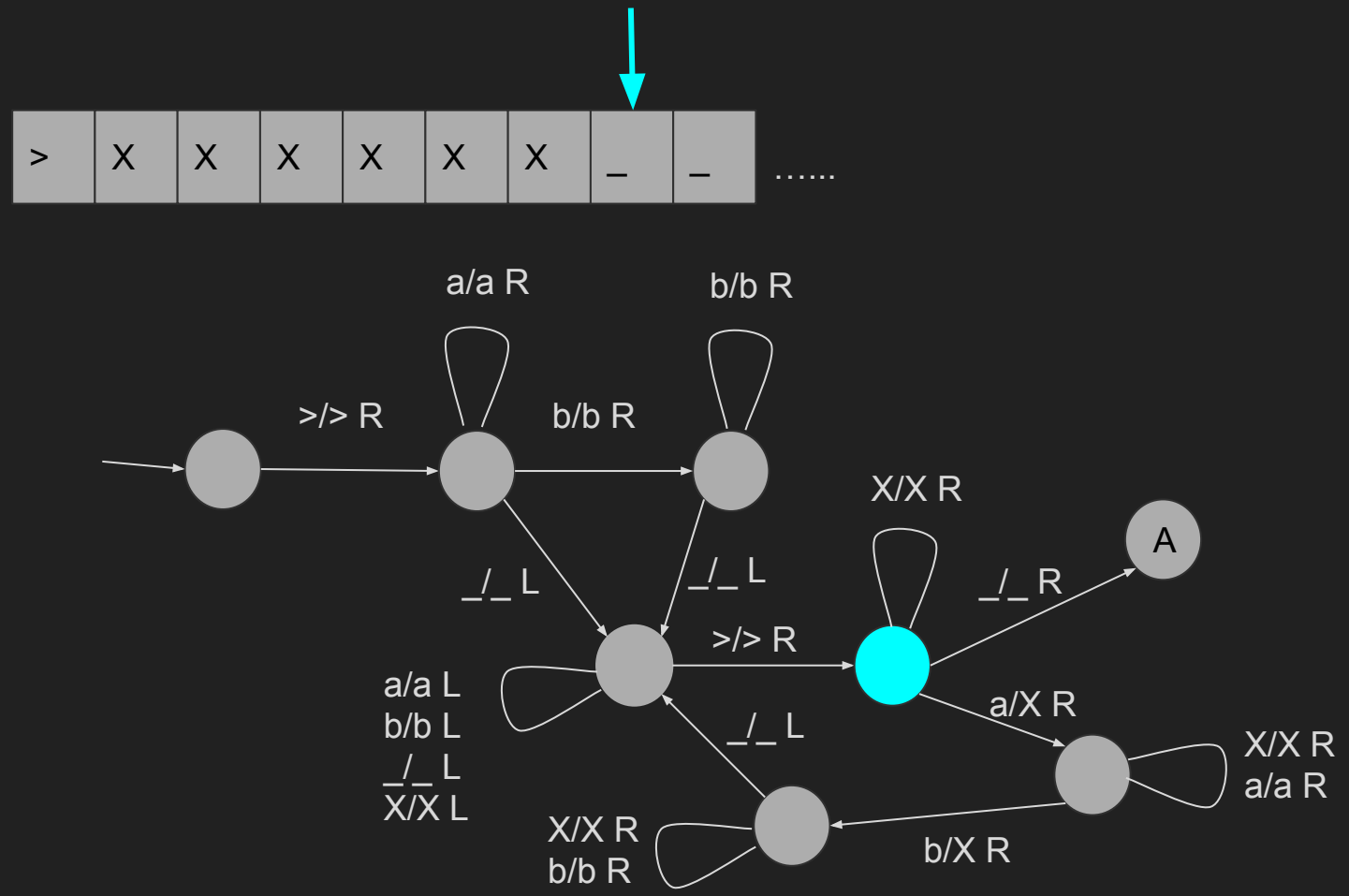
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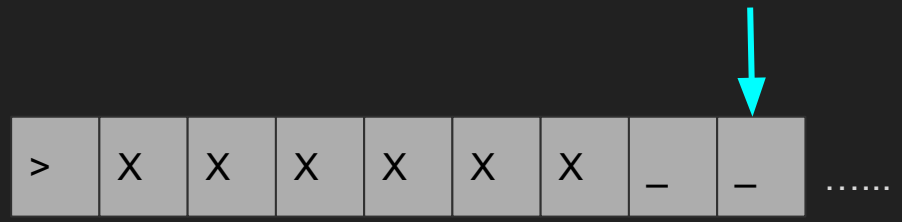
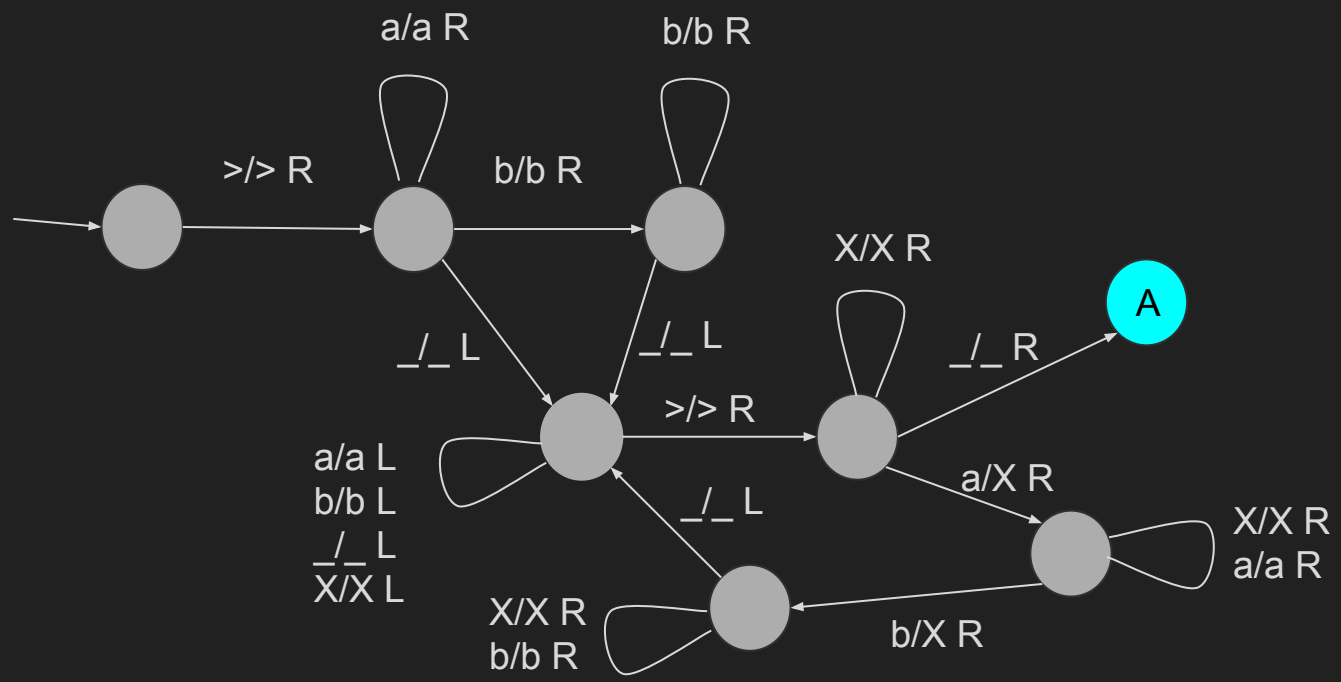


The Turing machine accepts strings *aaabbb*



Example — $\{u \in \{a,b\}^* \mid u = rev(u)\}$

Exercise! Answer next week